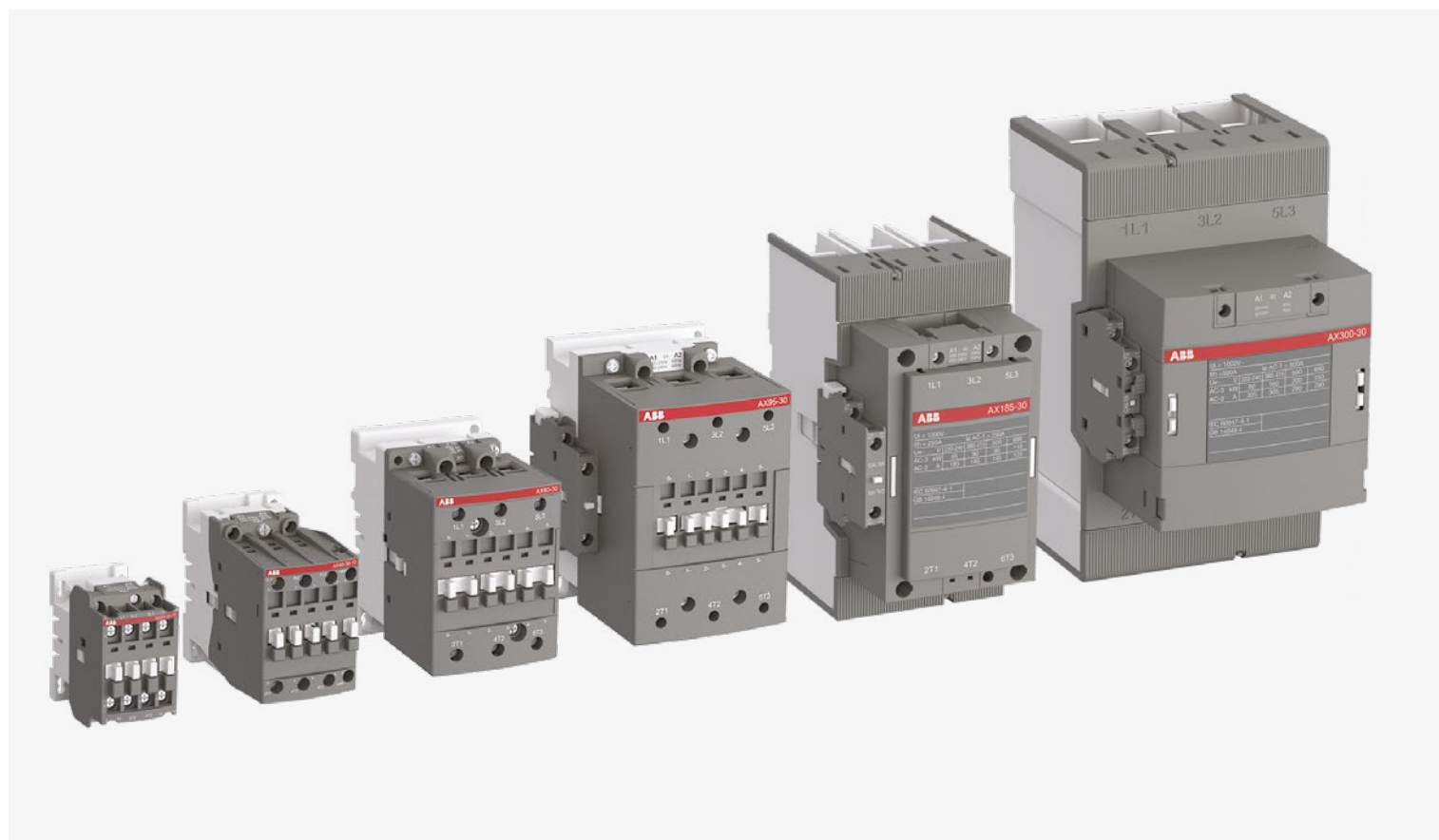


MAIN CATALOG

Motor protection and control

AX09 ... AX370 contactors

Overload relays



- Speed-up your projects
- Easy to install
- Continuous operation.

AX contactor range
Control made simple

Motor rated operational powers and currents

The currents given below concern standard three-phase four-pole cage motors (1500 r.p.m. at 50 Hz 1800 r.p.m. at 60 Hz). These values are given for guidance and may vary according to the motor manufacturer and depending on the number of poles.

IEC	Motor nominal current: standardized values in grey colour (according to IEC 60947-4-1 Annex G)									
Motor power	220 V	230 V	240 V	380 V	400 V	415 V	440 V	500 V	660 V	690 V
kW	A	A	A	A	A	A	A	A	A	A
0.06	0.37	0.35	0.34	0.21	0.2	0.19	0.18	0.16	0.13	0.12
0.09	0.54	0.52	0.50	0.32	0.3	0.29	0.26	0.24	0.18	0.17
0.12	0.73	0.7	0.67	0.46	0.44	0.42	0.39	0.32	0.24	0.23
0.18	1	1	1	0.63	0.6	0.58	0.53	0.48	0.37	0.35
0.25	1.6	1.5	1.4	0.9	0.85	0.82	0.74	0.68	0.51	0.49
0.37	2.0	1.9	1.8	1.2	1.1	1.1	1	0.88	0.67	0.64
0.55	2.7	2.6	2.5	1.6	1.5	1.4	1.3	1.2	0.91	0.87
0.75	3.5	3.3	3.2	2.0	1.9	1.8	1.7	1.5	1.15	1.1
1.1	4.9	4.7	4.5	2.8	2.7	2.6	2.4	2.2	1.7	1.6
1.5	6.6	6.3	6	3.8	3.6	3.5	3.2	2.9	2.2	2.1
2.2	8.9	8.5	8.1	5.2	4.9	4.7	4.3	3.9	2.9	2.8
3	11.8	11.3	10.8	6.8	6.5	6.3	5.7	5.2	4	3.8
4	15.7	15	14.4	8.9	8.5	8.2	7.4	6.8	5.1	4.9
5.5	20.9	20	19.2	12.1	11.5	11.1	10.1	9.2	7	6.7
7.5	28.2	27	25.9	16.3	15.5	14.9	13.6	12.4	9.3	8.9
11	39.7	38	36.4	23.2	22	21.2	19.3	17.6	13.4	12.8
15	53.3	51	48.9	30.5	29	28	25.4	23	17.8	17
18.5	63.8	61	58.5	36.8	35	33.7	30.7	28	22	21
22	75.3	72	69	43.2	41	39.5	35.9	33	25.1	24
30	100	96	92	57.9	55	53	48.2	44	33.5	32
37	120	115	110	69	66	64	58	53	40.8	39
45	146	140	134	84	80	77	70	64	49.1	47
55	177	169	162	102	97	93	85	78	59.6	57
75	240	230	220	139	132	127	116	106	81	77
90	291	278	266	168	160	154	140	128	97	93
110	355	340	326	205	195	188	171	156	118	113
132	418	400	383	242	230	222	202	184	140	134
160	509	487	467	295	280	270	245	224	169	162
200	637	609	584	368	350	337	307	280	212	203
250	782	748	717	453	430	414	377	344	261	250
315	983	940	901	568	540	520	473	432	327	313
355	1109	1061	1017	642	610	588	535	488	370	354
400	1255	1200	1150	726	690	665	605	552	418	400
500	1545	1478	1416	895	850	819	745	680	515	493
560	1727	1652	1583	1000	950	916	832	760	576	551
630	1928	1844	1767	1116	1060	1022	929	848	643	615
710	2164	2070	1984	1253	1190	1147	1043	952	721	690
800	2446	2340	2243	1417	1346	1297	1179	1076	815	780
900	2760	2640	2530	1598	1518	1463	1330	1214	920	880
1000	3042	2910	2789	1761	1673	1613	1466	1339	1014	970

UL/CSA	Motor nominal current: single and three phase (according to UL 60947-4-1A)									
Motor power	120 V 1ph	200 V 1ph	200 V 3ph	208 V 1ph	208 V 3ph	220- 240 V 1ph	220- 240 V 3ph	380- 415 V 3ph	440- 480 V 3ph	550- 600 V 3ph
hp	A	A	A	A	A	A	A	A	A	A
1/10	3	-	-	-	-	1.5	-	-	-	-
1/8	3.8	-	-	-	-	1.9	-	-	-	-
1/6	4.4	2.5	-	2.4	-	2.2	-	-	-	-
1/4	5.8	3.3	-	3.2	-	2.9	-	-	-	-
1/3	7.2	4.1	-	4	-	3.6	-	-	-	-
1/2	9.8	5.6	2.5	5.4	2.4	4.9	2.2	1.3	1.1	0.9
3/4	13.8	7.9	3.7	7.6	3.5	6.9	3.2	1.8	1.6	1.3
1	16	9.2	4.8	8.8	4.6	8	4.2	2.3	2.1	1.7
1-1/2	20	11.5	6.9	11	6.6	10	6	3.3	3	2.4
2	24	13.8	7.8	13.2	7.5	12	6.8	4.3	3.4	2.7
3	34	19.6	11	18.7	10.6	17	9.6	6.1	4.8	3.9
5	56	32.2	17.5	30.8	16.7	28	15.2	9.7	7.6	6.1
7-1/2	80	46	25.3	44	24.2	40	22	14	11	9
10	100	57.5	32.2	55	30.8	50	28	18	14	11
15	135	-	48.3	-	46.2	68	42	27	21	17
20	-	-	62.1	-	59.4	88	54	34	27	22
25	-	-	78.2	-	74.8	110	68	44	34	27
30	-	-	92	-	88	136	80	51	40	32
40	-	-	120	-	114	176	104	66	52	41
50	-	-	150	-	143	216	130	83	65	52
60	-	-	177	-	169	-	154	103	77	62
75	-	-	221	-	211	-	192	128	96	77
100	-	-	285	-	273	-	248	165	124	99
125	-	-	359	-	343	-	312	208	156	125
150	-	-	414	-	396	-	360	240	180	144
200	-	-	552	-	528	-	480	320	240	192
250	-	-	-	-	-	-	604	403	302	242
300	-	-	-	-	-	-	722	482	361	289
350	-	-	-	-	-	-	828	560	414	336
400	-	-	-	-	-	-	954	636	477	382
450	-	-	-	-	-	-	1030	-	515	412
500	-	-	-	-	-	-	1180	786	590	472



AX contactor range

Control made simple
The performance you need

OVERVIEW
P. 8

CONTACTORS
P. 10

**MANUAL MOTOR
STARTERS**
P. 82

OVERLOAD RELAYS
P. 98

**GENERAL
TECHNICAL DATA**
P. 117

INDEX
P. 124

01

02

03

04

05

06

AX contactors

The simplest way to get the control and performance you need

The AX contactor range offers exceptional reliability and performance in a brilliant, space-saving design. Use it for motor starting applications up to 370 A / 200 kW 400 V AC-3.



Speed-up your projects

Simpler selection process

Speed up your projects with ABB's simpler order codes, faster identification, easier connection, and a complete and flexible range of accessories.

Compliance process is faster too as AX contactors use environmentally friendly materials that comply with energy label.



Easy to install

Faster fitting by design

ABB's smart design saves time with every detail. AX contactors are smaller and easier to handle. All terminals are delivered in open position so wiring is faster.

ABB's broad range provides the best configuration for the job. Single or multiple pole blocks are no problem. Front or side mounted auxiliary contact blocks are available as well.



Continuous operation

Proven, secure, trusted

Trust a proven solution from a brand with 100 years of experience in contactors design and manufacture. ABB's AX range makes starting solutions that are more reliable - with type 2 coordination between contactors and short-circuit protection devices guaranteed. ABB's mechanically linked contacts and mirror contact functions make control circuits safe and reliable.

Save time when building motor starting solution with AX contactors

Complete range compatible with ABB low voltage solutions



Tested component combinations

Using ABB's coordination tables gives users a choice of fully tested assemblies and product combinations. It's quicker and easier to build DOL starters, reversing starters or star delta starters using ABB's range of AX contactors, manual motor starters, molded case breakers, fuses and overload relays.

Create smart starters

AX contactors look more professional and, together with connection kits, they provide a better finish than cables or bus bars.

Save time

ABB starters come with connection kits to make assembly simpler and faster. The kits save time on cable preparation and eliminate fitment and wiring error risks.

AX contactors

Features and benefits

Every detail designed for you

Smart packaging design makes it simpler to identify the product you need – the product type, coil voltage, order code and bar code are all clearly displayed. The same goes when the product is unboxed. A quick glance at the front tells you what product, contactor type and coil voltage you have. Terminal markings are also plainly visible. The rated values and main approvals are ready to read on the side.



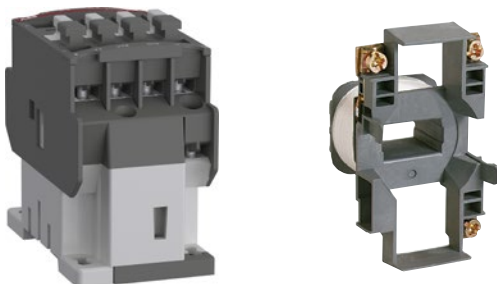
Certified, trusted contactors

ABB's AX contactors are designed in compliance with IEC 60947-4-1 and GB 14048-4 requirements. These trusted safety products have CB certification, CE marking as well as UL, CCC and CCS approval.



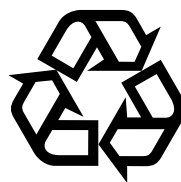
Easier to connect

ABB designed its AX contactors so that all screw-heads are accessible from the front. One Pozidriv #2 screwdriver fits every contactor terminal and the complete accessory range. All main and auxiliary terminals can take one or two cables and contactors up to AX150 have three coil terminals for connection from the top or the bottom. Right out of the box, all terminals are ready in the open position for wiring.



Environmentally sound

The design and production of ABB's AX contactor range follows ISO 14000 processes. The raw materials are free of red phosphorous, cadmium, mercury, brominated substances (PBB, PBPE) and other pollutants. AX contactors and main accessories also comply with the European directive ROHS 2011/65/EU incl. 2015/863/EU. The same goes for the packaging design. The box is fully recyclable and clearly marked to aid correct disposal.

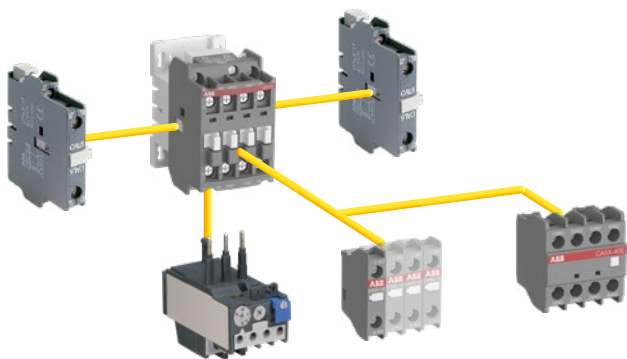


AX contactors

Features and benefits

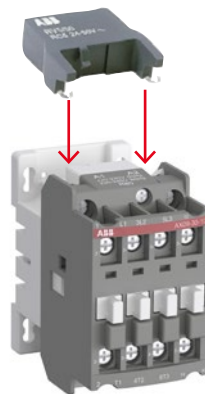
Auxiliaries made simple, secure and flexible

Common interfaces that are clip-on and self-locking make mounting ABB contactors simpler. Its AX contactor range uses the same front-mounted auxiliary contact blocks up to AX150 and the same side-mounted auxiliary contact blocks up to AX80. To maximize flexibility, users can front-mount any single- or four-pole auxiliary contact block. Two-pole auxiliary contact block can be side-mounted.



Protect control circuits and save space

ABB's AX range makes surge suppressors easy to snap on and connect. Designing contactor coils to them fit within their overall dimensions without additional space requirements. The smart design and proven technologies provide safe protection for circuits against over-voltages when the contactor opens.

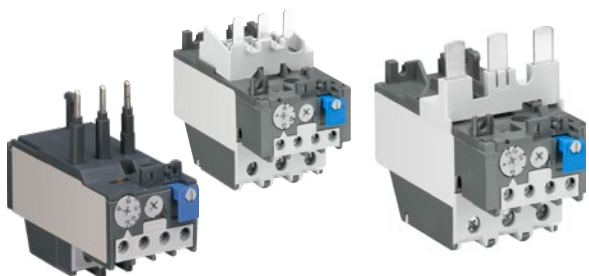


Complete range of multi-function overload relays

ABB's complete range of thermal overload relays provides class 10 protection and key functions including:

- Motor protection against overload and phase failure
- Automatic and manual reset both included
- Test and stop functions

ABB's thermal overload relays are suitable for three-phase or single-phase motor applications with temperature compensation between -25 °C and +55 °C.

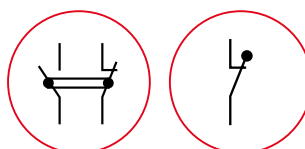


Safe and reliable control circuits

A white contact carrier ensures the contactor state is visible at all times, even with accessories mounted.

Built-in and add-on auxiliary contacts provide reliable low signal contacts for 12 V 3 mA. Failure rates are less than 10⁻⁶ according to IEC 60947-5-1.

Mechanically linked functions are available with 4-pole blocks CA5X. Mirror contacts are available with built-in NC contacts for AX09 ... AX40 and side-mounted blocks CAL5-11X and 4-pole blocks CA5X for AX09 ... AX80.





3-pole contactors for motor control and power switching



IEC	AC-3 Rated operational power $\theta \leq 55\text{ }^{\circ}\text{C}$, 400 V	kW	4	5.5	7.5	11	15	18.5	
AC Control supply 			Type	AX09	AX12	AX18	AX25	AX32	AX40
IEC	AC-3 Rated operational current $\theta \leq 55\text{ }^{\circ}\text{C}$, 400 V	A	9	12	18	25	32	40	
	AC-1 Rated operational current $\theta \leq 40\text{ }^{\circ}\text{C}$, 690 V	A	22	25	27	32	55	60	

Main accessories

Auxiliary contact blocks	Front mounting	CA5X-10 (1 x N.O.) CA5X-01 (1 x N.C.) CA5X-4 pole (add on block with 4 contacts N.O. or N.C. combination)
	Side mounting	CAL5X-11 (1 x N.O. + 1 x N.C.)
Timers	Electronic	TEF5-ON TEF5-OFF
Interlocking units	Mechanical	VM5-1
	Mechanical / Electrical	VE5-1
Surge suppressors	Varistor (AC / DC)	RV5 (24...440 V)
	RC type (AC)	RC5-1 (24...440 V)

Overload relays

Thermal relays 	Class 10A	TA25DU-M (0.1...32A)
		TA42DU-M (18...42 A)
Electronic relays 	Class 10E, 20E, 30E	

(1) The max. AC-3 operational current is 23 A for AX25 with TA25DU-25M.

(2) The max. AC-3 operational current is 74 A for AX80 with TA75DU-80M.

(3) The max. AC-3 operational current is 182 A for AX205 with TA200DU-200.

Manual motor starters

 	Thermal / magnetic protection Class 10	MS116 (0.10...32 A) Ics up to 50 kA for class 10A
	Accessories	
	For contactor mounting	BEA16/116 (4)
		BEA25/116 (4)
		BEA25/132 (5)

(4) AX.. with MS116-0.16 ... MS116-16

(5) AX25 with MS116-20 ... MS116-32



	22	30	37	45	55	75	90	110	132	160	200
	AX50	AX65	AX80	AX95	AX115	AX150	AX185	AX205	AX260	AX300	AX370
	50	65	80	96	115	150	185	205	265	305	370
	100	115	125	145	160	190	250	275	400	500	600

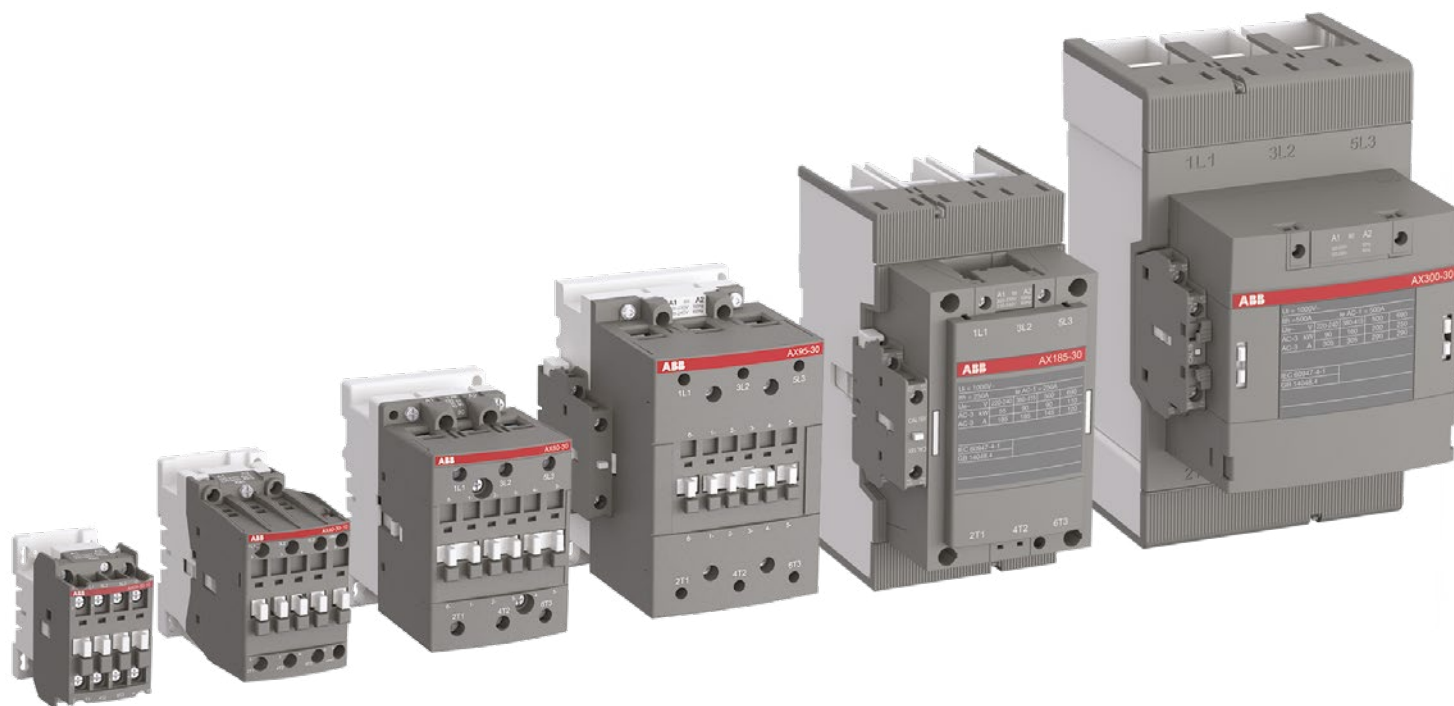
				CAL18X-11 (1 x N.O. + 1 x N.C.)			
VE5-2				VM300H			
RC5-2 (24...440 V)				RC5-3 (240...440 V)			

TA75DU-M (18...80 A) (2)		TA80DU-M		TA200DU (66...200 A)(3)	
		TA110DU-M			
				EF205 (63...210 A)	
				EF370 (115...380 A)	

Short-circuit protection devices

MCCB and switch fuses





AX contactors and NX contactor relays

	AX 3-pole contactors	
12	AX09 ... AX12	AC operated
13	AX18 ... AX25	AC operated
14	AX32, AX40	AC operated
15	AX50 ... AX80	AC operated
16	AX50 ... AX80	AC operated with 1 N.O. + 1 N.C.
17	AX95 ... AX150	AC operated with 1 N.O. + 1 N.C.
18	Main accessories	
21	AX185 ... AX205	AC operated with 1 N.O. + 1 N.C.
22	AX260... AX370	AC operated with 1 N.O. + 1 N.C.
23	Main accessories	
26	Technical data	
40	DC Circuit switching	
41	Electrical durability	
45	Star-delta starting	
47	Terminal marking and positioning	
48	Dimensions	
	NX contactor relays	
52	NX	AC operated
53	Main accessories	
54	Technical data	
56	Terminal marking and positioning	
57	Accessories for AX09 ... AX80 pole contactors contactor relays	
80	Voltage code table	

AX09 ... AX12 3-pole contactors

4 to 5.5 kW

AC operated



AX09 ... AX12

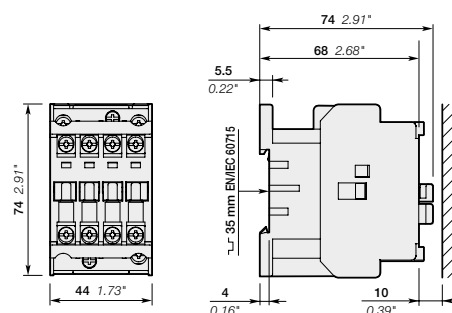
AX09 ... AX12 contactors are mainly used for controlling 3-phase motors and power circuits up to 690 V AC.

These contactors are of the block type design with:

- 3 main poles and 1 built-in auxiliary contact
- control circuit: AC operated
- add-on auxiliary contact blocks for front or side mounting and a wide range of accessories.

IEC		Rated control circuit voltage Uc (1)		Auxiliary contacts fitted	Type	Order code	Weight (1 pce) kg
Rated operational power 400 V AC-3 kW	current $\theta \leq 40^\circ\text{C}$ AC-1 A	V 50 Hz	V 60 Hz				
4	22	24	24	1 0	AX09-30-10-81	1SBL901074R8110	0.340
				0 1	AX09-30-01-81	1SBL901074R8101	0.340
		110	110...120	1 0	AX09-30-10-84	1SBL901074R8410	0.340
				0 1	AX09-30-01-84	1SBL901074R8401	0.340
		220...230	230...240	1 0	AX09-30-10-80	1SBL901074R8010	0.340
				0 1	AX09-30-01-80	1SBL901074R8001	0.340
		230...240	240...260	1 0	AX09-30-10-88	1SBL901074R8810	0.340
				0 1	AX09-30-01-88	1SBL901074R8801	0.340
5.5	25	24	24	1 0	AX09-30-10-86	1SBL901074R8610	0.340
				0 1	AX09-30-01-86	1SBL901074R8601	0.340
		110	110...120	1 0	AX12-30-10-81	1SBL911074R8110	0.340
				0 1	AX12-30-01-81	1SBL911074R8101	0.340
		220...230	230...240	1 0	AX12-30-10-84	1SBL911074R8410	0.340
				0 1	AX12-30-01-84	1SBL911074R8401	0.340
		230...240	240...260	1 0	AX12-30-10-80	1SBL911074R8010	0.340
				0 1	AX12-30-01-80	1SBL911074R8001	0.340
		230...240	240...260	1 0	AX12-30-10-88	1SBL911074R8810	0.340
				0 1	AX12-30-01-88	1SBL911074R8801	0.340
		400...415	415...440	1 0	AX12-30-10-85	1SBL911074R8501	0.340
				0 1	AX12-30-01-85	1SBL911074R8501	0.340
		400...415	415...440	1 0	AX12-30-10-86	1SBL911074R8610	0.340
				0 1	AX12-30-01-86	1SBL911074R8601	0.340

(1) For other voltage version see voltage code table.



AX09, AX12

Main dimensions mm, inches

AX18 ... AX25 3-pole contactors

7.5 to 11 kW

AC operated



AX18

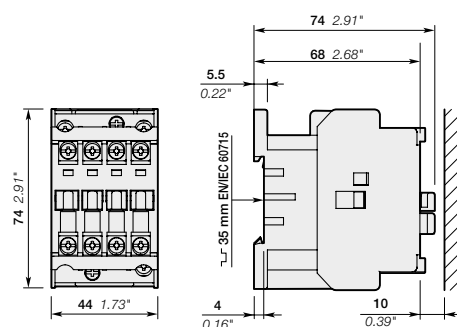
AX18 ... AX25 contactors are mainly used for controlling 3-phase motors and power circuits up to 690 V AC.

These contactors are of the block type design with:

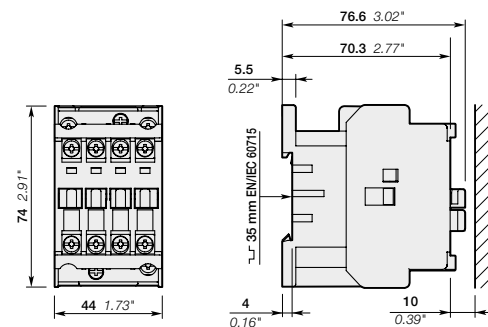
- 3 main poles and 1 built-in auxiliary contact
- control circuit: AC operated
- add-on auxiliary contact blocks for front or side mounting and a wide range of accessories.

IEC		Rated control circuit voltage Uc (1)		Auxiliary contacts fitted	Type	Order code	Weight (1 pce)
Rated operational power 400 V AC-3 kW	current $\theta \leq 40^\circ\text{C}$ AC-1 A	V 50 Hz	V 60 Hz				
7.5	27	24	24	1 0	AX18-30-10-81	1SBL921074R8110	0.340
				0 1	AX18-30-01-81	1SBL921074R8101	0.340
		110	110...120	1 0	AX18-30-10-84	1SBL921074R8410	0.340
				0 1	AX18-30-01-84	1SBL921074R8401	0.340
		220...230	230...240	1 0	AX18-30-10-80	1SBL921074R8010	0.340
				0 1	AX18-30-01-80	1SBL921074R8001	0.340
		230...240	240...260	1 0	AX18-30-10-88	1SBL921074R8810	0.340
				0 1	AX18-30-01-88	1SBL921074R8801	0.340
		400...415	415...440	1 0	AX18-30-10-86	1SBL921074R8610	0.340
				0 1	AX18-30-01-86	1SBL921074R8601	0.340
11	32	24	24	1 0	AX25-30-10-81	1SBL931074R8110	0.340
				0 1	AX25-30-01-81	1SBL931074R8101	0.340
		110	110...120	1 0	AX25-30-10-84	1SBL931074R8410	0.340
				0 1	AX25-30-01-84	1SBL931074R8401	0.340
		220...230	230...240	1 0	AX25-30-10-80	1SBL931074R8010	0.340
				0 1	AX25-30-01-80	1SBL931074R8001	0.340
		230...240	240...260	1 0	AX25-30-10-88	1SBL931074R8810	0.340
				0 1	AX25-30-01-85	1SBL931074R8501	0.340
		400...415	415...440	1 0	AX25-30-10-86	1SBL931074R8610	0.340
				0 1	AX25-30-01-86	1SBL931074R8601	0.340

(1) For other voltage version see voltage code table.



AX18



AX25

Main dimensions mm, inches

AX32, AX40 3-pole contactors

15 to 18.5 kW

AC operated



AX32, AX40

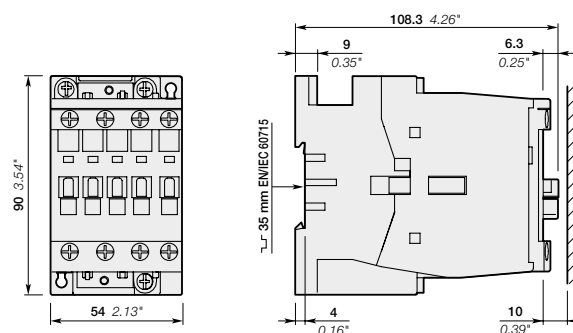
AX32, AX40 contactors are mainly used for controlling 3-phase motors and power circuits up to 690 V AC.

These contactors are of the block type design with:

- 3 main poles and 1 built-in auxiliary contact
- control circuit: AC operated
- add-on auxiliary contact blocks for front or side mounting and a wide range of accessories.

IEC		Rated control circuit voltage Uc (1)		Auxiliary contacts fitted	Type	Order code	Weight (1 pce)
Rated operational power 400 V AC-3 kW	current $\theta \leq 40^\circ\text{C}$ AC-1 A	V 50 Hz	V 60 Hz				kg
15	55	24	24	1 0	AX32-30-10-81	1SBL281074R8110	0.71
				0 1	AX32-30-01-81	1SBL281074R8101	0.71
		110	110...120	1 0	AX32-30-10-84	1SBL281074R8410	0.71
				0 1	AX32-30-01-84	1SBL281074R8401	0.71
		220...230	230...240	1 0	AX32-30-10-80	1SBL281074R8010	0.71
				0 1	AX32-30-01-80	1SBL281074R8001	0.71
		230...240	240...260	1 0	AX32-30-10-88	1SBL281074R8810	0.71
				0 1	AX32-30-01-88	1SBL281074R8801	0.71
		400...415	415...440	1 0	AX32-30-10-86	1SBL281074R8610	0.71
				0 1	AX32-30-01-86	1SBL281074R8601	0.71
18.5	60	24	24	1 0	AX40-30-10-81	1SBL321074R8110	0.71
				0 1	AX40-30-01-81	1SBL321074R8101	0.71
		110	110...120	1 0	AX40-30-10-84	1SBL321074R8410	0.71
				0 1	AX40-30-01-84	1SBL321074R8401	0.71
		220...230	230...240	1 0	AX40-30-10-80	1SBL321074R8010	0.71
				0 1	AX40-30-01-80	1SBL321074R8001	0.71
		230...240	240...260	1 0	AX40-30-10-88	1SBL321074R8810	0.71
				0 1	AX40-30-01-88	1SBL321074R8801	0.71
		400...415	415...440	1 0	AX40-30-10-86	1SBL321074R8610	0.71
				0 1	AX40-30-01-86	1SBL321074R8601	0.71

(1) For other voltage version see voltage code table.



AX32, AX40

Main dimensions mm, inches

AX50 ... AX80 3-pole contactors

22 to 37 kW

AC operated



AX50 ... AX80

AX50 ... AX80 contactors are mainly used for controlling 3-phase motors and power circuits up to 690 V AC.

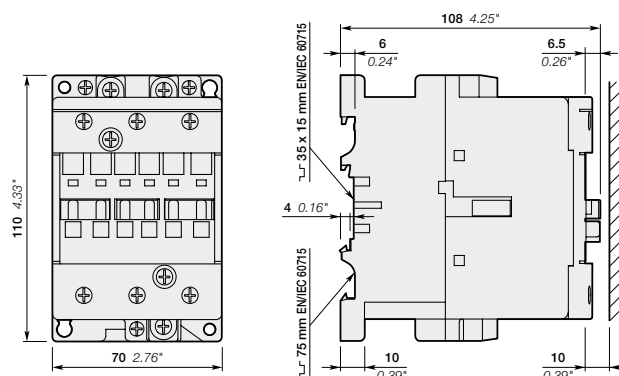
These contactors are of the block type design with:

- 3 main poles
- control circuit: AC operated
- add-on auxiliary contact blocks for front or side mounting and a wide range of accessories.

Ordering details (without auxiliary block)

IEC		Rated control circuit voltage U _c (1)		Auxiliary contacts fitted	Type	Order code	Weight (1 pce)
Rated operational power 400 V AC-3 kW	current θ ≤ 40 °C AC-1 A	V 50 Hz	V 60 Hz				
22	100	24	24	0 0	AX50-30-00-81	1SBL351074R8100	1.120
		110	110...120	0 0	AX50-30-00-84	1SBL351074R8400	1.120
		220...230	230...240	0 0	AX50-30-00-80	1SBL351074R8000	1.120
		230...240	240...260	0 0	AX50-30-00-88	1SBL351074R8800	1.120
		400...415	415...440	0 0	AX50-30-00-86	1SBL351074R8600	1.120
30	115	24	24	0 0	AX65-30-00-81	1SBL371074R8100	1.120
		110	110...120	0 0	AX65-30-00-84	1SBL371074R8400	1.120
		220...230	230...240	0 0	AX65-30-00-80	1SBL371074R8000	1.120
		230...240	240...260	0 0	AX65-30-00-88	1SBL371074R8800	1.120
		400...415	415...440	0 0	AX65-30-00-86	1SBL371074R8600	1.120
37	125	24	24	0 0	AX80-30-00-81	1SBL411074R8100	1.120
		110	110...120	0 0	AX80-30-00-84	1SBL411074R8400	1.120
		220...230	230...240	0 0	AX80-30-00-80	1SBL411074R8000	1.120
		230...240	240...260	0 0	AX80-30-00-88	1SBL411074R8800	1.120
		400...415	415...440	0 0	AX80-30-00-86	1SBL411074R8600	1.120

(1) For other voltage version see voltage code table.



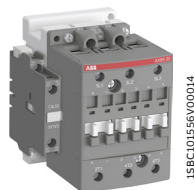
AX50, AX65, AX80

Main dimensions mm, inches

AX50 ... AX80 3-pole contactors

22 to 37 kW

AC operated with 1 N.O. + 1 N.C. auxiliary contacts



AX50 ... AX80

AX50 ... AX80 contactors are mainly used for controlling 3-phase motors and power circuits up to 690 V AC.

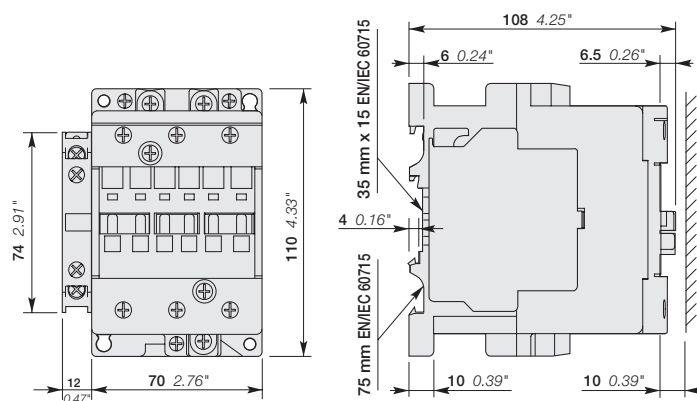
These contactors are of the block type design with:

- 3 main poles
- control circuit: AC operated
- add-on auxiliary contact blocks for front or side mounting and a wide range of accessories.

Ordering details (without auxiliary block)

IEC		Rated control circuit voltage U _c (1)		Auxiliary contacts fitted	Type	Order code	Weight (1 pce) kg
Rated operational power	current $\theta \leq 40^\circ\text{C}$	V 50 Hz	V 60 Hz				
400 V AC-3 kW	AC-1 A						
22	100	24	24	0 0	AX50-30-11-81	1SBL351074R8111	1.160
		110	110...120	0 0	AX50-30-11-84	1SBL351074R8411	1.160
		220...230	230...240	0 0	AX50-30-11-80	1SBL351074R8011	1.160
		230...240	240...260	0 0	AX50-30-11-88	1SBL351074R8811	1.160
		400...415	415...440	0 0	AX50-30-11-86	1SBL351074R8611	1.160
30	115	24	24	0 0	AX65-30-11-81	1SBL371074R8111	1.160
		110	110...120	0 0	AX65-30-11-84	1SBL371074R8411	1.160
		220...230	230...240	0 0	AX65-30-11-80	1SBL371074R8011	1.160
		230...240	240...260	0 0	AX65-30-11-88	1SBL371074R8811	1.160
		400...415	415...440	0 0	AX65-30-11-86	1SBL371074R8611	1.160
37	125	24	24	0 0	AX80-30-11-81	1SBL411074R8111	1.160
		110	110...120	0 0	AX80-30-11-84	1SBL411074R8411	1.160
		220...230	230...240	0 0	AX80-30-11-80	1SBL411074R8011	1.160
		230...240	240...260	0 0	AX80-30-11-88	1SBL411074R8811	1.160
		400...415	415...440	0 0	AX80-30-11-86	1SBL411074R8611	1.160

(1) For other voltage version see voltage code table.



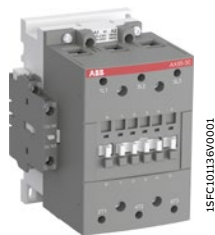
AX50, AX65, AX80

Main dimensions mm, inches

AX95 ... AX150 3-pole contactors

45 to 75 kW

AC operated with 1 N.O. + 1 N.C. auxiliary contacts



AX95 ... AX150

AX95 ... AX150 contactors are mainly used for controlling 3-phase motors and power circuits up to 690 V AC.

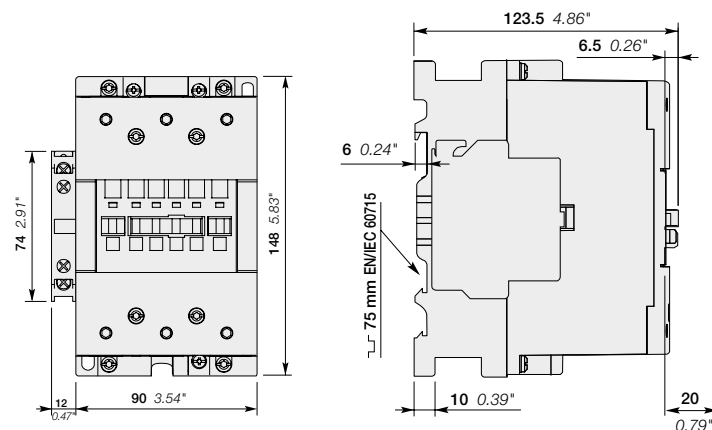
These contactors are of the block type design with:

- 3 main poles
- control circuit: AC operated
- add-on auxiliary contact blocks for front or side mounting and a wide range of accessories.

Ordering details (without auxiliary block)

IEC		Rated control circuit voltage U _c (1)		Auxiliary contacts fitted	Type	Order code	Weight (1 pce) kg
Rated operational power 400 V AC-3 kW	current $\theta \leq 40^\circ\text{C}$ AC-1 A	V 50 Hz	V 60 Hz				
45	145	24	24	0 0	AX95-30-11-81	1SFL431074R8111	2.080
		110	110...120	0 0	AX95-30-11-84	1SFL431074R8411	2.080
		220...230	230...240	0 0	AX95-30-11-80	1SFL431074R8011	2.080
		230...240	240...260	0 0	AX95-30-11-88	1SFL431074R8811	2.080
		400...415	415...440	0 0	AX95-30-11-86	1SFL431074R8611	2.080
55	160	24	24	0 0	AX115-30-11-81	1SFL981074R8111	2.080
		110	110...120	0 0	AX115-30-11-84	1SFL981074R8411	2.080
		220...230	230...240	0 0	AX115-30-11-80	1SFL981074R8011	2.080
		230...240	240...260	0 0	AX115-30-11-88	1SFL981074R8811	2.080
		400...415	415...440	0 0	AX115-30-11-86	1SFL981074R8611	2.080
75	190	24	24	0 0	AX150-30-11-81	1SFL991074R8111	2.080
		110	110...120	0 0	AX150-30-11-84	1SFL991074R8411	2.080
		220...230	230...240	0 0	AX150-30-11-80	1SFL991074R8011	2.080
		230...240	240...260	0 0	AX150-30-11-88	1SFL991074R8811	2.080
		400...415	415...440	0 0	AX150-30-11-86	1SFL991074R8611	2.080

(1) For other voltage version see voltage code table.



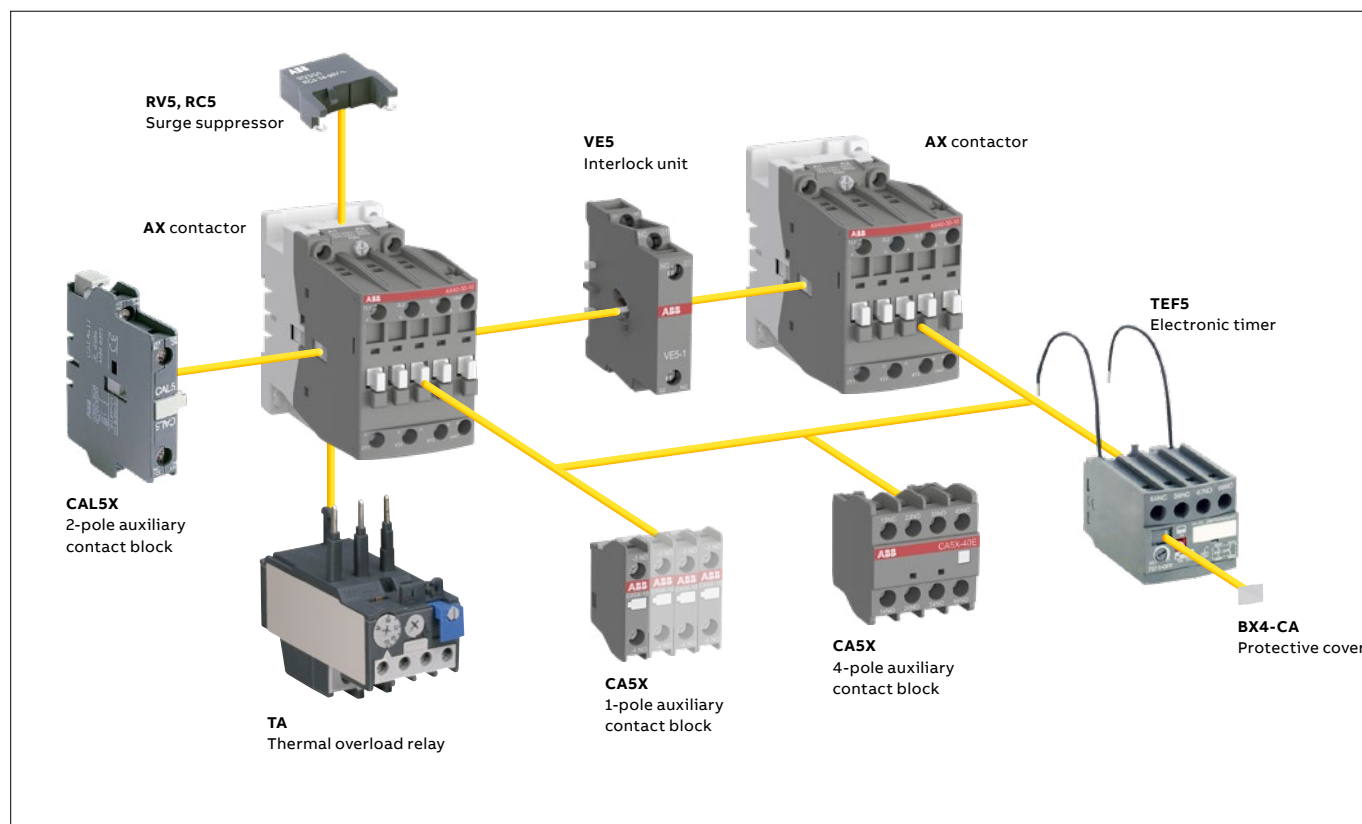
AX95, AX115, AX150

Main dimensions mm, inches

AX09 ... AX150 3-pole contactors

Main accessories

Contactor and main accessories (other accessories available)



Main accessory fitting details

Many configurations of accessories are possible depending on whether these are front-mounted or side-mounted.

Contactor types	Main poles	Built-in auxiliary contacts	Front-mounted accessories				Side-mounted accessories		
			Auxiliary contact blocks			Electronic timer	Auxiliary contact blocks		Interlock unit
									
			1-pole CA5X		4-pole CA5X		TEF5	2-pole CAL	VM5 or VE5
AX09 ... AX25	3 0 3 0	1 0 0 1 (1)	1 to 4 x CA5X 1 to 2 x CE5 max (2)	or	1 x CA5X (4-pole)	or	1 x TEF5	1 to 2 x CAL5X-11	or 1 x VM5-1 or VE5-1 + 1 x CAL5X-11
AX32, AX40	3 0 3 0	1 0 0 1 (1)	1 to 5 x CA5X 1 to 3 x CE5 max (3)	or	1 x CA5X (4-pole) + 3 x 1-pole CA5X or CE5 (3)	or	1 x TEF5 + 1 x CA5X (1-pole)	1 to 2 x CAL5X-11	or 1 x VM5-1 or VE5-1 + 1 x CAL5X-11
AX50 ... AX80	3 0	0 0	1 to 6 x CA5X 1 to 5 x CE5 max (4)	or	1 x CA5X (4-pole) + 2 x 1-pole CA5X or CE5 (4)	or	1 x TEF5 + 2 x CA5X (1-pole)	2 x CAL5X-11	or 1 x VE5-2
AX50 ... AX80	3 0	1 1	1 to 6 x CA5X 1 to 5 x CE5 max (4)	or	1 x CA5X (4-pole) + 2 x 1-pole CA5X or CE5 (4)	or	1 x TEF5 + 2 x CA5X (1-pole)	1 x CAL5X-11	or 1 x VE5-2
AX95 ... AX150	3 0	1 1	1 to 6 x CA5X	or	1 x CA5X (4-pole) + 2 x 1-pole CA5X	–	+	1 x CAL18X-11	or 1 x VE5-2

(1) 2 N.C. CA5X auxiliary contacts maximum in mounting position 5. for mounting position refer technical data page.

(2) The total number of N.O. or N.C. CE5 and other N.C. CA5X is limited to 2. CE5 not allowed in mounting position 5.

(3) The total number of N.O. or N.C. CE5 and other N.C. CA5X is limited to 3. CE5 not allowed in mounting position 5.

(4) The total number of N.O. or N.C. CE5 and other N.C. CA5X is limited to 5. CE5 not allowed in mounting position 5.

Overload relays fitting details (1)

Contactor types	Thermal overload relays	Electronic overload relays
AX09 ... AX25	TA25DU-M (0.1...32 A)	–
AX32, AX40	TA25DU-M (0.1...32 A) or TA42DU-M (18...42 A)	–
AX50 ... AX80	TA75DU-M (18...80 A)	–
AX95 ... AX150	TA80DU-M (29...80 A) or TA110DU-M (66...110 A)	–

The addition of a thermal overload relay on the contactor does not prevent fitting of many other accessories as shown above.

(1) Direct mounting - No kit required.

AX09 ... AX150 3-pole contactors

Main accessories



CA5X-10



CA5X-4P



CAL5X-11



VE5-1

For contactors	Auxiliary contacts	Type	Order code	Pkg qty	Weight (1 pce)
					kg

Front-mounted instantaneous auxiliary contact blocks

AX09 ... AX150 and NX 4-pole	1 -	CA5X-10	1SBN019010R1010	10	0.014
	- 1	CA5X-01	1SBN019010R1001	10	0.014
AX50 ... AX150	2 2	CA5X-22E	1SBN019040R1022	2	0.060
AX09 ... AX40-30-10	2 2	CA5X-22M	1SBN019040R1122	2	0.060

Side-mounted instantaneous auxiliary contact block, 2 poles

AX09 ... AX80 and NX - 4 pole	1 1	CAL5X-11	1SBN019020R1011	2	0.050
AX95 ... AX150 (1)	1 1	CAL18X-11	1SFN019820R1011	2	0.050

Mechanical interlock units for two horizontal mounted contactors (2)

Left side contactor	Right side contactor	Mounting					
AX09 ... AX40	AX09 ... AX40	Horizontal	- -	VM5-1	1SBN030100R1000	1	0.066
AX95 ... AX205	AX185 ... AX205	Horizontal	- -	VM300H	1SFN034700R1000	1	0.150

Mechanical interlock units for two vertical mounted contactors

Up contactor	Down contactor	Mounting					
AX95 ... AX150	AX150 ... AX370	Vertical	- -	VM300V	1SFN034701R1000	1	0.150

Mechanical and electrical interlock units for two horizontal mounted contactors

Left side contactor	Right side contactor	Mounting					
AX09 ... AX40	AX09 ... AX40	Horizontal	0 2	VE5-1	1SBN030110R1000	1	0.076
AX32 ... AX80	AX50 ... AX80	Horizontal	0 2	VE5-2	1SBN030210R1000	1	0.146
AX50 ... AX80	AX32 ... AX80	Horizontal	0 2	VE5-2	1SBN030210R1000	1	0.146
AX50 ... AX80	AX95 ... AX150	Horizontal	0 2	VE5-2 (3)	1SBN030210R1000	1	0.146
AX95 ... AX150	AX50 ... AX80	Horizontal	0 2	VE5-2 (3)	1SBN030210R1000	1	0.146
AX95 ... AX150	AX95 ... AX150	Horizontal	0 2	VE5-2	1SBN030210R1000	1	0.146

(1) See "Main accessory fitting details".

(2) Mechanical durability: VM5-1 = 5 millions cycles, VM300H = 1 million cycles.

(3) The combination of AX50 ... AX80 contactors interlocked with AX95 ... AX150 contactors cannot be mounted on symmetrical rail (75 mm, IEC/EN 60715).

AX09 ... AX150 3-pole contactors

Main accessories



1SBCE01396F0014

TEF5-OFF



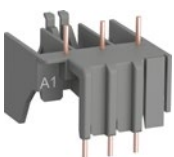
1SBCE56548F0301

WB75-A



1SBCE57400F0301

RV5/50



1SBCE582813F0301

BEA

For contactors	Time delay range selected by switch	Delay type	Auxiliary contacts	Type	Order code	Pkg qty	Weight (1 pce)
							kg

Electronic timers

AX09 ... AX80	0.1...1 s	ON-delay	1 1	TEF5-ON	1SBN020312R1000	1	0.065
NX 4 pole	1...10 s 10...100 s	OFF-delay	1 1	TEF5-OFF	1SBN020314R1000	1	0.065

Note: Rated control circuit voltage Uc 24...240 V 50/60 Hz or DC.

For contactors	Rated control circuit voltage Uc		Type	Order code	Pkg qty	Weight (1 pce)
	50Hz	60 Hz				kg

Mechanical latching units

AX09 ... AX80	24	24...28	WB75-A	FPTN372726R1001	1	0.120
	220...230	220...255	WB75-A	FPTN372726R1006	1	0.120

(1) See "Main accessory fitting details".

For contactors	Rated control circuit voltage Uc	Type	Order code	Pkg qty	Weight (1 pce)
	V AC				kg

Surge suppressors

AX09 ... AX150	24...50	RV5/50	1SBN050010R1000	2	0.015
	50...133	RV5/133	1SBN050010R1001	2	0.015
	110...250	RV5/250	1SBN050010R1002	2	0.015
	250...440	RV5/440	1SBN050010R1003	2	0.015
AX09 ... AX40	24...50	RC5-1/50	1SBN050100R1000	2	0.012
	50...133	RC5-1/133	1SBN050100R1001	2	0.012
	110...250	RC5-1/250	1SBN050100R1002	2	0.012
	250...440	RC5-1/440	1SBN050100R1003	2	0.012
AX50 ... AX150	24...50	RC5-2/50	1SBN050200R1000	2	0.015
	50...133	RC5-2/133	1SBN050200R1001	2	0.015
	110...250	RC5-2/250	1SBN050200R1002	2	0.015
	250...440	RC5-2/440	1SBN050200R1003	2	0.015

For contactors	MMS type	Type	Order code	Pkg qty	Weight (1 pce)
					kg

Connecting links with manual motor starters

AX09 ... AX18	MS116-0.16 ... MS116-16	BEA16/116	1SBN081406R1000	10	0.020
AX25	MS116-0.16 ... MS116-16	BEA25/116	1SBN089306T1000	10	0.020
AX25	MS116-20 ... MS116-32	BEA25/132	1SBN089306T1001	10	0.020

AX185, AX205 3-pole contactors

90 to 110 kW

AC operated with 1 N.O. + 1 N.C. auxiliary contacts



AX185, AX205

1SFL01137V0001

AX185, AX205 contactors are mainly used for controlling 3-phase motors and power circuits up to 1000 V AC.

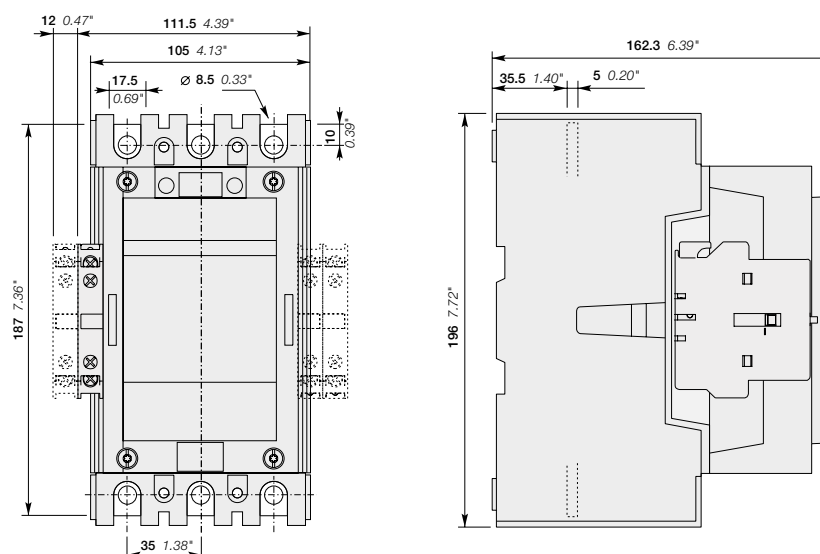
These contactors are of the block type design with:

- 3 main poles
- control circuit: AC operated
- add-on auxiliary contact blocks for front or side mounting and a wide range of accessories.

Ordering details (without auxiliary block)

IEC		Rated control circuit voltage U _c (1)		Auxiliary contacts fitted	Type	Order code	Weight (1 pce)
Rated operational power	current θ ≤ 40 °C	V 50 Hz	V 60 Hz				
400 V	AC-3						kg
kW	A						
90	250	24	24	1 1	AX185-30-11-81	1SFL491074R8111	3.800
		110	110...120	1 1	AX185-30-11-84	1SFL491074R8411	3.800
		220...230	230...240	1 1	AX185-30-11-80	1SFL491074R8011	3.800
		230...240	240...260	1 1	AX185-30-11-88	1SFL491074R8811	3.800
		400...415	415...440	1 1	AX185-30-11-86	1SFL491074R8611	3.800
110	275	24	24	1 1	AX205-30-11-81	1SFL501074R8111	3.800
		110	110...120	1 1	AX205-30-11-84	1SFL501074R8411	3.800
		220...230	230...240	1 1	AX205-30-11-80	1SFL501074R8011	3.800
		230...240	240...260	1 1	AX205-30-11-88	1SFL501074R8811	3.800
		400...415	415...440	1 1	AX205-30-11-86	1SFL501074R8611	3.800

(1) For other voltage version see voltage code table.



AX185, AX205

Main dimensions mm, inches

AX260 ... AX370 3-pole contactors

132 to 200 kW

AC operated with 1 N.O. + 1 N.C. auxiliary contacts



AX260 ... AX370

1SFLC0114/V00001

AX260 ... AX370 contactors are mainly used for controlling 3-phase motors and power circuits up to 1000 V AC.

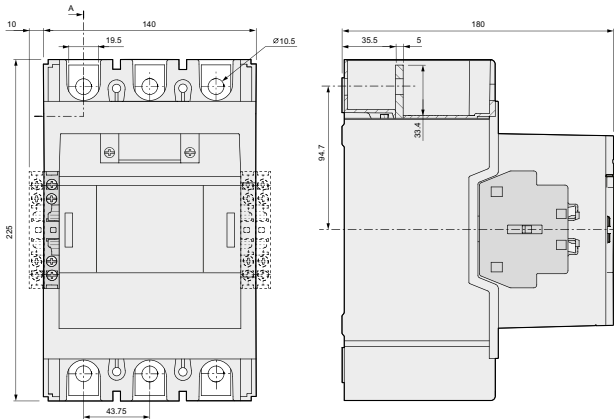
These contactors are of the block type design with:

- 3 main poles
- control circuit: AC operated
- add-on auxiliary contact blocks for front or side mounting and a wide range of accessories.

Ordering details (without auxiliary block)

IEC		Rated control circuit voltage U _c (1)		Auxiliary contacts fitted	Type	Order code	Weight (1 pce)
Rated operational power	current						
400 V	θ ≤ 40 °C						
AC-3	AC-1						
kW	A	V 50 Hz	V 60 Hz				kg
132	400	24	24	1 1	AX260-30-11-81	1SFL547074R8111	5.400
		110	110...120	1 1	AX260-30-11-84	1SFL547074R8411	5.400
		220...230	230...240	1 1	AX260-30-11-80	1SFL547074R8011	5.400
		230...240	240...260	1 1	AX260-30-11-88	1SFL547074R8811	5.400
		400...415	415...440	1 1	AX260-30-11-86	1SFL547074R8611	5.400
160	500	24	24	1 1	AX300-30-11-81	1SFL587074R8111	5.400
		110	110...120	1 1	AX300-30-11-84	1SFL587074R8411	5.400
		220...230	230...240	1 1	AX300-30-11-80	1SFL587074R8011	5.400
		230...240	240...260	1 1	AX300-30-11-88	1SFL587074R8811	5.400
		400...415	415...440	1 1	AX300-30-11-86	1SFL587074R8611	5.400
200	600	24	24	1 1	AX370-30-11-81	1SFL607074R8111	5.400
		110	110...120	1 1	AX370-30-11-84	1SFL607074R8411	5.400
		220...230	230...240	1 1	AX370-30-11-80	1SFL607074R8011	5.400
		230...240	240...260	1 1	AX370-30-11-88	1SFL607074R8811	5.400
		400...415	415...440	1 1	AX370-30-11-86	1SFL607074R8611	5.400

(1) For other voltage version see voltage code table.

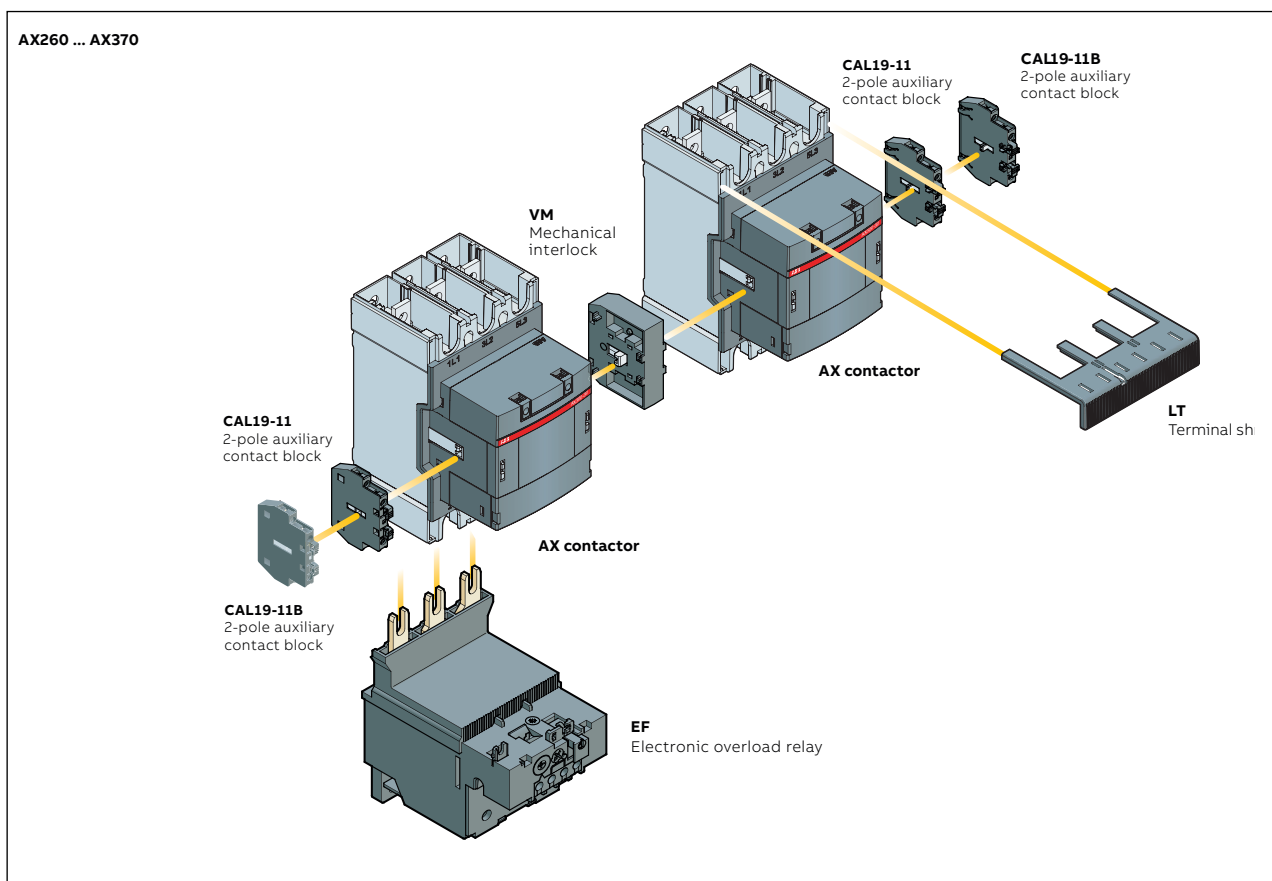
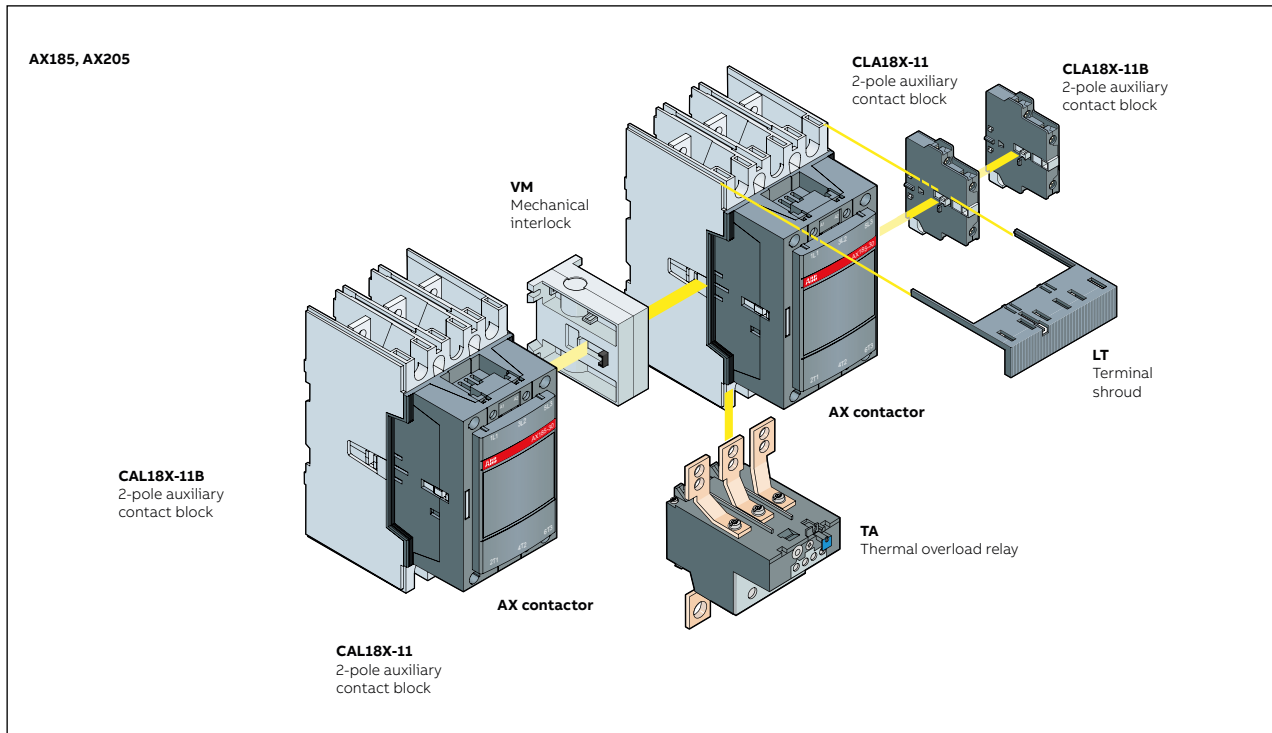


AX260... AX370

Main dimensions mm, inches

AX185 ... AX370 3-pole contactors

Main accessories



AX185 ... AX370 3-pole contactors

Main accessories

Contactor types	Main poles	Auxiliary contacts	Sided-mounted accessories	Mechanical interblock units	Mounting and positioning
			Add-on auxiliary contact blocks CAL18X11-11, CAL18X-11B, CAL19-11, CAL19-11B		 Factory mounted auxiliary contacts Add-on CAL18-11 auxiliary contacts Add-on CAL18-11 auxiliary contacts

Contactors + auxiliary contact blocks

AX185 ... AX205	3	0	1	1	1 x CAL18X-11	+ 2 x CAL18X-11B	-	
AX260 ... AX370	3	0	1	1	1 x CAL19-11	+2 x CAL19-11B	-	

Contactors with mechanical interlocking + auxiliary contact blocks

AX185 ... AX205	3	0	1	1	1 x CAL18X-11 (1)	+ 2 x CAL18X-11B (1)	+ VM...H (2)	
AX260 ... AX370	3	0	1	1	1 x CAL19-11	+2 x CAL19-11B	+ VM... (2)	

(1) Total number of auxiliary contact blocks for the two contactors

(2) Interlock type, according to the contactor ratings (see "Accessories")

Contactor types	Thermal overload relays	Electronic overload relays

Overload relays details (3)

AX185 ... AX205	TA200DU (66 ... 200 A)	EF205 (63 ... 210 A)
AX260 ... AX370	-	EF370 (115 ... 380 A)

The addition of a thermal overload relay on the contactor does not prevent fitting of many other accessories as shown above.

(3) TDirect mounting - No kit required.

AX185 ... AX370 3-pole contactors

Main accessories



CAL18X-11

1SFC101143V0201



VM300H

1SFC58041F0301



LT ... AC

1SFT98099-019C3



LT ... AL

1SFT9099-125



LW

1SFT98000-011C3



LX

1SFT98000_012C3

For contactors	Auxiliary contacts	Type	Order code	Pkg qty	Weight (1 pce)
					kg

Side-mounted instantaneous auxiliary contact blocks

AX185 ... AX205	1 1	CAL18X-11	1SFN019820R1011	2	0.050
	1 1	CAL18X-11B	1SFN019820R3311	2	0.050
AX260 ... AX370	1 1	CAL19-11	1SFN010820R1011	2	0.040
	1 1	CAL19-11B	1SFN010820R3311	2	0.040

Left side contactor	Right side contactor	Mounting	Auxiliary contacts	Type	Order code	Pkg qty	Weight (1 pce)
							kg

Mechanical interlock unit for two horizontal mounted contactors

AX95 ... AX205	AX185 ... AX205	Horizontal	- -	VM300H	1SFN034700R1000	1	0.150
AX260 ... AX370	AX260 ... AX370	Horizontal	- -	VM19	1SFN030300R1000	1	0.054
AX185 ... AX205	AX260 ... AX370	Horizontal	- -	VM205/260	1SFN035003R1000	1	0.075
AX260 ... AX370	AX185 ... AX205	Horizontal	- -	VM205/260	1SFN035003R1000	1	0.075
AX260 ... AX370	AF400 ... AF460	Horizontal	- -	VM370/400	1SFN035403R1000	1	0.239
AF400 ... AF460	AX260 ... AX370	Horizontal	- -	VM370/400	1SFN035403R1000	1	0.239

Up contactor	Down contactor	Mounting	Auxiliary contacts	Type	Order code	Pkg qty	Weight (1 pce)
							kg

Mechanical interlock units for two vertical mounted contactors

AX95 ... AX150	AX150 ... AX205	Vertical	- -	VM300V	1SFN034701R1000	1	0.150
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Terminal shrouds

AX185 ... AX205 with connectors	LT185-AC	1SFN124701R1000	2	0.050
AX185 ... AX205 with lugs	LT185-AL	1SFN124703R1000	2	0.220
AX185 ... AX205 with shorting bar or between contactor and TOL/EOL in DOL starters	LT185-AY	1SFN124704R1000	2	0.050
AX260 ... AX370, with cable clamps	LT370-30C	1SFN125401R1000	2	0.035
AX260 ... AX370, with compression lugs	LT370-30L	1SFN125403R1000	2	0.280
AX260 ... AX370, with shorting bar or between contactor and TOL/EOL in DOL starters	LT370-30Y	1SFN125404R1000	1	0.075

Up contactor	Dimensions		Type	Order code	Pkg qty	Weight (1 pce)
	hole Ø	Bar				kg
	mm	mm				

Terminal extension

AX185 ... AX205	10.5	20 x 5	LX185	1SFN074710R1000	1	0.250
AX260 ... AX370	10.5	25 x 5	LX370	1SFN075410R1000	1	0.234

Terminal shrouds

AX185 ... AX205	8.5	20 x 5	LW185	1SFN074707R1000	1	0.250
AX260 ... AX370	10.5	25 x 5	LW370	1SFN075407R1000	1	0.340

For contactors	Rated control circuit voltage Uc V AC	Type	Order code	Pkg qty	Weight (1 pce)
					kg

Surge suppressors

AX185 ... AX205	250...440	RC5-3/440	1SFN050300R1003	2	0.028
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AX09 ... AX40 3-pole contactors

Technical data

Main pole - Utilization characteristics according to IEC

Contactor types	AC operated	AX09	AX12	AX18	AX25	AX32	AX40
Standards		IEC 60947-1 / 60947-4-1 and EN 60947-1 / 60947-4-1					
Rated operational voltage U _e max.		690 V					
Rated frequency (without derating)		50 / 60 Hz					
Conventional free-air thermal current I _{th} acc. to IEC 60947-4-1, open contactors, $\theta \leq 40^\circ\text{C}$		24 A	26 A	28 A	32 A	65 A	65 A
With conductor cross-sectional area		4 mm ²	4 mm ²	4 mm ²	6 mm ²	16 mm ²	16 mm ²
AC-1 Utilization category							
For air temperature close to contactor							
I _e / Rated operational current AC-1	$\theta \leq 40^\circ\text{C}$	22 A	25 A	27 A	32 A	55 A	60 A
U _e max. $\leq 690\text{ V}, 50/60\text{ Hz}$	$\theta \leq 55^\circ\text{C}$	22 A	22 A	25 A	27 A	55 A	60 A
	$\theta \leq 70^\circ\text{C}$	18 A	18 A	20 A	23 A	39 A	42 A
With conductor cross-sectional area		2.5 mm ²	2.5 mm ²	4 mm ²	6 mm ²	10 mm ²	16 mm ²
AC-3 Utilization category							
For air temperature close to contactor $\theta \leq 55^\circ\text{C}$							
I _e / Max. rated operational current AC-3 (1)							
 3-phase motors	220-230-240 V	9 A	12 A	18 A	25 A	32 A	40 A
	380-400 V	9 A	12 A	18 A	25 A	32 A	40 A
	415 V	9 A	12 A	18 A	25 A	32 A	40 A
	440 V	9 A	9 A	12 A	16 A	32 A	37 A
	500 V	9 A	9 A	12 A	14 A	28 A	33 A
	690 V	7 A	7 A	9 A	10 A	21 A	25 A
 1500 r.p.m. 50 Hz 1800 r.p.m. 60 Hz 3-phase motors	220-230-240 V	2.2 kW	3 kW	4 kW	6.5 kW	9 kW	11 kW
	380-400 V	4 kW	5.5 kW	7.5 kW	11 kW	15 kW	18.5 kW
	415 V	4 kW	5.5 kW	9 kW	11 kW	15 kW	18.5 kW
	440 V	4 kW	4 kW	5.5 kW	9 kW	18.5 kW	22 kW
	500 V	5.5 kW	5.5 kW	7.5 kW	9 kW	18.5 kW	22 kW
	690 V	5.5 kW	5.5 kW	7.5 kW	9 kW	18.5 kW	22 kW
Rated making capacity AC-3		10 x I _e AC-3 acc. to IEC 60947-4-1					
Rated breaking capacity AC-3		8 x I _e AC-3 acc. to IEC 60947-4-1					
AC-8a Utilization category (without thermal overload relay - U _e 400 V 50/60 Hz - $\theta \leq 40^\circ\text{C}$)							
I _e / Rated operational current AC-8a		12 A	16 A	22 A	30 A	40 A	50 A
Rated operational power AC-8a		5.5 kW	7.5 kW	11 kW	15 kW	20 kW	25 kW
Short-circuit protection device for contactors without thermal overload relay - Motor protection excluded (2) U _e $\leq 500\text{ V AC}$ - gG type fuse		25 A	25 A	32 A	32 A	63 A	63 A
Rated short-time withstand current I _{cw} at 40 °C ambient temperature, in free air from a cold state	1 s	250 A	280 A	300 A	300 A	600 A	600 A
	10 s	100 A	120 A	145 A	200 A	400 A	400 A
	30 s	60 A	70 A	80 A	105 A	225 A	225 A
	1 min	50 A	55 A	60 A	85 A	150 A	150 A
	15 min	26 A	26 A	28 A	32 A	65 A	65 A
Maximum breaking capacity cos $\phi = 0.45$	at 440 V	250 A	250 A	250 A	250 A	820 A	820 A
	at 690 V	90 A	90 A	90 A	90 A	340 A	340 A
Power dissipation per pole	I _e / AC-1	0.8 W	0.8 W	1 W	1.2 W	2.5 W	3 W
	I _e / AC-3	0.1 W	0.1 W	0.2 W	0.35 W	0.9 W	1.3 W
Max. electrical switching frequency	AC-1	600 cycle/h					
	AC-3	1200 cycle/h					
Mechanical durability							
Number of operating cycles		10 millions operating cycles					
Max. switching frequency		3600 cycles/h					

(1) For the corresponding kW/A values of 1500 r.p.m. 50 Hz or 1800 r.p.m. 60 Hz, 3-phase motors, see "Motor rated operational powers and currents".

(2) For the protection of motor starters against short circuits, see "Coordination with short-circuit protection devices".

AX50 ... AX150 3-pole contactors

Technical data

Main pole - Utilization characteristics according to IEC

Contactor types	AC operated	AX50	AX65	AX80	AX95	AX115	AX150
Standards		IEC 60947-1 / 60947-4-1 and EN 60947-1 / 60947-4-1					
Rated operational voltage U _e max.		690 V			1000 V		
Rated frequency (without derating)		50 / 60 Hz					
Conventional free-air thermal current I _{th} acc. to IEC 60947-4-1, open contactors, θ ≤ 40 °C		100 A	125 A	125 A	145 A	160 A	190 A
With conductor cross-sectional area		35 mm ²	50 mm ²	50 mm ²	50 mm ²	70 mm ²	95 mm ²
AC-1 Utilization category							
For air temperature close to contactor							
le / Rated operational current AC-1	θ ≤ 40 °C	100 A	115 A	125 A	145 A	160 A	190 A
U _e max. ≤ 690 V, 50/60 Hz	θ ≤ 55 °C	85 A	95 A	105 A	135 A	145 A	145 A
	θ ≤ 70 °C	70 A	80 A	85 A	115 A	130 A	130 A
With conductor cross-sectional area		35 mm ²	50 mm ²	50 mm ²	50 mm ²	70 mm ²	95 mm ²
AC-3 Utilization category							
For air temperature close to contactor θ ≤ 55 °C							
le / Max. rated operational current AC-3 (1)							
 3-phase motors	220-230-240 V	53 A	65 A	80 A	96 A	115 A	150 A
	380-400 V	50 A	65 A	80 A	96 A	115 A	150 A
	415 V	50 A	65 A	80 A	96 A	115 A	150 A
	440 V	45 A	65 A	70 A	93 A	100 A	100 A
	500 V	45 A	55 A	65 A	80 A	100 A	100 A
	690 V	35 A	43 A	46 A	65 A	82 A	82 A
 Rated operational power AC-3 (1) 1500 r.p.m. 50 Hz 1800 r.p.m. 60 Hz 3-phase motors	220-230-240 V	15 kW	18.5 kW	22 kW	25 kW	30 kW	45 kW
	380-400 V	22 kW	30 kW	37 kW	45 kW	55 kW	75 kW
	415 V	25 kW	37 kW	40 kW	55 kW	59 kW	75 kW
	440 V	25 kW	37 kW	40 kW	55 kW	59 kW	59 kW
	500 V	30 kW	37 kW	45 kW	55 kW	59 kW	59 kW
	690 V	30 kW	37 kW	40 kW	55 kW	75 kW	75 kW
Rated making capacity AC-3		10 x I _e AC-3 acc. to IEC 60947-4-1					
Rated breaking capacity AC-3		8 x I _e AC-3 acc. to IEC 60947-4-1					
AC-8a Utilization category (without thermal overload relay - U _e 400 V 50/60 Hz - θ ≤ 40 °C)							
le / Rated operational current AC-8a		63 A	85 A	95 A	120 A	140 A	-
Rated operational power AC-8a		30 kW	45 kW	50 kW	59 kW	75 kW	-
Short-circuit protection device for contactors without thermal overload relay - Motor protection excluded (2)							
U _e ≤ 500 V AC - gG type fuse		100 A	125 A	160 A	160 A	200 A	315 A
Rated short-time withstand current I _{cw} at 40 °C ambient temperature, in free air from a cold state	1 s	1000 A	1000 A	1000 A	1320 A	1320 A	1320 A
	10 s	650 A	650 A	650 A	800 A	800 A	800 A
	30 s	370 A	370 A	370 A	500 A	500 A	500 A
	1 min	250 A	250 A	250 A	350 A	350 A	350 A
	15 min	110 A	135 A	135 A	160 A	160 A	175 A
Maximum breaking capacity cos φ = 0.45		at 440 V	1300 A	1300 A	1300 A	1160 A	1160 A
	at 690 V	630 A	630 A	630 A	800 A	800 A	800 A
Power dissipation per pole		I _e / AC-1	5 W	6.5 W	7 W	6.5 W	10.5 W
	I _e / AC-3	1.3 W	1.5 W	2.3 W	2.7 W	3.9 W	6.5 W
Max. electrical switching frequency		AC-1	600 cycle/h		300 cycles/h		
	AC-3	600 cycle/h		300 cycles/h			
Mechanical durability							
Number of operating cycles		10 millions operating cycles					
Max. switching frequency		3600 cycles/h					

(1) For the corresponding kW/A values of 1500 r.p.m. 50 Hz or 1800 r.p.m. 60 Hz, 3-phase motors, see "Motor rated operational powers and currents".

(2) For the protection of motor starters against short circuits, see "Coordination with short-circuit protection devices".

AX185 ... AX370 3-pole contactors

Technical data

Main pole - Utilization characteristics according to IEC

Contactor types	AC operated	AX185	AX205	AX260	AX300	AX370
Standards		IEC 60947-1 / 60947-4-1 and EN 60947-1 / 60947-4-1				
Rated operational voltage U _e max.		1000 V				1000 V
Rated frequency (without derating)		50 / 60 Hz				
Conventional free-air thermal current I _{th} acc. to IEC 60947-4-1, open contactors, θ ≤ 40 °C		250 A	275 A	400 A	500 A	600 A
With conductor cross-sectional area		120 mm²	150 mm²	240 mm²	300 mm²	2x185 mm²(2)
AC-1 Utilization category						
For air temperature close to contactor						
I _e / Rated operational current AC-1	θ ≤ 40 °C	250 A	275 A	400 A	500 A	600 A
U _e max. ≤ 690 V, 50/60 Hz	θ ≤ 55 °C	230 A	250 A	350 A	400 A	500 A
	θ ≤ 70 °C	180 A	180 A	290 A	325 A	400 A
With conductor cross-sectional area		120 mm²	150 mm²	240 mm²	300 mm²	2x185 mm²(2)
AC-3 Utilization category						
For air temperature close to contactor θ ≤ 55 °C						
I _e / Max. rated operational current AC-3 (1)						
 3-phase motors	220-230-240 V	185 A	205 A	265 A	305 A	370 A
	380-400 V	185 A	205 A	265 A	305 A	370 A
	415 V	185 A	205 A	265 A	305 A	370 A
	440 V	145 A	185 A	265 A	305 A	370 A
	500 V	145 A	170 A	250 A	290 A	135 A
	690 V	145 A	170 A	250 A	290 A	135 A
 Rated operational power AC-3 (1) 1500 r.p.m. 50 Hz 1800 r.p.m. 60 Hz 3-phase motors	220-230-240 V	55 kW	59 kW	75 kW	90 kW	110 kW
	380-400 V	90 kW	110 kW	132 kW	160 kW	200 kW
	415 V	90 kW	110 kW	132 kW	160 kW	200 kW
	440 V	75 kW	90 kW	160 kW	160 kW	200 kW
	500 V	90 kW	110 kW	160 kW	200 kW	250 kW
	690 V	110 kW	132 kW	200 kW	250 kW	315 kW
Rated making capacity AC-3		10 x I _e AC-3 acc. to IEC 60947-4-1				
Rated breaking capacity AC-3		8 x I _e AC-3 acc. to IEC 60947-4-1				
AC-8a Utilization category (without thermal overload relay - U _e 400 V 50/60 Hz - θ ≤ 40 °C)						
I _e / Rated operational current AC-8a		-	-	-	-	-
Rated operational power AC-8a		-	-	-	-	-
Short-circuit protection device for contactors without thermal overload relay - Motor protection excluded (2)						
U _e ≤ 500 V AC - gG type fuse		315 A	355 A	500 A	500 A	630 A
Rated short-time withstand current I _{cw} at 40 °C ambient temperature, in free air from a cold state	1 s	1800 A	2000 A	2650 A	3050 A	3700 A
	10 s	1200 A	1500 A	2120 A	2440 A	2960 A
	30 s	800 A	1000 A	1224 A	1409 A	1709 A
	1 min	600 A	800 A	865 A	996 A	1208 A
	15 min	280 A	320 A	400 A	500 A	600 A
Maximum breaking capacity cos φ = 0.45	at 440 V	1500 A	2000 A	3800 A	4600 A	5000 A
	at 690 V	1200 A	1600 A	3300 A	3800 A	4000 A
Power dissipation per pole	I _e / AC-1	16 W	17 W	32 W	50 W	72 W
	I _e / AC-3	8 W	10 W	14 W	19 W	27 W
Max. electrical switching frequency	AC-1	300 cycle/h			300 cycles/h	
	AC-3	300 cycle/h			300 cycles/h	
Mechanical durability						
Number of operating cycles		5 millions operating cycles				
Max. switching frequency		3600 cycles/h			300 cycles/h	

(1) For current above 275 A use terminal enlargements or terminal extensions

(2) For current above 450 A use terminal enlargements or terminal extensions

AX09 ... AX40 3-pole contactors

Technical data

Main pole - Utilization characteristics according to UL / NEMA / CSA

Contactor types	AC operated	AX09	AX12	AX18	AX25	AX32	AX40
Standards		UL 60947-4-1, CSA C22.2 NO. 60947-4-1-07					
Maximum operational voltage		600 V					
UL / CSA general use rating							
600 V AC		21 A	25 A	30 A	30 A	50 A	60 A
With conductor cross-sectional area		AWG 10	AWG 10	AWG 10	AWG 10	AWG 8	AWG 6
UL / CSA maximum 1-phase motor rating							
Full load current	120 V AC	9.8 A	13.8 A	16 A	24 A	34 A	34 A
	240 V AC	10 A	12 A	17 A	17 A	40 A	40 A
Horse power rating	120 V AC	1/2 hp	3/4 hp	1 hp	2 hp	3 hp	3 hp
	240 V AC	1.5 hp	2 hp	3 hp	3 hp	7.5 hp	7.5 hp
UL / CSA maximum 3-phase motor rating							
Full load current (1)	200-208 V AC	7.8 A	11 A	17.5 A	25.3 A	32.2 A	32.2 A
	220-240 V AC	6.8 A	9.6 A	15.2 A	22 A	28 A	42 A
	440-480 V AC	7.6 A	11 A	14 A	21 A	34 A	40 A
	550-600 V AC	9 A	11 A	17 A	17 A	32 A	41 A
Horse power rating (1)	200-208 V AC	2 hp	3 hp	5 hp	7.5 hp	10 hp	10 hp
	220-240 V AC	2 hp	3 hp	5 hp	7.5 hp	10 hp	15 hp
	440-480 V AC	5 hp	7.5 hp	10 hp	15 hp	25 hp	30 hp
	550-600 V AC	7.5 hp	10 hp	15 hp	15 hp	30 hp	40 hp
Short-circuit protection device for contactors without thermal overload relay - Motor protection excluded							
High fault current		100 kA	100 kA	100 kA	100 kA	100 kA	100 kA
Fuse rating		30 A	30 A	30 A	45 A	200 A	200 A
Fuse type, 600 V		J	J	J	J	J	J
Maximum electrical switching frequency							
For general use		600 cycles/h					
For motor use		1200 cycles/h					

(1) For the corresponding kW/A or hp/A values of 1500 r.p.m, 50 Hz or 1800 r.p.m, 60 Hz, 3-phase motors, see "Motor rated operational powers and currents".

AX50 ... AX80 3-pole contactors

Technical data

Main pole - Utilization characteristics according to UL / NEMA / CSA

Contactor types	AC operated	AX50	AX65	AX80
Standards		UL 60947-4-1, CSA C22.2 NO. 60947-4-1-07		
Maximum operational voltage		600 V		
UL / CSA general use rating				
600 V AC		80 A	90 A	105 A
With conductor cross-sectional area		AWG 4	AWG 3	AWG 2
UL / CSA maximum 1-phase motor rating				
Full load current	120 V AC	34 A	56 A	80 A
	240 V AC	50 A	50 A	68 A
Horse power rating	120 V AC	3 hp	5 hp	7-1/2 hp
	240 V AC	10 hp	10 hp	15 hp
UL / CSA maximum 3-phase motor rating				
Full load current (1)	200-208 V AC	48.3 A	62.1 A	78.2 A
	220-240 V AC	54 A	68 A	80 A
	440-480 V AC	52 A	77 A	77 A
	550-600 V AC	52 A	77 A	77 A
Horse power rating (1)	200-208 V AC	15 hp	20 hp	25 hp
	220-240 V AC	20 hp	25 hp	30 hp
	440-480 V AC	40 hp	60 hp	60 hp
	550-600 V AC	50 hp	75 hp	75 hp
Short-circuit protection device for contactors without thermal overload relay - Motor protection excluded				
High fault current		100 kA	100 kA	100 kA
Fuse rating		100 A	200 A	200 A
Fuse type, 600 V		J	J	J
Maximum electrical switching frequency				
For general use		600 cycles/h		
For motor use		600 cycles/h		

(1) For the corresponding kW/A or hp/A values of 1500 r.p.m, 50 Hz or 1800 r.p.m, 60 Hz, 3-phase motors, see "Motor rated operational powers and currents".

AX50 ... AX80 3-pole contactors

Technical data

Main pole - Utilization characteristics according to UL / NEMA / CSA

Contactor types	AC operated	AX95	AX115	AX150	AX185	AX205	AX260	AX300	AX370
Standards	UL 508, CSA C22.2 N°14, UL 60947-1 / 60947-4-1A and CSA 60947-1 / 60947-4-1A								
Maximum operational voltage	600 V								
UL / CSA general use rating 600 V AC		150 A	150 A	170 A	250 A	260 A	350 A	400 A	520 A
With conductor cross-sectional area		AWG 1	AWG 1/0	AWG 2/0	MCM 250	MCM 300	MCM 500	2//AWG 3/0	2//MCM 300
UL / CSA maximum 3-phase motor rating									
Full load current (1)	200-208 V AC	92 A	92 A	120 A	150 A	169 A	221 A	285 A	359 A
	220-240 V AC	80 A	104 A	130 A	145 A	192 A	248 A	312 A	360 A
	440-480 V AC	77 A	96 A	124 A	156 A	180 A	240 A	302 A	361 A
	550-600 V AC	77 A	99 A	125 A	144 A	192 A	242 A	298 A	336 A
Horse power rating (1)	200-208 V AC	30 hp	30 hp	40 hp	50 hp	60 hp	75 hp	100 hp	125 hp
	220-240 V AC	30 hp	40 hp	50 hp	60 hp	75 hp	100 hp	125 hp	150 hp
	440-480 V AC	60 hp	75 hp	100 hp	125 hp	150 hp	200 hp	250 hp	300 hp
	550-600 V AC	75 hp	100 hp	125 hp	150 hp	200 hp	250 hp	300 hp	350 hp
Short-circuit protection device for contactors without thermal overload relay - Motor protection excluded									
High fault current		100 kA	100 kA	100 kA	100 kA	100 kA	100 kA	100 kA	100 kA
Fuse rating		200 A	200 A	(2)	400 A	400 A	500 A	600 A	800 A
Fuse type, 600 V		J	J	(2)	J	J	J	J	J
Maximum electrical switching frequency									
For general use		300 cycles/h							
For motor use		300 cycles/h							

(1) For the corresponding kW/A or hp/A values of 1500 r.p.m, 50 Hz or 1800 r.p.m, 60 Hz, 3-phase motors, see "Motor rated operational powers and currents".

(2) In preparation.

AX09 ... AX40 3-pole contactors

Technical data

General technical data

Contactor types	AC operated	AX09	AX12	AX18	AX25	AX32	AX40
Rated insulation voltage U_i acc. to IS / IEC 60947-4-1		690 V					
acc. to UL / CSA		600 V					
Rated impulse withstand voltage U_{imp} .		6 kV					
Ambient air temperature close to contactor							
Operation	Fitted with thermal overload relay	-25...+55 °C (1)					
	Without thermal overload relay	-40...+70 °C					
Storage		-60...+80 °C					
Climatic withstand		acc. to IEC 60068-2-30 and 60068-2-11 - UTE C 63-100 specification II					
Maximum operating altitude (without derating)		3000 m					
Shock withstand acc. to IEC 60068-2-27 and EN 60068-2-27							
Mounting position 1							
	Shock direction	1/2 sinusoidal shock for 11 ms: no change in contact position, closed or open position					
	A	20 g					
	B1	10 g closed position / 5 g open position					
	B2	15 g					
	C1	20 g					
	C2	20 g					

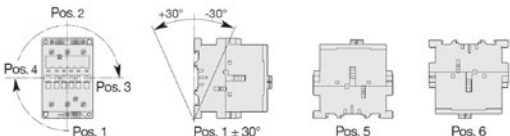
(1) The max. operational current is 23 A for AX25 with TA25DU-25M.

Magnet system characteristics

Contactor types	AC operated	AX09	AX12	AX18	AX25	AX32	AX40
Coil operating limits acc. to IS / IEC 60947-4-1	AC supply	at $\theta \leq 55$ °C 0.85...1.1 x U_c Please also refer to "Mounting characteristics and conditions for use"					
AC control voltage 50/60 Hz							
Rated control circuit voltage U_c	at 50 Hz	24...440 V					
	at 60 Hz	24...440 V					
Coil consumption	Average pull-in value	50 Hz	70 VA			120 VA	
		60 Hz	80 VA			140 VA	
		50/60 Hz (1)	74 VA / 70 VA			125 VA / 120 VA	
	Average holding value	50 Hz	8 VA / 2 W			12 VA / 3 W	
		60 Hz	8 VA / 2 W			12 VA / 3 W	
		50/60 Hz (1)	8 VA / 2 W			12 VA / 3 W	
Drop-out voltage		approx. 40...65 % of U_c					
Operating time							
Between coil energization and:	N.O. contact closing	10...26 ms				8...21 ms	
	N.C. contact opening	7...21 ms				6...18 ms	
Between coil de-energization and:	N.O. contact opening	4...15 ms				4...11 ms	
	N.C. contact closing	9...20 ms				7...14 ms	

(1) 50/60 Hz coils: see "Coil voltage code table".

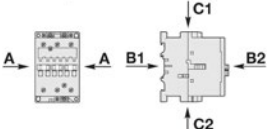
Mounting characteristics and conditions for use

Contactor types	AC operated	AX09	AX12	AX18	AX25	AX32	AX40
Mounting positions		 Max. N.O. or N.C. built-in and add-on N.O. or N.C. auxiliary contacts: see accessory fitting details for 3-pole contactor AX09 ... AX80					
Control voltage / Ambient temperature							
Mounting positions 1, 1±30°, 2, 3, 4, 5	at $\theta \leq 55$ °C	0.85...1.1 x U_c					
	at 55 °C < $\theta \leq 70$ °C	U_c					
6	at $\theta \leq 55$ °C	0.95...1.1 x U_c					
	at $\theta > 55$ °C	Unauthorized					
Mounting distances		The contactors can be assembled side by side					
Fixing							
On rail according to IEC 60715, EN 60715		35 x 7.5 mm or 35 x 15 mm					
By screws (not supplied)		2 x M4 screws placed diagonally					

AX50 ... AX150 3-pole contactors

Technical data

General technical data

Contactor types	AC operated	AX50	AX65	AX80	AX95	AX115	AX150
Rated insulation voltage U_i acc. to IEC 60947-4-1		690 V			1000 V		
acc. to UL / CSA		600 V			–		
Rated impulse withstand voltage U_{imp} .		6 kV			8 kV		
Ambient air temperature close to contactor							
Operation	Fitted with thermal overload relay	-25...+55 °C (1)					
	Without thermal overload relay	-40...+70 °C					
Storage		-60...+80 °C			-40 to +70 °C		
Climatic withstand		acc. to IEC 60068-2-30 and 60068-2-11 - UTE C 63-100 specification II			acc. to IEC 60068-2-30		
Maximum operating altitude (without derating)		3000 m					
Shock withstand acc. to IEC 60068-2-27 and EN 60068-2-27							
Mounting position 1							
							
Shock direction		1/2 sinusoidal shock for 11 ms: no change in contact position, closed or open position (2)					
A		20 g					
B1		10 g closed position / 5 g open position					
B2		15 g					
C1		20 g					
C2		20 g					

(1) The max. operational current is 74A for AX80 with TA75DU-80M.

(2) These values are not valid for rail mounting with contactors AX95 ... AX150.

Magnet system characteristics

Contactor types		AC operated	AX50	AX65	AX80	AX95	AX115	AX150
Coil operating limits acc. to IEC 60947-4-1		AC supply	at $\theta \leq 55^\circ\text{C}$ 0.85...1.1 x U_c			At $\theta \leq 70^\circ\text{C}$ 0.85 ... 1.1 x U_c		
			Please also refer to "Mounting characteristics and conditions for use"					
AC control voltage 50/60 Hz								
Rated control circuit voltage U_c		at 50 Hz	24...440 V					
		at 60 Hz	24...440 V					
Coil consumption	Average pull-in value	50 Hz	180 VA			350 VA		
		60 Hz	210 VA			450 VA		
		50/60 Hz (1)	190 VA / 180 VA			410 VA / 365 VA		
	Average holding value	50 Hz	18 VA / 5.5 W			22 VA / 6.5 W		
		60 Hz	18 VA / 5.5 W			26 VA / 8 W		
		50/60 Hz (1)	18 VA / 5.5 W			27 VA / 7.5 W		
Drop-out voltage			approx. 40...65 % of U_c					
Operating time								
Between coil energization and:	N.O. contact closing	8...27 ms				10...25 ms		
	N.C. contact opening	7...22 ms				7...22 ms		
Between coil de-energization and:	N.O. contact opening	4...11 ms				7...15 ms		
	N.C. contact closing	7...14 ms				10...18 ms		

(1) 50/60 Hz coils: see "Coil voltage code table".

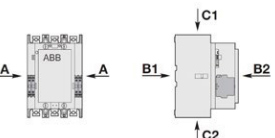
Mounting characteristics and conditions for use

Contactor types	AC operated	AX50	AX65	AX80	AX95	AX115	AX150
Mounting positions							
		Add on max. N.O. or N.C. auxiliary contacts: see accessory fitting details for 3-pole contactor AX50 ... AX150					
Control voltage / Ambient temperature							
Mounting positions	1, 1±30°, 2, 3, 4, 5	at $\theta \leq 55\text{ }^{\circ}\text{C}$	0.85...1.1 x U_c				
		at $55\text{ }^{\circ}\text{C} \leq \theta \leq 70\text{ }^{\circ}\text{C}$	U_c			0.85...1.1 x U_c	
	6	at $\theta \leq 55\text{ }^{\circ}\text{C}$	0.95...1.1 x U_c				
		at $\theta \leq 55\text{ }^{\circ}\text{C}$	Unauthorized				
Mounting distances		The contactors can be assembled side by side					
Fixing							
On rail according to IEC 60715, EN 60715		35 x 15 mm or 75 x 25 mm			-		
By screws (not supplied)		2 x M6 screws placed diagonally			2 x M6 screws placed diagonally		

AX185 ... AX370 3-pole contactors

Technical data

General technical data

Contactor types		AC operated	AX185	AX205	AX260	AX300	AX370
Rated insulation voltage U_i acc. to IEC 60947-4-1			1000 V				
Rated impulse withstand voltage U_{imp} .			8 kV				
Ambient air temperature close to contactor							
Operation		Fitted with thermal overload relay	-25...+55 °C (1)				
		Without thermal overload relay	-40...+70 °C				
Storage			-40 to +70 °C				
Climatic withstand			Category B according to IEC60947-1 / EN 60947-1 Annex Q				
Maximum operating altitude (without derating)			3000 m				
Shock withstand acc. to IEC 60068-2-27 and EN 60068-2-27			1/2 sinusoidal shock for 11 ms: no change in contact position, closed or open position (2)			2 sinusoidal shock for 30 ms	
Mounting position 1							
			Shock direction				
			A	5 g	20 g		
			B1	5 g	15 g closed position / 3 g open position		
			B2	5 g	15 g closed position / 3 g open position		
			C1	5 g	20 g		
			C2	5 g	20 g		

(1) The max. operational current is 74 A for AX80 with TA75DU-80M.

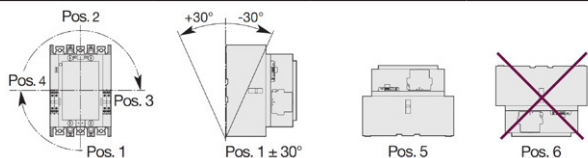
(2) These values are not valid for rail mounting with contactors AX95 ... AX150.

Magnet system characteristics

Contactor types		AC operated	AX185	AX205	AX260	AX300	AX370
Coil operating limits acc. to IEC 60947-4-1		AC supply	at $\theta \leq 70\text{ }^{\circ}\text{C}$ 0.85...1.1 x U_c		At $\theta \leq 70\text{ }^{\circ}\text{C}$ 0.85 ... 1.1 x U_c		
			Please also refer to "Mounting characteristics and conditions for use"				
AC control voltage 50/60 Hz							
Rated control circuit voltage U_c		at 50 Hz	24...440 V				
		at 60 Hz	24...440 V				
Coil consumption	Average pull-in value	50 Hz				-	
		60 Hz				-	
		50/60 Hz (1)				475 VA	
	Average holding value	50 Hz				-	
		60 Hz				-	
		50/60 Hz (1)				17.5 VA	
Drop-out voltage			approx. 40...65 % of U_c		55% of U_c min.		
Operating time							
Between coil energization and:	N.O. contact closing	8...27 ms				30...60 ms	
	N.C. contact opening	7...22 ms				-	
Between coil de-energization and:	N.O. contact opening	4...11 ms				45...80 ms	
	N.C. contact closing	7...14 ms				-	

(1) 50/60 Hz coils: see "Coil voltage code table".

Mounting characteristics and conditions for use

Contactor types	AC operated	AX185	AX205	AX260	AX300	AX370
Mounting positions		 Max. add-on N.O. or N.C. auxiliary contacts: see accessory fitting details for 3-pole contactor AX185 ... AX370				
Control voltage / Ambient temperature						
Mounting positions	1, 1±30°, 2, 3, 4, 5	at $\theta \leq 70\text{ °C}$	0.85...1.1 x U_c			
	6	at $\theta \leq 55\text{ °C}$	Unauthorized			
Mounting distances		The contactors can be assembled side by side				
Fixing		-				
On rail according to IEC 60715, EN 60715		-				
By screws (not supplied)		4 x M5				

AX09 ... AX40 3-pole contactors

Technical data

Connecting characteristics

Contactor types	AC operated	AX09	AX12	AX18	AX25	AX32	AX40
Main terminals		 Screw terminals with cable clamp				 Screw terminals with double connector 2 x (5.6 x 6.5 mm)	
Connection capacity (min. ... max.)							
Main conductors (poles)							
 Rigid	Solid ($\leq 4 \text{ mm}^2$)	} 1 x	1...4 mm ²		1...6 mm ²	2.5...16 mm ²	
 Stranded ($\geq 6 \text{ mm}^2$)			1...4 mm ²		1...6 mm ²	2.5...16 mm ²	
 Flexible with non insulated ferrule		1 x	0.75...2.5 mm ²		0.75...4 mm ²	2.5...10 mm ²	
 Flexible with insulated ferrule		2 x	0.75...2.5 mm ²		0.75...4 mm ²	2.5...10 mm ²	
 Bars or lugs		1 x	-		0.75...4 mm ²	2.5...10 mm ²	
		2 x	-		0.75...2.5 mm ²	2.5...10 mm ²	
		L <	7.7 mm		9.6 mm	-	
		l >	3.7 mm		3.7 mm	-	
Connection capacity acc. to UL/CSA		1 or 2 x	AWG 8...1		AWG 16...10	AWG 8...4	
Stripping length		10 mm				14 mm	
Tightening torque		1 Nm / 9 lb.in			1.2 Nm / 11 lb.in	2.3 Nm / 20 lb.in	
Auxiliary conductors (built-in auxiliary terminals + coil terminals)							
 Rigid solid		1 x	1...4 mm ²				
		2 x	1...4 mm ²				
 Flexible with non insulated ferrule		1 x	0.75...2.5 mm ²				
 Lugs		2 x	0.75...2.5 mm ²				
		L <	7.7 mm		8 mm		
		l >	3.7 mm		3.7 mm		
Connection capacity acc. to UL/CSA		1 or 2 x	AWG 18...14				
Stripping length		10 mm					
Tightening torque							
Coil terminals		1 Nm / 9 lb.in					
Built-in auxiliary terminals		1 Nm / 9 lb.in					
Degree of protection							
acc. to IEC 60947-1 / EN 60947-1 and IEC 60529							
Main terminals		IP20 (only front side)					
Coil terminals		IP20					IP20
Built-in auxiliary terminals		IP20 (only front side)					IP20
Screw terminals		Delivered in open position, screws of unused terminals must be tightened					
Main terminals		M3.5					M5
Screwdriver type		Flat Ø 5.5 / Pozidriv 2					Flat Ø 6.5 / Pozidriv 2
Coil terminals		M3.5					
Screwdriver type		Flat Ø 5.5 / Pozidriv 2					
Built-in auxiliary terminals		M3.5					
Screwdriver type		Flat Ø 5.5 / Pozidriv 2					

AX50 ... AX150 3-pole contactors

Technical data

Connecting characteristics

Contactor types	AC operated	AX50	AX65	AX80	AX95	AX115	AX150
Main terminals		 Screw terminals with single connector (13 x 10 mm)			 Screw terminals with single connector (14 x 14 mm)		
Connection capacity (min. ... max.)							
Main conductors (poles)							
	Rigid Solid (≤ 4 mm²)	}	1 x	6...50 mm²	10...95 mm²		
	Stranded (≥ 6 mm²)		2 x	6...25 mm²	6...35 mm²		
	Flexible with ferrule	1 x	6...35 mm²	10...70 mm² (1)			
	Flexible with insulated ferrule	2 x	6...16 mm²	6...35 mm² (1)			
	Flexible with insulated ferrule	1 x	6...35 mm²	10...70 mm² (1)			
	Flexible with insulated ferrule	2 x	6...16 mm²	6...35 mm² (1)			
	Bars or lugs	L <	-	30 mm (2)			
		l >	-	6 mm			
Connection capacity acc. to UL / CSA		AWG 8 ... 1			AWG 6 ... 2/0		
Stripping length		16 mm			9 mm		
Tightening torque		4.00 Nm / 35 lb.in			8 Nm / 71 lb.in		
	Recommended				9 Nm		
	Max.	4.50 Nm					
Auxiliary conductors (built-in auxiliary terminals + coil terminals)							
	Rigid solid	1 x	1...4 mm²	0.75...2.5 mm²			
	Flexible with ferrule	2 x	1...4 mm²	0.75...2.5 mm²			
	Flexible with ferrule	1 x	1...2.5 mm²	0.75...2.5 mm²			
	Flexible with ferrule	2 x	0.75...2.5 mm²				
	Lugs	L <	8 mm²				
		l >	3.7 mm²				
Connection capacity acc. to UL / CSA		1 or 2x	AWG 18 ... 14			AWG 18 ... 14	
Stripping length							
Coil terminals		9 mm			9 mm		
Built-in auxiliary terminals		10 mm					
Tightening torque							
Coil terminals		Recommended	1 Nm / 9 lb.in				
	Max.	1.2 Nm					
Built-in auxiliary terminals		Recommended	-				
	Max.	-					
Degree of protection							
acc. to IEC 60947-1 / EN 60947-1 and IEC 60529							
Main terminals		IP10					
Coil terminals		IP20					
Built-in auxiliary terminals		-					
Screw terminals		Delivered in open position, screws of unused terminals must be tightened					
Main terminals			M6			M8	
	Screwdriver type	Flat Ø 6.5 / Pozidriv 2			Hexagon socket (s = 4 mm)		
Coil terminals			M3.5				
	Screwdriver type	Flat Ø 5.5 / Pozidriv 2					
Built-in auxiliary terminals			-				
	Screwdriver type	-					

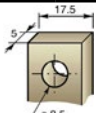
(1) AX95 - AX150: use flexible without ferrule.

(2) With LW110 enlargement piece, see "Accessories".

AX185 ... AX205 3-pole contactors

Technical data

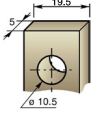
Connecting characteristics

Contactor types	AC operated	AX185	AX205
Main terminals			
Flat type			
Connection capacity (min. ... max.)			
Main conductors (poles)			
 Rigid with connector	Single for Cu cable	6...185 mm ²	
	Single for Al/Cu cable	25...150 mm ²	
	Double for Al/Cu cable	-	
 Bars or lugs		L < 24 mm	
		Ø > 8 mm	
Connection capacity acc. to UL / CSA		6 ... 250 MCM	
Stripping length		9 mm	
Tightening torque	Recommended	18 Nm / 160 lb.in	
	Max.	20 Nm	
Auxiliary conductors (built-in auxiliary terminals + coil terminals)			
 Rigid solid		1 x 1...4 mm ²	
		2 x 1...4 mm ²	
 Flexible with ferrule		1 x 0.75...2.5 mm ²	
		2 x 0.75...2.5 mm ²	
 Lugs		L < 8 mm ²	
		I > 3.7 mm ²	
Connection capacity acc. to UL / CSA	1 or 2x	AWG 18 ... 14	
Stripping length		9 mm	
Tightening torque			
Coil terminals	Recommended	1 Nm / 9 lb.in	
	Max.	1.2 Nm	
Degree of protection acc. to IEC 60947-1 / EN 60947-1 and IEC 60529			
Main terminals		IP00	
Coil terminals		IP20	
Screw terminals			
Main terminals		M8	
		Screw and bolts	
Coil terminals		M3.5	
	Screwdriver type	Flat Ø 5.5 / Pozidriv 2	

AX260 ... AX370 3-pole contactors

Technical data

Connecting characteristics

Contactor types	AC operated	AX260	AX300	370
Main terminals				
Flat type				
Connection capacity (min. ... max.)				
Main conductors (poles)				
 Rigid with connector	Cu cable Stranded	1 x 16...300 mm ²		
	Clamp type	1SDA055016R1		
	Tightening torque	25 Nm		
 Cu cable Stranded	2 x 70...185 mm ²			
 Clamp type	1SCA022194R0890 (OZXB4)			
	Tightening torque	22 Nm		
 Al cable Stranded	1 x 185...240 mm ²			
	Clamp type	1SDA055020R1		
	Tightening torque	43 Nm		
 Cu cable Flexible	1 x 16...240 mm ²			
	Clamp type	1SDA055016R1		
	Tightening torque	25 Nm		
 Cu cable Flexible	2 x 70...185 mm ²			
 Clamp type	1SCA022194R0890 (OZXB4)			
	Tightening torque	22 Nm		
 Bars or lugs	Double for Al/Cu cable	70...185 mm ²		
	W <	32 mm (1.260 in)		
	Ø >	10 mm (0.394 in)		
	Socket type	LL...included		
	Tightening torque	28 Nm / 248 lb.in		
Connection capacity acc. to UL / CSA	1 x	4 ... 400 MCM		
Tightening torque		42 Nm / 372 lb.in		
Auxiliary conductors (Coil terminals)				
 Rigid / Stranded	1 x 1...4 mm ²			
	2 x 1...4 mm ²			
 Flexible	1 x 0.75...2.5 mm ²			
	2 x 0.75...2.5 mm ²			
 Flexible with non insulated	1 x 0.75...2.5 mm ²			
	2 x 0.75...2.5 mm ²			
 Flexible with insulated ferrule	1 x 0.75...2.5 mm ²			
	2 x 0.75...2.5 mm ²			
 Lugs	L <	8 mm		
	l >	3.5 mm		
Connection capacity acc. to UL / CSA	1 or 2x	AWG 18 ... 14		
Stripping length		9 mm		
Tightening torque		1.00 Nm / 9 lb.in		
Degree of protection acc. to IEC 60947-1 / EN 60947-1 and IEC 60529				
Main terminals		IP00		
Coil terminals		IP20		
Screw terminals				
Main terminals		M10		
	Screwdriver type	Screws and bolts		
Coil terminals (delivered in open position)		M3.5		
	Screwdriver type	Flat Ø 5.5 mm / Pozidriv 2		

AX09 ... AX40 3-pole contactors

Technical data

Built-in auxiliary contacts according to IEC - Other auxiliary contacts see "Accessories"

Contactor types	AC operated	AX09	AX12	AX18	AX25	AX32	AX40
Rated operational voltage U _e max.		690 V					
Rated frequency (without derating)		50 / 60 Hz					
Conventional free air thermal current I _{th} - θ ≤ 40 °C		16 A					
I _e / Rated operational current AC-15 acc. to IEC 60947-5-1	24-127 V 50/60 Hz	6 A					
	220-240 V 50/60 Hz	4 A					
	380-440 V 50/60 Hz	3 A					
	500 V 50/60 Hz	2 A					
	690 V 50/60 Hz	2 A					
Making capacity AC-15		10 x I _e AC-15 acc. to IEC 60947-5-1					
Breaking capacity AC-15		10 x I _e AC-15 acc. to IEC 60947-5-1					
I _e / Rated operational current DC-13 acc. to IEC 60947-5-1	24 V DC	6 A / 144 W					
	48 V DC	2.8 A / 134 W					
	72 V DC	2 A / 144 W					
	110 V DC	1.1 A / 121 W					
	125 V DC	1.1 A / 138 W					
	220 V DC	0.55 A / 121 W					
	250 V DC	0.55 A / 138 W					
Short-circuit protection device gG type fuse		10 A					
Rated short-time withstand current I _{cw}	for 1.0 s	100 A					
	for 0.1 s	140 A					
Minimum switching capacity with failure rate acc. to IEC 60947-5-4		12 V / 3 mA					
Non-overlapping time between N.O. and N.C. contacts		≥ 2 ms					
Power dissipation per pole at 6 A		0.1 W					
Max. electrical switching frequency	AC-15	1200 cycles/h					
	DC-13	900 cycles/h					
Mechanically linked contacts acc. to annex L of IEC 60947-5-1		Built-in N.O. or N.C. auxiliary contacts and additional N.O. or N.C. auxiliary contacts of 4-pole CA5X are mechanically linked contacts.					
Mirror contacts acc. to annex F of IEC 60947-4-1		Built-in N.C. auxiliary contacts or additional N.C. auxiliary contacts (CA5X, CAL5X-11) are mirror contacts.					

Built-in auxiliary contacts according to UL / CSA

Contactor types	AC operated	AX09	AX12	AX18	AX25	AX32	AX40
Max. operational voltage		600 V AC, 600 V DC					
Pilot duty		A600, P300					
AC thermal rated current		10 A					
AC maximum volt-ampere making		7200 VA					
AC maximum volt-ampere breaking		720 VA					
DC thermal rated current		5 A					
DC maximum volt-ampere making-breaking		138 VA					

AX09 ... AX80 contactors

DC circuit switching

General

- The arc switching on DC is more difficult than on AC.
- For selecting a contactor it is essential to determine the current, the voltage and the L/R time constant of the controlled load.
- For information, typical time constant values are quoted hereafter: non inductive loads such as resistance furnaces ($L/R \approx 1$ ms), inductive loads such as shunt motors ($L/R \approx 2$ ms) or series motors ($L/R \approx 7.5$ ms).
- The addition of a resistor in parallel with an inductive winding helps in the elimination of the arcs.
- All the poles required for breaking must be connected in series between the load and the source polarity not linked to earth (or chassis).

Technical data

- The tables indicate for the standard contactors the I_e max. operating currents depending on: the utilization category (i.e. L/R) DC-1, DC-3, DC-5 as defined in the IEC 60947-4-1 publication, the operating voltage U_e and the pole coupling details.
- Ampere values quoted in these tables are valid for a $-25...+70$ °C temperature close to the contactors, as long as these values do not exceed the AC-1 Ampere values for the corresponding ambient temperature.
- Max. switching frequency: 300 cycles/h.

Selection table

Contactor types	AX09	AX12	AX18	AX25	AX32	AX40	AX50	AX65	AX80
-----------------	------	------	------	------	------	------	------	------	------

Utilization category DC-1, $L/R \leq 1$ ms

	≤ 72 V	22	25	27	30	50	60	100	110	120
	110 V	8	10	15	20	-	-	-	-	-
	220 V	-	-	-	-	-	-	-	-	-
	≤ 72 V	22	25	27	30	50	60	100	110	120
	110 V	22	25	27	30	50	60	100	110	120
	220 V	8	10	15	20	-	-	-	-	-
	≤ 72 V	22	25	27	30	50	60	100	110	120
	110 V	22	25	27	30	50	60	100	110	120
	220 V	22	25	27	30	50	60	100	110	120

Utilization category DC-3, $L/R \leq 2$ ms

	≤ 72 V	22	25	27	30	50	60	100	110	120
	110 V	5	6	7	8	-	-	-	-	-
	220 V	-	-	-	-	-	-	-	-	-
	≤ 72 V	22	25	27	30	50	60	100	110	120
	110 V	22	25	27	30	50	60	100	110	120
	220 V	5	6	7	8	-	-	-	-	-
	≤ 72 V	22	25	27	30	50	60	100	110	120
	110 V	22	25	27	30	50	60	100	110	120
	220 V	22	25	27	30	50	60	100	110	120

Utilization category DC-5, $L/R \leq 7.5$ ms

	≤ 72 V	7	9	12	16	30	40	50	63	75
	110 V	3	4	4	4	-	-	-	-	-
	220 V	-	-	-	-	-	-	-	-	-
	≤ 72 V	22	25	27	30	50	60	100	110	120
	110 V	8	10	15	20	45	50	80	90	100
	220 V	3	4	4	4	-	-	-	-	-
	≤ 72 V	22	25	27	30	50	60	100	110	120
	110 V	22	25	27	30	50	60	100	110	120
	220 V	7	9	12	16	30	40	50	63	75

3-pole contactors

Electrical durability and utilization categories

General

Utilization categories determine the current making and breaking conditions relating to the characteristics of the loads to be controlled by the contactors. International standard IEC 60947-4-1 and European standard EN 60947-4-1 are the standards to be referred to.

If I_c is the current to be broken by the contactor and I_e the rated operational current normally drawn by the load, then:

- Categories AC-1 and AC-3: $I_c = I_e$
- Category AC-2: $I_c = 2.5 \times I_e$
- Category AC-4: $I_c = 6 \times I_e$

Generally speaking $I_c = m \times I_e$ where m is a multiple of the load operational current.

On next pages, the curves corresponding to categories AC-1, AC-3 and AC-4 represent the electrical durability variation of standard contactors in relation to the breaking current I_c .

Electrical durability is expressed in millions of operating cycles.

Curve utilization mode

Electrical durability forecast and contactor selection for categories AC-1, AC-2, AC-3 or AC-4

- Note the characteristics of the load to be controlled:
 - Operational voltage U_e
 - Current normally drawn I_e (U_e / I_e / kW relation for motors, see "Motor rated operational powers and currents").
 - Utilization category AC-1, AC-2, AC-3 or AC-4
 - Breaking current $I_c = I_e$ for AC-1 and for AC-3 ; $I_c = 2.5 \times I_e$ for AC-2 ; $I_c = 6 \times I_e$ for AC-4
- Define the number of operating cycles N required.
- On the diagram corresponding to the operational category, select the contactor with the curve immediately above the intersection point (I_c ; N).

Electrical durability forecast and contactor selection for mixed duty motor control: AC-3 ($I_c = I_e$) type switching off while "motor running" and, occasionally, AC-4 ($I_c = 6 \times I_e$) type switching off while "motor accelerating"

- Note the characteristics of the motor to be controlled:
 - Operational voltage U_e
 - Current normally drawn while "motor running" I_e (U_e / I_e / kW relation for motors, see "Motor rated operational powers and currents")
 - Breaking current for AC-3 $I_c = I_e$
 - Breaking current for AC-4 while "motor accelerating" $I_c = 6 \times I_e$
 - Percentage of AC-4 operating cycles K (on the basis of the total number of operating cycles)
- Define the total number of operating cycles N required.
- Note the smallest contactor rating compatible for AC-3 (U_e / I_e) on Main pole utilization characteristic table (see "Technical data").
- For the selected contactor make a note of the following in relation to the voltage using diagram AC-3 in next pages:
 - The number of operating cycles A for $I_c = I_e$ (AC-3)
 - The number of operating cycles B for $I_c = 6 \times I_e$ (AC-4)
- Calculate the estimated number of cycles N' (N' is always below A)

$$N' = \frac{A}{1 + 0.01 K (A/B - 1)}$$

- If N' is too low in relation to the target N , calculate the estimated number of cycles for a higher contactor rating.

Case of uninterrupted duty

For uninterrupted duty, some verifications of preventing maintenance are necessary to check the functionality of the concerned product (consult us).

The combined effect of environmental conditions and the proper temperature of the product may require some disposals. As a matter of fact, for this duty, the use duration prevails over the number of operating cycles.

3-pole contactors

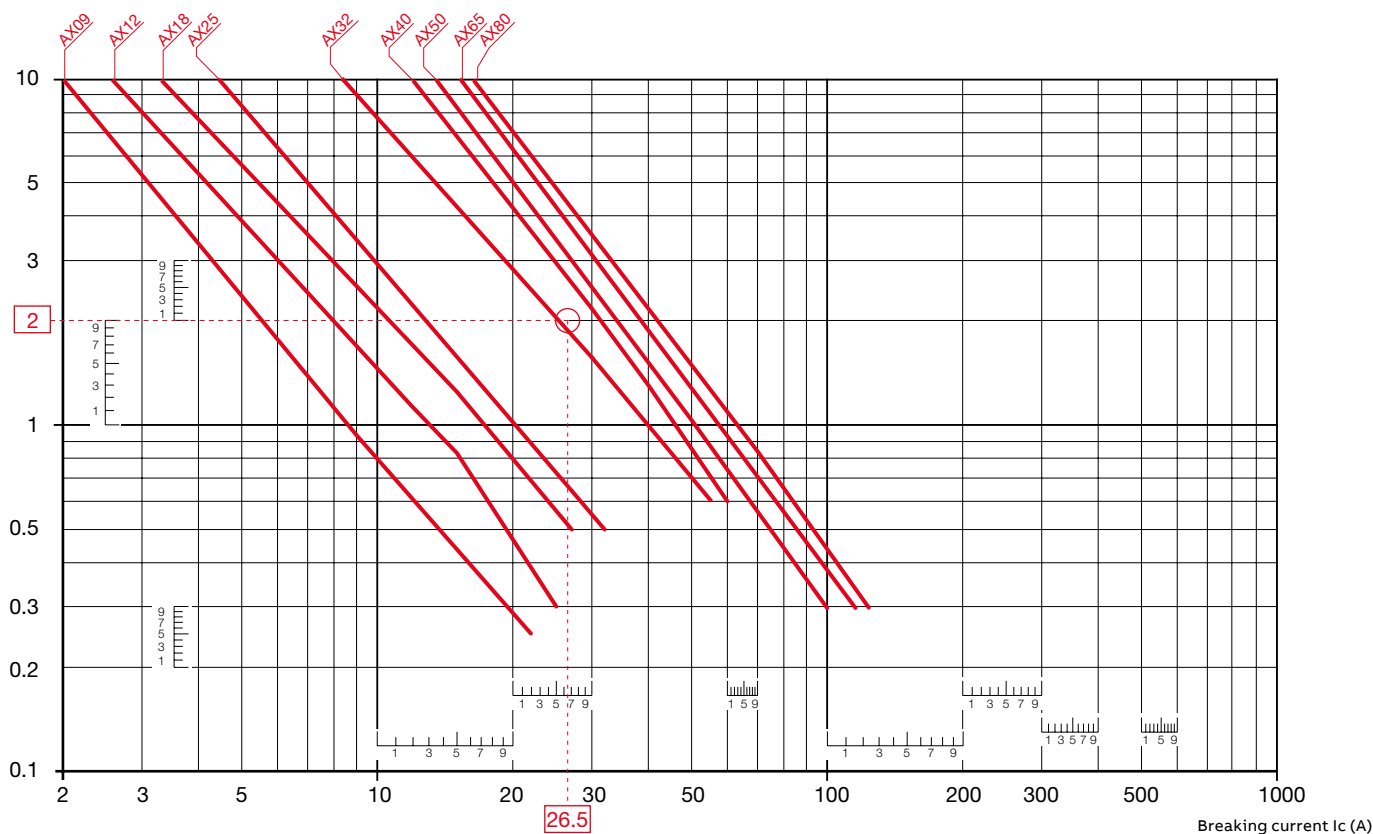
Electrical durability

Electrical durability for AC-1 utilization category - $U_e \leq 690\text{ V}$

Switching non-inductive or slightly inductive loads. The breaking current I_c for AC-1 is equal to the rated operational current of the load.

Ambient temperature and maximum electrical switching frequency: see "Technical data".

Millions of operating cycles



Example:

$I_c / AC-1 = 26.5\text{ A}$ – Electrical durability required = 2 millions operating cycles.

Using the AC-1 curves above select the AX32 contactor at intersection "O" (26.5 A / 2 millions operating cycles).

3-pole contactors

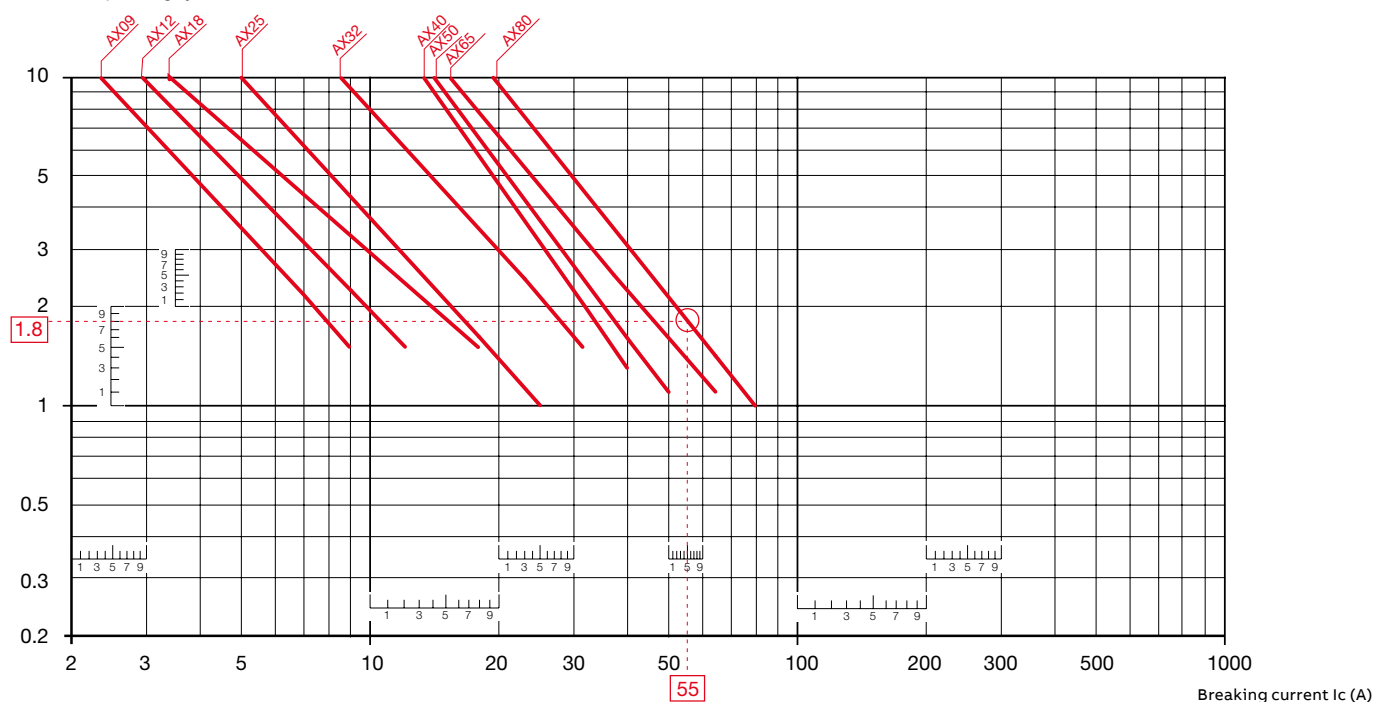
Electrical durability

Electrical durability for AC-3 utilization category - $U_e \leq 440$ V.

Switching cage motors: starting and switching off running motors. The breaking current I_c for AC-3 is equal to the rated operational current I_e (I_e = motor full load current).

Ambient temperature and maximum electrical switching frequency: see "Technical data".

Millions of operating cycles



Example:

Motor power 30 kW for AC-3 - $U_e = 400$ V and $I_e = 55$ A utilization – Electrical durability required = 1.8 million operating cycles. For AC-3: $I_c = I_e$. Select the AX80 contactor at intersection "O" (55 A / 1.8 million operating cycles) on the curves (AC-3 - $U_e \leq 440$ V).

3-pole contactors

Electrical durability

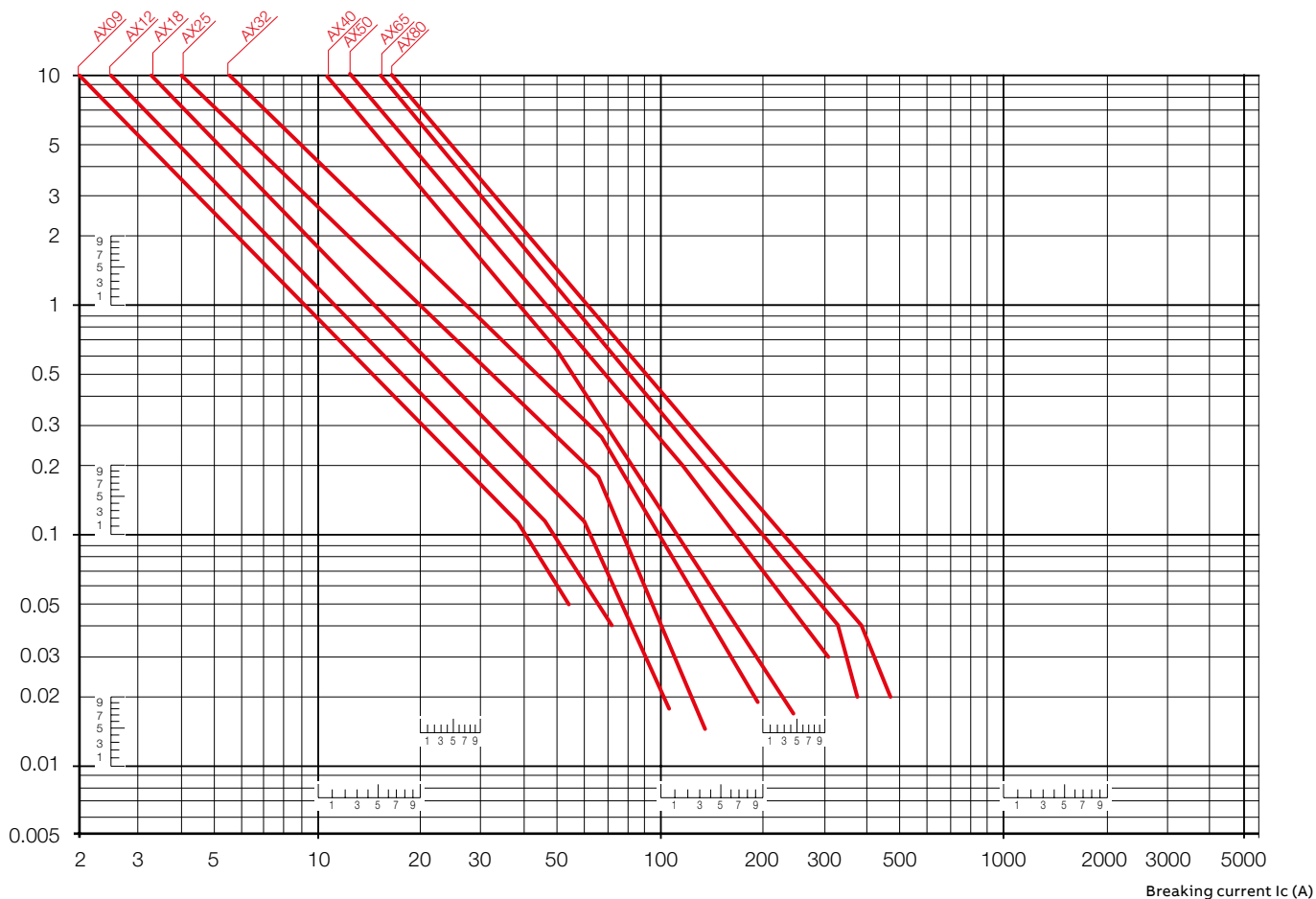
Electrical durability for AC-4 utilization category - $U_e \leq 440 \text{ V}$

Ambient temperature $\leq 55^\circ\text{C}$

Switching cage motors: starting, reverse operation and step-by-step operation. The breaking current I_c is equal to $6 \times I_e$ for AC-4, keeping in mind that I_e is the motor rated operational current (I_e = motor full-load current).

Maximum electrical switching frequency: see "Technical data".

Millions of operating cycles



AC-4		AX09	AX12	AX18	AX25	AX32	AX40	AX50	AX65	AX80
380 V (200000 operating cycles)	kW	1.8	2.2	2.9	4.5	5.6	6.2	8.8	10.5	12.5
400 V (200000 operating cycles)	kW	1.9	2.3	3.1	4.8	5.9	6.5	9.3	11.5	13.1
440 V (200000 operating cycles)	kW	2.1	2.7	3.5	5.3	6.6	7.6	10.7	13.1	15.1

Star-delta starting of three-phase asynchronous motors

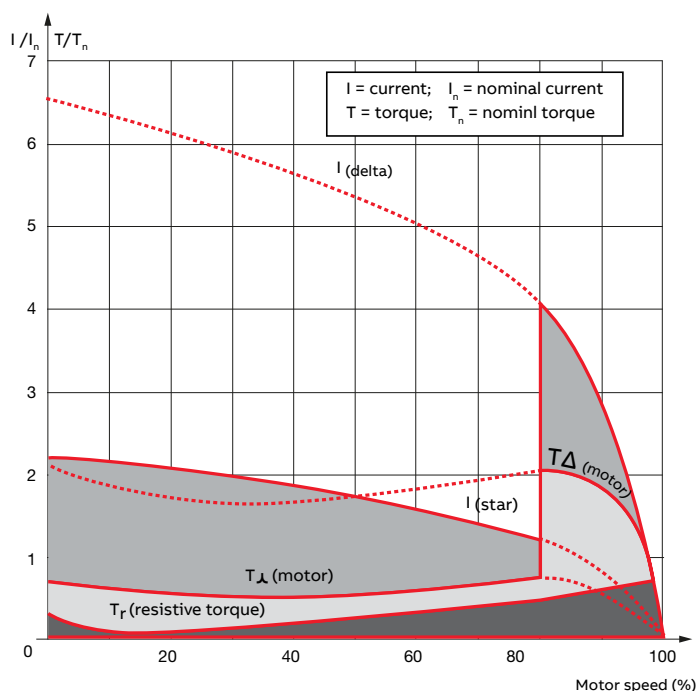
Contactor selection

General

Star-delta starting is the most common method to reduce the starting current of a motor. This system can be used on all the squirrel cage motors, which are normally used in delta connection.

In this type of starting, it is recommended to choose motors having high starting torque i.e. much higher than the resistive torque in order to reach sufficient high speed when the motor is connected in star.

Star-delta starting



Technical Data

When starting:

inrush current is reduced to a third of direct starting current
motor torque is reduced to a third or even less of direct starting torque.

Transient current is generated when switching from star to delta connection.

Utilization

During the initial starting phase ("star" connection), the resistive torque of the driven load must remain, irrespective of speed, less than the "star" motor torque until "star-delta" switching occurs.

This starting mode is therefore ideal for machines having high starting torque such as:

- pumps
- centrifugal compressors
- wood-working machines, etc.

In order to prevent a high current peak, at least 85% of nominal speed must be reached before switching from star to delta.

Precautions

Motor nominal voltage in delta connection must be equal to that of the mains.

Example:

A motor for 415 V star-delta starting must be designed for 415 V in "delta" connection. Its usual designation is "415 V / 690 V motor". The motor must be constructed with 6 terminal windings.

Sequence

Starting is a three-stage process:

1st stage - "Star" connection

Press the "On" button on the control circuit to close the KM2 "star" contactor. The KM1 "line" contactor then closes and the motor starts. Countdown of programmed starting time (normally 6 to 16 s) then begins.

2nd stage - "Star" to "Delta" switching

When the programmed starting time is over, the KM2 "star" contactor opens.

3rd stage - "Delta" connection

A transition time (or dwelling time) of 50 ms is fixed between opening of the "star" contactor and closing of the "delta" contactor by the use of CT-SDS timer. This prevents short circuit between "star" and "delta".

Note: An electrical interlock between star and delta is mandatory such as VE 5 or through auxiliary contacts.

Furthermore, in open transition, the current interruption may reach up to 95 ms: it shall be checked that this duration is compatible with the application i.e. mainly if the decreasing in rotation speed is acceptable during the starting phase.

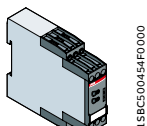
Star-delta starting of three-phase asynchronous motors

Contactor selection

Rated operational power - AC-3									Max. starting time from cold state (1) seconds	Line contactor	Delta contactor	Star contactor	Overload relay (2)	Timer
220-230 V kW	240 V kW	380 V kW	400 V kW	415 V kW	440 V kW	500 V kW	660 V kW	690 V kW						
4	4	5.5	7.5	7.5	7.5	9	9	9	15	AX09	AX09	AX09	TA25DU M	CT-SDS.22S
5.5	5.5	9	9	9	9	9	9	9	15	AX12	AX12	AX09	TA25DU M	CT-SDS.22S
7.5	9	15	15	15	11	11	11	11	15	AX18	AX18	AX12	TA25DU M	CT-SDS.22S
11	11	22	22	22	15	15	15	15	15	AX25	AX25	AX18	TA25DU M	CT-SDS.22S
15	15	25	30	30	30	30	30	30	15	AX32	AX32	AX25	TA42DU M	CT-SDS.22S
18.5	22	37	37	37	37	37	37	37	30	AX40	AX40	AX32	TA42DU M	CT-SDS.22S
25	30	45	45	45	45	55	55	55	30	AX50	AX50	AX32	TA75DU M	CT-SDS.22S
30	37	55	55	55	55	55	55	55	30	AX65	AX65	AX40	TA75DU M	CT-SDS.22S
37	45	55	75	75	75	75	75	75	30	AX80	AX80	AX50	TA75DU M	CT-SDS.22S
45	55	75	90	90	90	90	90	90	20	AX95	AX95	AX65	TA110DU-M	CT-SDS.22S
55	55	90	110	110	110	110	132	132	20	AX115	AX115	AX80	TA110DU-M	CT-SDS.22S
75	75	132	132	132	110	110	132	132	20	AX150	AX150	AX95	TA110DU-M	CT-SDS.22S
90	90	160	160	160	160	160	160	200	20	AX185	AX185	AX115	TA200DU	CT-SDS.22S
110	110	160	200	200	200	200	250	250	20	AX205	AX205	AX185	TA200DU / EF205	CT-SDS.22S
132	132	250	250	250	250	315	400	400	20	AX260	AX260	AX205	EF370	CT-ERS.21S
160	160	250	250	315	315	355	400	500	20	AX300	AX300	AX205	EF370	CT-ERS.21S
200	200	355	355	355	400	355	500	500	20	AX370	AX370	AX260	EF370	CT-ERS.21S

(1) Usual time value = 6...16 s.

(2) The setting current value is : nominal motor current x 0,58



CT-SDS...

Ordering details - Electronic timer

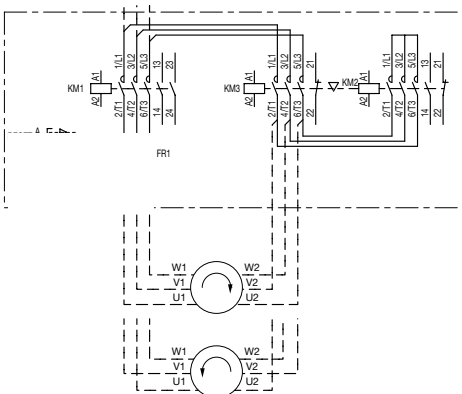
Timing function	Time ranges	Output	Rated control supply voltage	Type	Order code	Pkg qty	Weight (1 pce) kg
Star-delta change-over (3)	7 (0.05 s - 10 min)	2 n/o contacts, 3 LEDs	28-48 V DC	CT-SDS.22S	1SVR730210R3300	1	0.114
			24-240 V AC				
			380-440 V AC	CT-SDS.23S	1SVR730211R2300	1	0.118
ON-delay (4)	10 (0.05 s - 300h)	2 c/o SPDT contacts	24-240 V AC/DC	CT-ERS.21S	1SVR730100R0300	1	0.152

(3) 50 ms transition time

(4) No transition time

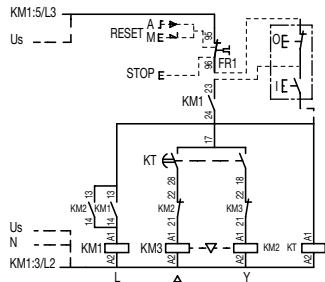
Power circuit diagram

AX09 ... AX370 contactors

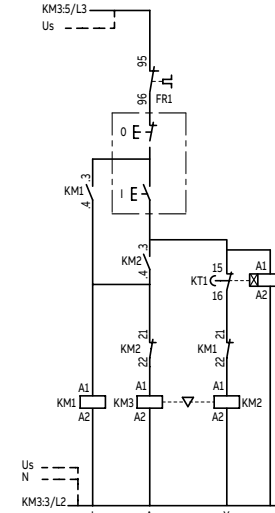


Control circuit diagrams - Remote control

AX09 ... AX205 contactors



AX260 ... AX370 contactors

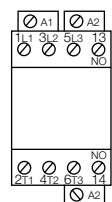


AX09 ... AX370 3-pole contactors

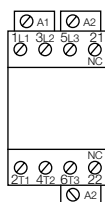
Terminal marking and positioning

AX09 ... AX150 contactors - AC operated

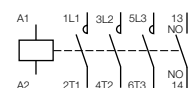
Standard devices without addition of auxiliary contacts



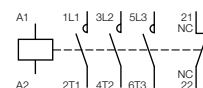
AX09 ... AX40-30-10



AX09 ... AX40-30-01

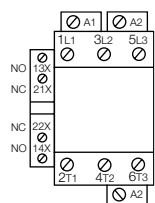


AX09 ... AX40-30-10

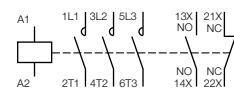


AX09 ... AX40-30-01

Standard devices with factory mounted auxiliary contacts

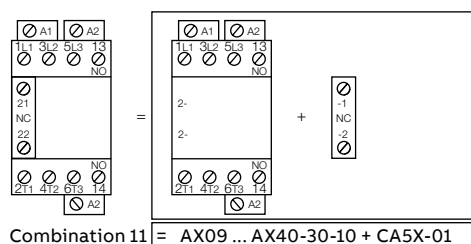


AX50 ... AX150-30-11

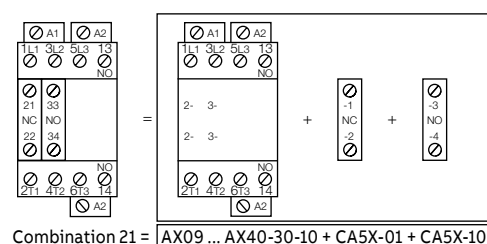


AX50 ... AX150-30-11

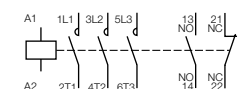
Other possible contact combinations with auxiliary contacts added by the user



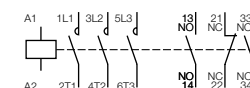
Combination 11 = AX09 ... AX40-30-10 + CA5X-01



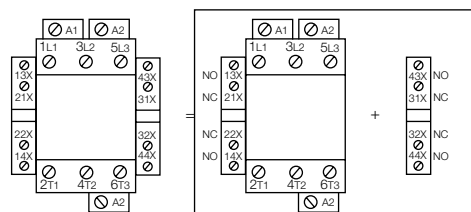
Combination 21 = AX09 ... AX40-30-10 + CA5X-01 + CA5X-10



Combination 11



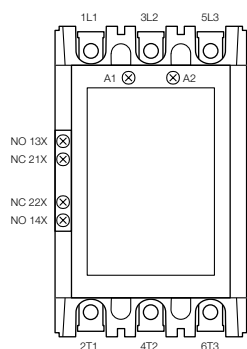
Combination 21



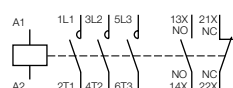
Combination 22 = AX50 ... AX80-30-11 + CAL5X-11

AX185 ... AX370 contactors - AC operated

Standard devices with factory mounted auxiliary contacts



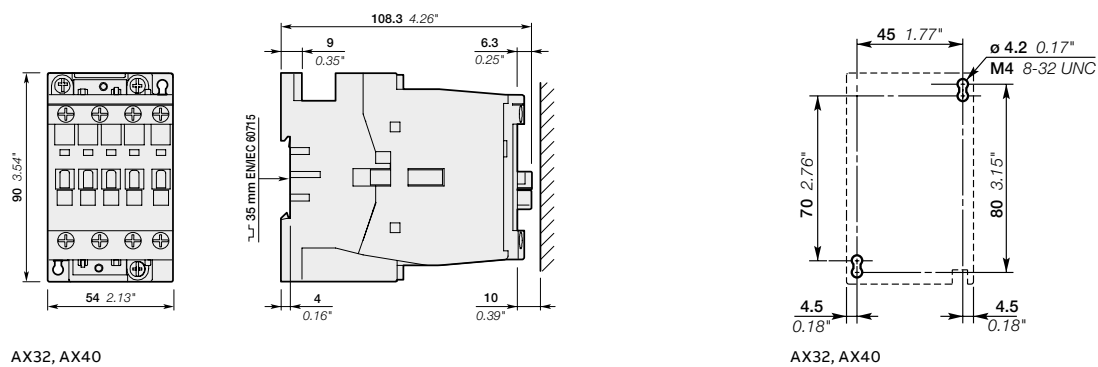
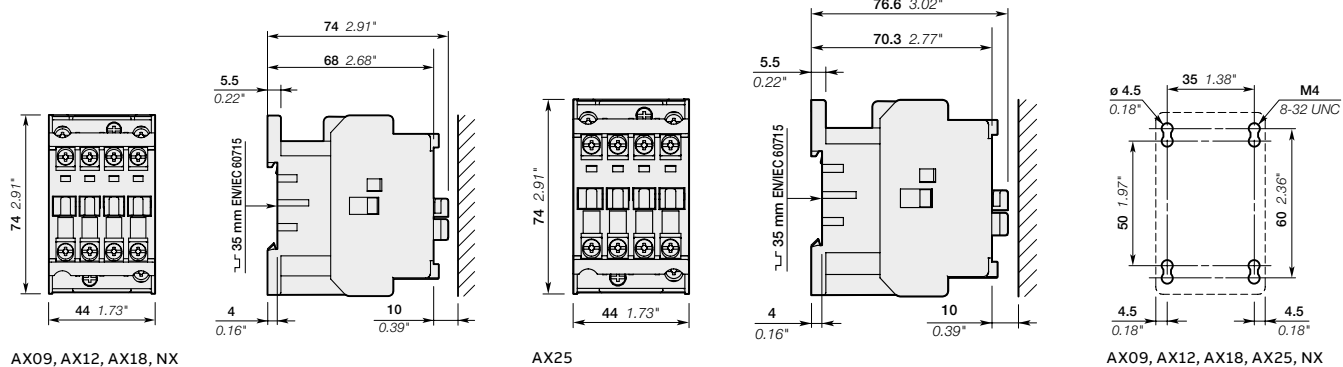
AX185 ... AX370-30-11



AX185 ... AX370-30-11

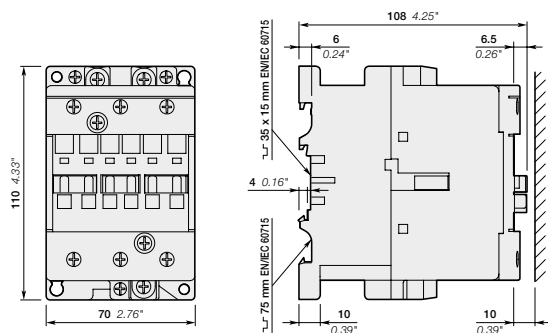
AX09 ... AX40 3-pole contactors

Dimensions

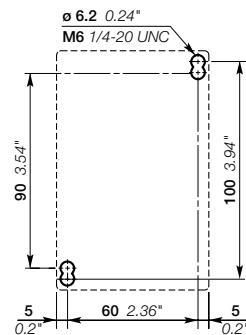


AX50 ... AX150 3-pole contactors

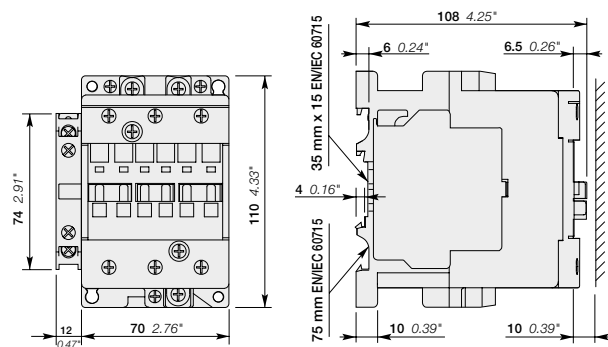
Dimensions



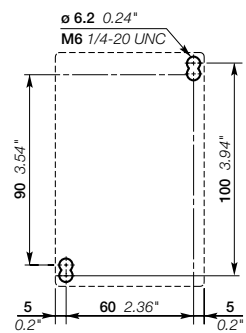
AX50, AX65, AX80



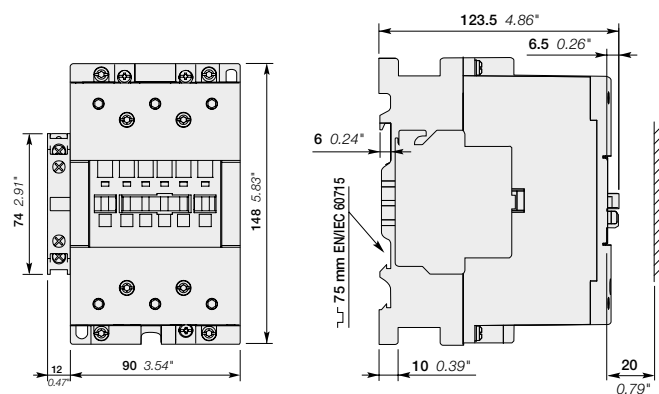
AX50, AX65, AX80



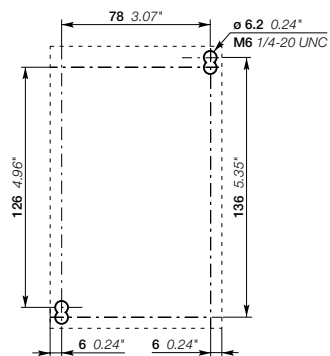
AX50, AX65, AX80 + CAL5X-11



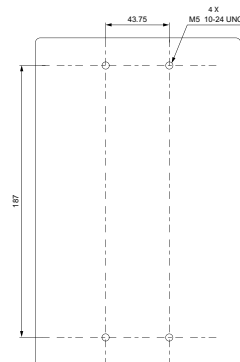
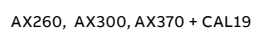
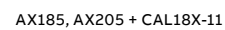
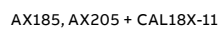
AX50, AX65, AX80 + CAL5X-11



AX95, AX115, AX150 + CAL18X-11



AX95, AX115, AX150 + CAL18X-11



NX contactor relays

Ordering details

NX contactor relays

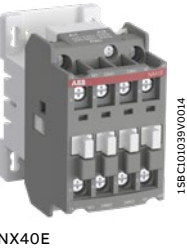
- 52 AC operated
- 53 Main accessories

Technical data

Terminal marking and positioning

NX contactor relays

AC operated

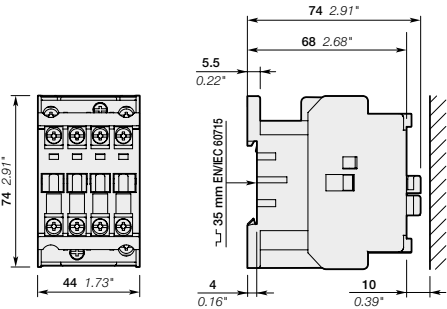


NX contactor relays are used for switching auxiliary circuits and control circuits. These contactor relays are of the block type design with:

- 4 poles. Contactor relays have mechanically linked auxiliary contact elements
- control circuit: AC operated
- add-on auxiliary contact blocks for front or side mounting and a wide range of accessories.

Number of contacts	Rated control circuit voltage Uc (1)		Type	Order code	Weight (1 pce) kg
	V 50 Hz	V 60 Hz			
	24	24	NX22E-81	1SBH901074R8122	0.326
	110	110...120	NX22E-84	1SBH901074R8422	0.326
	220...230	230...240	NX22E-80	1SBH901074R8022	0.326
	230...240	240...260	NX22E-88	1SBH901074R8822	0.326
	400...415	415...440	NX22E-86	1SBH901074R8622	0.326
	24	24	NX31E-81	1SBH901074R8131	0.326
	110	110...120	NX31E-84	1SBH901074R8431	0.326
	220...230	230...240	NX31E-80	1SBH901074R8031	0.326
	230...240	240...260	NX31E-88	1SBH901074R8831	0.326
	400...415	415...440	NX31E-86	1SBH901074R8631	0.326
	24	24	NX40E-81	1SBH901074R8140	0.326
	110	110...120	NX40E-84	1SBH901074R8440	0.326
	220...230	230...240	NX40E-80	1SBH901074R8040	0.326
	230...240	240...260	NX40E-88	1SBH901074R8840	0.326
	400...415	415...440	NX40E-86	1SBH901074R8640	0.326

(1) Other control voltages see voltage code table.



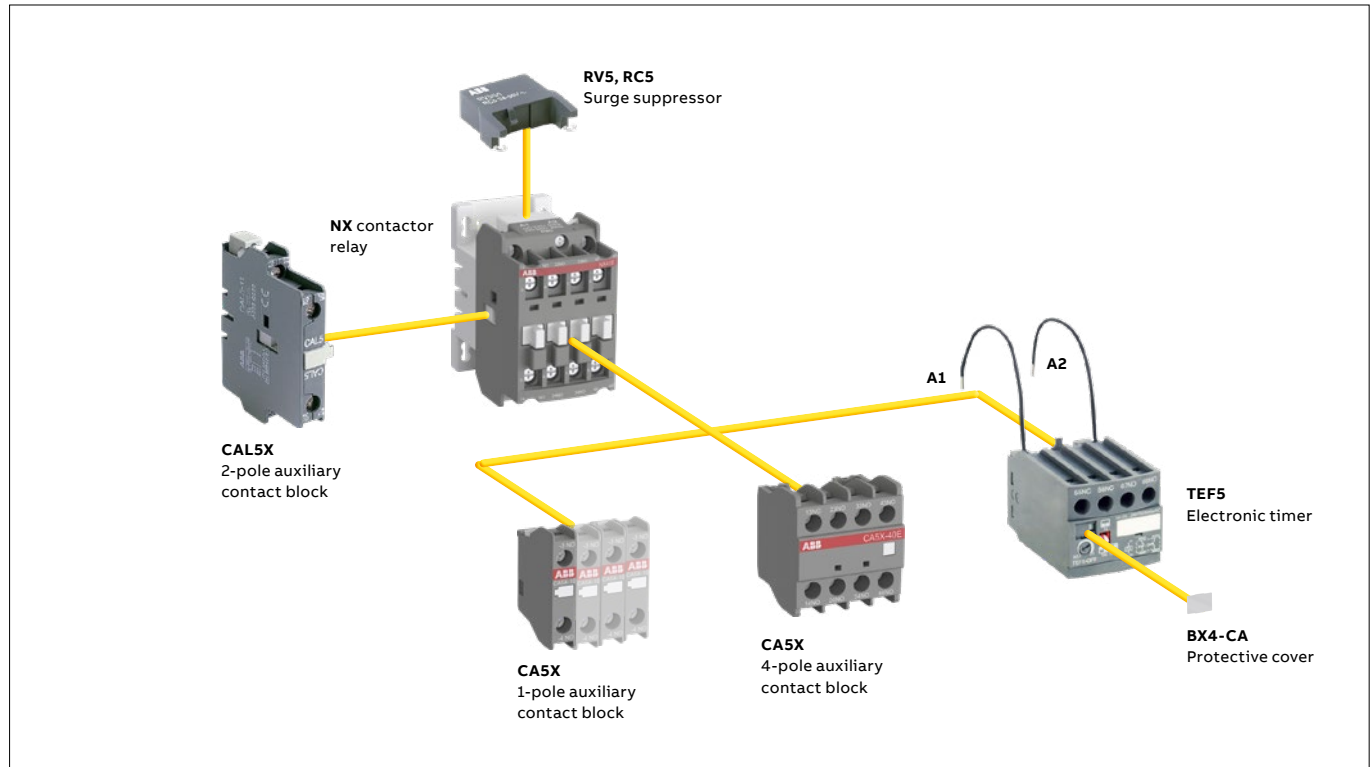
NX22E, NX31E, NX40E

Main dimensions mm, inches

NX contactor relays

Main accessories

Contactor relay and main accessories (other accessories available)



Main accessory fitting details

Many configurations of accessories are possible depending on whether these are front-mounted or side-mounted.

Contactor types	Main poles 	Front-mounted accessories				Side-mounted accessories	
		Auxiliary contact blocks		Electronic timer		Auxiliary contact blocks	
		1-pole CA5X	4-pole CA5X	TEF5		2-pole CAL5X-11	
NX	2 2 E (1)	1 to 4 x CA5X (or 1 x CE5) (2)	or 1 x CA5X (4-pole)	or 1 x TEF5	+	1 to 2 x CAL5X-11	
	3 1 E (1)	1 to 4 x CA5X	or 1 x CA5X (4-pole)	or 1 x TEF5	+	1 to 2 x CAL5X-11	
	4 0 E	(1 to 2 x CE5 max) (3)					

(1) 2 N.C. front mounted auxiliary contacts maximum in mounting position 5.

(2) CE5 auxiliary contacts not allowed in mounting position 5.

(3) The total number of N.O. or N.C. CE5 and other N.C. CA5X auxiliary contacts is limited to 2.

NX contactor relays

Technical data

Contact utilization characteristics according to IEC

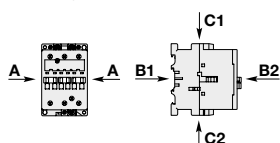
Contactor relay types	AC operated	NX
Standards		IEC 60947-1 / 60947-5-1 and EN 60947-1 / 60947-5-1
Rated operational voltage U_e max.		690 V
Rated frequency (without derating)		50 / 60 Hz
Conventional free-air thermal current $I_{th} \theta \leq 40^\circ\text{C}$		16 A
I_e / Rated operational current AC-15 acc. to IEC 60947-5-1	24-127 V 50/60 Hz	6 A
	220-230 V 50/60 Hz	4 A
	380-415 V 50/60 Hz	3 A
	500 V 50/60 Hz	2 A
	690 V 50/60 Hz	2 A
Rated making capacity AC-15		10 x I_e AC-15 acc. to IEC 60947-5-1
Rated breaking capacity AC-15		10 x I_e AC-15 acc. to IEC 60947-5-1
I_e / Rated operational current DC-13 acc. to IEC 60947-5-1	24 V DC	6 A / 144 W
	48 V DC	2.8 A / 134 W
	72 V DC	1 A / 72 W
	110 V DC	0.55 A / 60 W
	125 V DC	0.55 A / 69 W
	220 V DC	0.30 A / 66 W
	250 V DC	0.30 A / 75 W
Short-circuit protection device for contactors $U_e \leq 500$ V AC - gG type fuse		10 A
Rated short-time withstand current I_{cw} at 40°C ambient temperature, in free air from a cold state	for 1.0 s	100 A
	for 0.1 s	140 A
Minimum switching capacity with failure rate acc. to IEC 60947-5-4		17 V / 5 mA
Non-overlapping time between N.O. and N.C. contacts		≥ 2 ms
Power dissipation per pole at 6 A		0.1 W
Max. electrical switching frequency	AC-15	1200 cycles/h

Contact utilization characteristics according to UL / CSA

Contactor relay types	AC operated	NX
Standards		UL 508, CSA C22.2 N°14-05
Max. operational voltage		600 V AC
Pilot duty		A600, Q300

General technical data

Contactor relay types	AC operated	NX
Rated insulation voltage U_i acc. to IEC 60947-5-1		690 V
acc. to UL / CSA		600 V
Rated impulse withstand voltage U_{imp} .		6 kV
Ambient air temperature		
Operation in free air		$-40...+70^\circ\text{C}$
Storage		$-60...+80^\circ\text{C}$
Climatic withstand		acc. to IEC 60068-2-30 and 60068-2-11 - UTE C 63-100 specification II
Maximum operating altitude (without derating)		3000 m
Mechanical durability		
Number of operating cycles		≥ 20 millions operating cycles
Max. switching frequency		6000 cycles/h
Shock withstand acc. to IEC 60068-2-27 and EN 60068-2-27		
Mounting position 1		
	Shock direction	1/2 sinusoidal shock for 11 ms: no change in contact position, closed or open position
	A	20 g
	B1	5 g
	B2	15 g
	C1	20 g
	C2	20 g



NX contactor relays

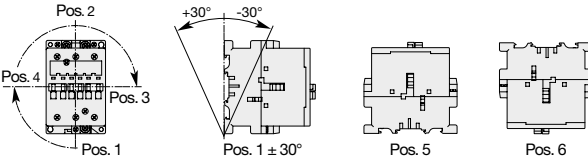
Technical data

Magnet system characteristics

Contactor relay types	AC operated	NX
Coil operating limits acc. to IEC 60947-4-1	AC supply	at $\theta \leq 55^\circ\text{C}$ 0.85...1.1 x U_c Please also refer to "Mounting characteristics and conditions for use"
AC control voltage 50/60 Hz		
Rated control circuit voltage U_c	at 50 Hz	24...440 V
	at 60 Hz	24...440 V
Coil consumption	Average pull-in value	50 Hz 70 VA
		60 Hz 80 VA
		50/60 Hz (1) 74 VA / 70 VA
	Average holding value	50 Hz 8 VA / 2 W
		60 Hz 8 VA / 2 W
		50/60 Hz (1) 8 VA / 2 W
Drop-out voltage		approx. 40...65 % of U_c
Operating time		
Between coil energization and:	N.O. contact closing	10...26 ms
	N.C. contact opening	7...21 ms
Between coil de-energization and:	N.O. contact opening	4...11 ms
	N.C. contact closing	9...16 ms

(1) 50/60 Hz coils: see "Coil voltage code table".

Mounting characteristics and conditions for use

Contactor relay types	AC operated	NX
Mounting positions		 Add-on N.O. or N.C. auxiliary contacts: see accessory fitting details for contactor relays
Control voltage / Ambient temperature		
Mounting positions	1, 1±30°, 2, 3, 4, 5	at $\theta \leq 55^\circ\text{C}$ 0.85...1.1 x U_c
		at $\theta \leq 70^\circ\text{C}$ U_c
	6	at $\theta \leq 55^\circ\text{C}$ 0.95...1.1 x U_c
		at $\theta \leq 70^\circ\text{C}$ unauthorized
Mounting distances		The contactors can be assembled side by side
Fixing		
On rail according to IEC 60715, EN 60715		35 x 7.5 mm or 35 x 15 mm
By screws (not supplied)		2 x M4 screws placed diagonally

Connecting characteristics

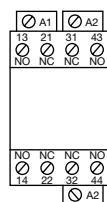
Contactor relay types	AC operated	NX
Main terminals		 Screw terminals with cable clamp
Connection capacity (min. ... max.)		
Main conductors (poles + coil terminals)		
Rigid	1 x	1...4 mm ²
	2 x	1...4 mm ²
Flexible with ferrule	1 x	0.75...2.5 mm ²
	2 x	0.75...2.5 mm ²
Bars or lugs	Pole terminals	L < 7.7 mm
		I < 3.7 mm
	Coil terminals	L < 8 mm
		I < 3.7 mm
Connection capacity acc. to UL / CSA	1 or 2 x	AWG 18...14
Stripping length		10 mm
Tightening torque		1 Nm / 9 lb.in
Degree of protection acc. to IEC 60947-1 and IEC 60529		
All terminals		IP20 (only front side)
Screw terminals		Delivered in open position, screws of unused terminals must be tightened
All terminals		M3.5
Screwdriver type		Flat Ø 5.5 / Pozidriv 2

NX contactor relays

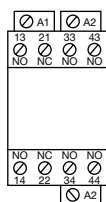
Terminal marking and positioning

NX contactor relays - AC operated

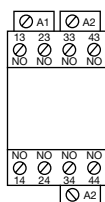
Standard devices without addition of auxiliary contacts



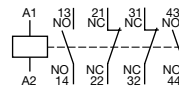
NX22E



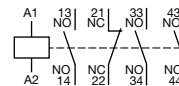
NX31E



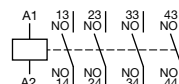
NX40E



NX22E

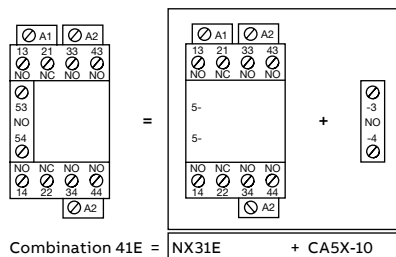


NX31E



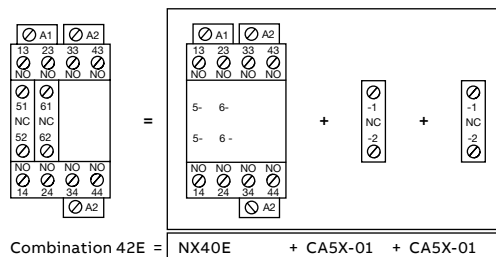
NX40E

Other possible contact combinations with auxiliary contacts added by the user



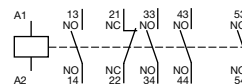
Combination 41E =

NX31E + CA5X-10



Combination 42E =

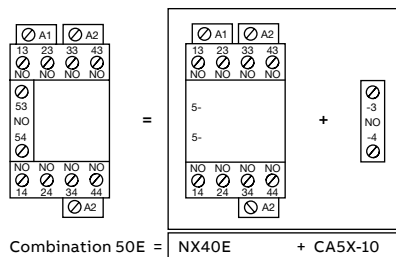
NX40E + CA5X-01 + CA5X-01



Combination 41E

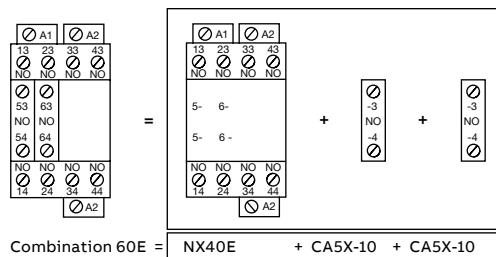


Combination 42E



Combination 50E =

NX40E + CA5X-10

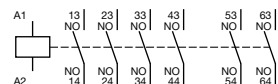


Combination 60E =

NX40E + CA5X-10 + CA5X-10



Combination 50E



Combination 60E

Accessories for AX09 ... AX370 3-pole contactors and NX contactor relays

	Auxiliary contact blocks
58	Ordering details
59	Technical data
62	Terminal marking and positioning
	Electronic timers
63	Ordering details
64	Technical data
	Mechanical and electrical interlock units
66	Ordering details
67	Technical data
	Interface relays
68	Interface relays
69	Technical data
	Mechanical latching units
70	Ordering details
71	Technical data
	Surge suppressors for contactor coils
72	Ordering details
73	Technical data
74	Additional terminal blocks
75	Function markers - Terminal for control lead connections
76	Terminal shrouds - Terminal engagements and extension
77	Terminal connecting strips and shorting bars
78	Connection accessories for starting solutions
79	Phase to phase connections - Connection sets for delta starters
80	Voltage code table

Auxiliary contact blocks



CA5X-10

AX07015



CA5X-4P

AX07013 CA5X 4P



CAL5X-11

1SBC57375P0301



CAL18X-11

1SFCL0003P0201

The auxiliary contact blocks are used for the operation of auxiliary circuits and control circuits for standard industrial environments.

Types of auxiliary contact blocks for front mounting:

- CA5X 1 or 4-pole block, instantaneous with N.O., N.C. contacts.

Select the 4-pole auxiliary contact blocks CA5X-..E, CA5X-..M, CA5X-..U, CA5X-..N, according to the contactor type for compliance with the standard requirements (see "Terminal Marking and Positioning").

Types of auxiliary contact blocks for side mounting:

- CAL... 2-pole block instantaneous N.O. + N.C. contacts.

For clipping onto the right- and/or left-hand side of the contactors.

The CAL...-11B is a second block for mounting in addition to a first CAL...-11 block, right- and/or left-hand of the AX185 ... AX370 contactors.

The auxiliary contact blocks are equipped with screw type connecting terminals delivered open, protected against accidental direct contact and bear the corresponding function marking.

For contactors	Auxiliary contacts	Type	Order code	Pkg qty	Weight (1 pce)
					kg
Front-mounted instantaneous auxiliary contact blocks					
AX09 ... AX150 and NX 4-pole	1 –	CA5X-10	1SBN019010R1010	10	0.014
	– 1	CA5X-01	1SBN019010R1001	10	0.014
AX50 ... AX150	2 2	CA5X-22E	1SBN019040R1022	2	0.060
	3 1	CA5X-31E	1SBN019040R1031	2	0.060
	4 0	CA5X-40E	1SBN019040R1040	2	0.060
	0 4	CA5X-04E	1SBN019040R1004	2	0.060
AX09...AX40-30-10	2 2	CA5X-22M	1SBN019040R1122	2	0.060
	3 1	CA5X-31M	1SBN019040R1131	2	0.060
	1 3	CA5X-13M	1SBN019040R1113	2	0.060
	0 4	CA5X-04M	1SBN019040R1104	2	0.060
AX09...AX40-30-01	2 2	CA5X-22U	1SBN019040R1322	2	0.060
	3 1	CA5X-31U	1SBN019040R1331	2	0.060
	4 0	CA5X-40U	1SBN019040R1340	2	0.060
	0 4	CA5X-04U	1SBN019040R1304	2	0.060
NX 4-pole	2 2	CA5X-22N	1SBN019040R1222	2	0.060
	3 1	CA5X-31N	1SBN019040R1231	2	0.060
	0 4	CA5X-04N	1SBN019040R1204	2	0.060
	4 0	CA5X-40N	1SBN019040R1240	2	0.060
	1 3	CA5X-13N	1SBN019040R1213	2	0.060
Side-mounted instantaneous auxiliary contact block, 2 pole					
AX09 ... AX80 and NX - 4 pole	1 1	CAL5X-11	1SBN019020R1011	2	0.050
AX95 ... AX205	1 1	CAL18X-11	1SFN019820R1011	2	0.050
AX185 ... AX205	1 1	CAL18X-11	1SFN019820R1011	2	0.050
	1 1	CAL18X-11B	1SFN019820R3311	2	0.050
AX260 ... AX370	1 1	CAL19-11	1SFN010820R1011	2	0.040
	1 1	CAL19-11B	1SFN010820R3311	2	0.040

For each contactor or contactor relay, refer to "Accessories fitting details" table.

Auxiliary contact blocks

Technical data

Contact utilization characteristics according to IEC

Types	Front mounted	Side mounted		
	1-pole CA5X, 4-pole CA5X	CAL5X-11	CAL18X-11, CAL18X-11B	CAL19-11, CAL19-11B
Standards	IEC 60947-5-1 and EN 60947-5-1			
Rated insulation voltage Ui acc. to IEC 60947-5-1	690 V			
Rated operational voltage Ue max.	24...690 V AC			
Conventional thermal current Ith - $\theta \leq 40^\circ\text{C}$	16 A			
Ie / Rated operational current AC-15 acc. to IEC 60947-5-1	24-127 V 50/60 Hz 6 A 220-240 V 50/60 Hz 4 A 380-440 V 50/60 Hz 3 A 500-690 V 50/60 Hz 2 A			
Making capacity	10 x Ie AC-15 acc. to IEC 60947-5-1			
Breaking capacity	10 x Ie AC-15 acc. to IEC 60947-5-1			
Ie / Rated operational current DC-13 acc. to IEC 60947-5-1	24 V DC 6 A / 144 W 48 V DC 2.8 A / 134 W 72 V DC 1 A / 72 W 110 V DC 0.55 A / 60 W 125 V DC 0.55 A / 69 W 220 V DC 0.3 A / 66 W 250 V DC 0.3 A / 75 W			
Short-circuit protection device gG type fuse	10 A			
Rated short-time withstand current Icw $\theta = 40^\circ\text{C}$	for 1.0 s 100 A for 0.1 s 140 A			
Minimum switching capacity	AX09 ... AX80 contactors with failure rate acc. to IEC 60947-5-4 AX95 ... AX150 contactors 24 V / 3 mA $\leq 10^{-6}$ 24 V / 50 mA (0.5 million of operating cycles) with failure rate acc. to IEC 60947-5-4 AX185 ... AX205 contactors 24 V / 50 mA (0.5 million of operating cycles) with failure rate acc. to IEC 60947-5-4 AX260 ... AX370 contactors with failure rate acc. to IEC 60947-5-4 24 V / 50 mA (0.5 million of operating cycles) with failure rate acc. to IEC 60947-5-4 AX370 ... AX400 contactors with failure rate acc. to IEC 60947-5-4 24 V / 50 mA (0.5 million of operating cycles)			
Power dissipation per pole at 6 A	0.1 W			
Mechanical durability	Number of operating cycles 10 millions (AX09 ... AX80) 3 millions (AX95 ... AX150) Max. switching frequency 3600 cycles/h			
Max. electrical switching frequency	AC-15 1200 cycles/h DC-13 900 cycles/h			

Contact utilization characteristics according to UL / CSA

Standards	UL 508, CSA C22.2 N°14
Max. operational voltage	600 V AC, 250 V DC
Pilot duty	A600, Q300
AC thermal taed current	10 A

Connecting characteristics

Connection capacity (min. ... max.)	
 Rigid solid	1 x 1...4 mm ² 2 x 1...4 mm ²
 Flexible with ferrule	1 x 0.75...2.5 mm ² 2 x 0.75...2.5 mm ²
 Lugs	L ≤ 7.7 mm L > 3.7 mm
Tightening torque	1 Nm / 9 lb.in
Connection capacity acc. to UL / CSA	1 or 2 x AWG 18...14
Stripping length	1-pole: 11 mm 4-pole: 10 mm
Degree of protection	IP20
acc. to IEC 60947-1 / EN 60947-1 and IEC 60529 / EN 60529	
Screw terminals	Delivered in open position, screws of unused terminals must be tightened
All terminals	M3.5
Screwdriver type	Flat Ø 5.5 / Pozidriv 2

Auxiliary contact blocks

for severe industrial environments



The auxiliary contact blocks are used for the operation of auxiliary and control circuits for severe industrial environments.

Types of auxiliary contact blocks for front mounting:

1-pole block, instantaneous with N.O. contact or N.C. contact, designed in 2 protection versions:

CE5-..D with built-in microswitch IP40, degree of protection (IP20 on terminals)

CE5-..W with built-in microswitch IP67, degree of protection (IP20 on terminals).

The auxiliary contact blocks are equipped with screw type connecting terminals delivered open, protected against accidental direct contact and bear the corresponding function marking each side of the mechanical latch).

For contactors	Auxiliary contacts	Type	Order code	Pkg qty	Weight (1 pce)
	 				kg
Front-mounting instantaneous auxiliary contact blocks, 1-pole					
AX09 ... AX80, NX	1 - - -	CE5-10D0.1	1SBN010015R1010	1	0.020
	- 1 - -	CE5-01D0.1	1SBN010015R1001	1	0.020
	1 - - -	CE5-10D2	1SBN010017R1010	1	0.020
	- 1 - -	CE5-01D2	1SBN010017R1001	1	0.020
	1 - - -	CE5-10W0.1	1SBN010016R1010	1	0.020
	- 1 - -	CE5-01W0.1	1SBN010016R1001	1	0.020
	1 - - -	CE5-10W2	1SBN010018R1010	1	0.020
	- 1 - -	CE5-01W2	1SBN010018R1001	1	0.020

For each contactor type, refer to "Accessory fitting details" table.

Auxiliary contact blocks

Technical data

Contact utilization characteristics according to IEC

Types	Front-mounted	1-pole CE5-..0.1	1-pole CE5-..2
Standards	IEC 60947-5-1 and EN 60947-5-1		
Rated insulation voltage U_i acc. to IEC 60947-5-1	250 V		
Rated operational voltage U_e max.	125 V		250 V
Conventional thermal current I_{th} - $\theta \leq 40^\circ\text{C}$	0.1 A		2 A
I_e / Rated operational current acc. to IEC 60947-5-1	AC-14		AC-15
	24-127 V 50/60 Hz	0.1 A	2 A
	220-240 V 50/60 Hz	–	2 A
Making capacity acc. to IEC 60947-5-1	6 x I_e AC-14		10 x I_e AC-15
Breaking capacity acc. to IEC 60947-5-1	6 x I_e AC-14		10 x I_e AC-15
I_e / Rated operational current DC-12 acc. to IEC 60947-5-1			
	24 V DC	0.1 A	2 A
	48 V DC	0.1 A	1 A
	72 V DC	0.1 A	0.3 A
	110 V DC	0.1 A	0.2 A
	125 V DC	–	0.2 A
	220 V DC	–	0.1 A
Short-circuit protection device	0.1 A (FF type fuses) (1)		
Minimum switching capacity			
AX09 ... AX80, NX contactors	3 V / 1 mA		17 V / 1 mA
With failure rate acc. to IEC 60947-5-4	–		$\leq 10^{-7}$
Mechanical durability	Number of operating cycles	5 millions for CE5-..D0.1	5 millions for CE5-..D2
		2.5 millions for CE5-..W0.1	2.5 millions for CE5-..W2
	Max. switching frequency	3600 cycles/h	
Electrical durability	Number of operating cycles	2.5 millions for CE5-..D0.1	1 million for CE5-..D2
		0.7 millions for CE5-..W0.1	0.3 millions for CE5-..W2
	Max. switching frequency	AC-14 1200 cycles/h	
		AC-15 1200 cycles/h	
		DC-12 900 cycles/h	

Contact utilization characteristics according to UL / CSA

Standards	UL 508, CSA C22.2 N°14	
Max. operational voltage	125 V AC / 110 V DC	250 V AC / 220 V DC
Pilot duty		
AC thermal rated current	0.1 A	2 A

Connecting characteristics

Connection capacity (min. ... max.)			
	Rigid solid	1 x	1...4 mm ²
		2 x	1...4 mm ²
	Flexible with ferrule	1 x	0.75...2.5 mm ²
		2 x	0.75...2.5 mm ²
	Lugs	L <	7.7 mm
		L >	3.7 mm
Connection capacity acc. to UL / CSA	1 or 2 x	AWG 18...14	
Tightening torque		1 Nm	
Degree of protection	Terminals	IP20	
acc. to IEC 60947-1 / EN 60947-1 and IEC 60529 / EN 60529	Microswitches	IP40 for CE5-..D0.1	IP40 for CE5-..D2
		IP67 for CE5-..W0.1	IP67 for CE5-..W2
Screw terminals	Delivered in open position, screws of unused terminals must be tightened		
All terminals	M3.5		
Screwdriver type	Flat Ø 5.5 / Pozidriv 2		

(1) or HRC fuses for very fast action (6.3 x 32 mm size).

Add-on auxiliary contacts

Terminal marking and positioning

1-pole auxiliary contacts



CA5X-01



CA5X-10

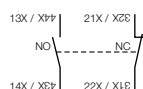
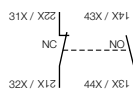


CE5-01

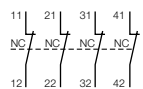


CE5-10

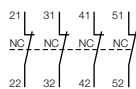
2-pole auxiliary contacts

CAL5X-11, CAL18X-11
(L. h. s. mounted)CAL5X-11, CAL18X-11
(R. h. s. mounted)

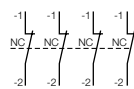
4-pole auxiliary contacts



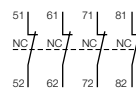
CA5X-04E



CA5X-04M



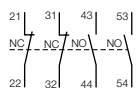
CA5X-04U



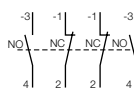
CA5X-04N



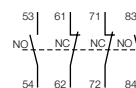
CA5X-22E



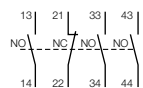
CA5X-22M



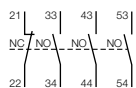
CA5X-22U



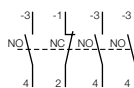
CA5X-22N



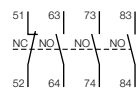
CA5X-31E



CA5X-31M



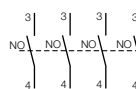
CA5X-31U



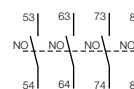
CA5X-31N



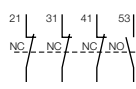
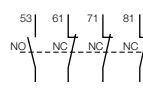
CA5X-40E



CA5X-40U



CA5X-40N



Electronic timers



TEF5-OFF

TEF5 frontal electronic timers are used for realizing timing function and are available in ON-delay and OFF-delay versions.

Compact solution in cabinet compared to separate timers

TEF5 electronic timers are front-mounted and locked on AX contactors or NX contactor relays. A mechanical indicator allows to show the state of the contactor. TEF5 electronic timers are supplied by direct wiring to the coil terminals A1 - A2 of the contactor or contactor relay. A varistor is integrated on the timer to offer a built-in protection against surges in the contactor coil.

Available for a wide control voltage range 24...240 V AC / DC

TEF5-ON or TEF5-OFF allow time-delayed functions up to 100 s in 3 distinct time ranges, independently of the control system. The time delay ranges are selected by a switch and the time delay can be adjusted by means of a rotary switch. The timing function is activated by closing or opening the device on which the timer is mounted. The OFF-delay version operates without additional control supply.

For contactors, contactor relays	Time delay range selected by switch	Delay type	Rated control circuit voltage Uc V 50/60 Hz or DC	Auxiliary contacts 	Type	Order code	Weight Pkg (1 pce) kg
AX09 ... AX80	0.1...1 s	ON-delay	24...240	1 1	TEF5-ON	1SBN020312R1000	0.065
NX 4-pole	1...10 s 10...100 s	OFF-delay	24...240	1 1	TEF5-OFF	1SBN020314R1000	0.065

Electronic timers

Technical data

Contact utilization characteristics according to IEC

Types		TEF5-ON	TEF5-OFF
Standards		IEC 60947-5-1 and EN 60947-5-1	
Rated insulation voltage U_i acc. to IEC 60947-5-1		400 V	
Rated impulse withstand voltage U_{imp}		4 kV	
Rated operational voltage U_e max.		240 V	
Rated frequency (without derating)		50 / 60 Hz	
Conventional thermal current I_{th} - $\theta \leq 40^\circ\text{C}$		5 A	
I _e / Rated operational current AC-15 acc. to IEC 60947-5-1	24-127 V 50/60 Hz	3 A	
	220-240 V 50/60 Hz	1.5 A	
Making capacity		10 x I _e AC-15 acc. to IEC 60947-5-1	
Breaking capacity		10 x I _e AC-15 acc. to IEC 60947-5-1	
I _e / Rated operational current DC-13 acc. to IEC 60947-5-1	24 V DC	1 A / 24 W	
Short-circuit protection device gG type fuse		6 A	
Rated short-time withstand current I_{cw} $\theta = 40^\circ\text{C}$	for 1.0 s	8 A	
	for 0.1 s	8 A	
Minimum switching capacity with failure rate acc. to IEC 60947-5-4	24 V DC	12 V / 3 mA	
		10^{-7}	
Power dissipation per pole at 3 A		0.1 W	
Function diagram		ON-delay	OFF-delay
		Bistable relay inside. Before use, once apply U_c then switch it off in order to initialize position of the contacts.	
Control circuit voltage			
AC control voltage 50/60 Hz	Rated control circuit voltage U_c	24...240 V AC	
	Average consumption	1.5 mA RMS	1 mA RMS
DC control voltage	Rated control circuit voltage U_c	24...240 V DC	
	Average consumption	1.5 mA	1 mA
Rated frequency limits		50 / 60 Hz	
Supply voltage range		0.85...1.1 x U_c (at $\theta \leq 70^\circ\text{C}$)	
Overvoltage protection		Varistor included	
Time delay range (t) selected by switch	0.1...1 s		
	1...10 s		
	10...100 s		
On-load reiteration accuracy under constant conditions		$\leq 1\%$	
Minimum ON period		0.15 s	1 s
Recovery time		0.15 s	0.1 s
Ambient air temperature	Operation	-25 °C ... +70 °C	
	Storage	-40 °C ... +80 °C	
Climatic withstand		Category B according to IEC 60947-1 Annex Q	
Maximum operating altitude		2000 m	
Mounting positions		Acc. to mounting positions permitted on contactors or contactor relays	
Shock withstand		1/2 sinusoidal shock for 11 ms: no change in contact position	
acc. to IEC 60068-2-27 and EN 60068-2-27 (Mounting position 1)		Same as contactor or contactor relay	
Mechanical durability			
	Number of operating cycles	5 millions operating cycles	
	Max. switching frequency	3600 cycles/h	1800 cycles/h
Max. electrical switching frequency			
	AC-15	1200 cycles/h	
	DC-13	900 cycles/h	

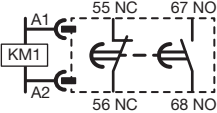
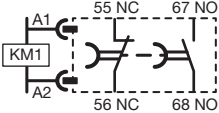
Electronic timers

Technical data

Contact utilization characteristics according to UL / CSA

Types	TEF5-ON	TEF5-OFF
Standards	UL 508, CSA C22.2 No. 14-18	
Rated insulation voltage Ui acc. to UL / CSA	300 V	
Max. operational voltage	240 V	
Pilot duty	B300, R300	
AC thermal rated current	5 A	
AC maximum volt-ampere making	3600 VA	
AC maximum volt-ampere breaking	360 VA	
DC thermal rated current	1 A	
DC maximum volt-ampere making-breaking	28 VA	

Connecting characteristics

Connection capacity (min. ... max.)	
 Rigid solid	1 x 1...2.5 mm ²
 Flexible with non insulated ferrule	2 x 1...2.5 mm ²
 Flexible with non insulated ferrule	1 x 0.75...2.5 mm ²
 Flexible with non insulated ferrule	2 x 0.75...2.5 mm ²
 Flexible with insulated ferrule	1 x 0.75...2.5 mm ²
 Flexible with insulated ferrule	2 x 0.75...1.5 mm ²
 Lugs	L ≤ 8 mm
	l > 3.7 mm
Connection capacity acc. to UL / CSA	1 or 2 x AWG 18...14
Stripping length	10 mm
Tightening torque	1 N.m / 9 lb.in
Degree of protection acc. to IEC 60947-1 / EN 60947-1 and IEC 60529 / EN 60529	IP20
Screw terminals All terminals	Delivered in open position, screws of unused terminals should be tightened M3.5
Screwdriver type	Flat Ø 5.5 / Pozidriv 2
Terminal Marking	 

Mechanical and electrical interlock units



1SBC58041LF0301

VM300H

Mechanical interlock units

The VM mechanical interlock units are designed for the interlocking of two AX contactors. When mounted between two contactors, the VM mechanical interlock unit prevents one of the contactors from closing as long as the other contactor is closed.

Left side contactor	Right side contactor	Mounting	Auxiliary contacts	Type	Order code	Pkg qty	Weight (1 pce)
							kg

Mechanical interlock units for two horizontal mounted contactors (1)

AX09 ... AX40	AX09 ... AX40	Horizontal	- -	VM5-1	1SBN030100R1000	1	0.066
AX95 ... AX205	AX185 ... AX205	Horizontal	- -	VM300H	1SFN034700R1000	1	0.150
AX260 ... AX370	AX260 ... AX370	Horizontal	- -	VM19	1SFN030300R1000	1	0.054
AX185 ... AX205	AX260 ... AX370	Horizontal	- -	VM205/260	1SFN035003R1000	1	0.075
AX260 ... AX370	AX185 ... AX205	Horizontal	- -	VM205/260	1SFN035003R1000	1	0.075
AX260 ... AX370	AF400 ... AF460	Horizontal	- -	VM370/400	1SFN035403R1000	1	0.239
AF400 ... AF460	AX260 ... AX370	Horizontal	- -	VM370/400	1SFN035403R1000	1	0.239

Mechanical interlock units for two vertical mounted contactors

Up contactor	Down contactor	Mounting					
AX95 ... AX150	AX150 ... AX205	Vertical	- -	VM300V	1SFN034701R1000	1	0.150



1SBC100095V0014

VE5-1

Mechanical and electrical interlock sets

The VM mechanical interlock units are designed for the interlocking of two AX contactors. When mounted between two contactors, the VM mechanical interlock unit prevents one of the contactors from closing as long as the other contactor is closed. The VE units include 2 N.C. contacts for electrical interlocking function.

Left side contactor	Right side contactor	Mounting	Auxiliary contacts	Type	Order code	Pkg qty	Weight (1 pce)
							kg

Mechanical interlock units for two horizontal mounted contactors

AX09 ... AX40	AX09 ... AX40	Horizontal	- 2	VE5-1	1SBN030110R1000	1	0.076
AX32 ... AX80	AX50 ... AX80	Horizontal	- 2	VE5-2	1SBN030210R1000	1	0.146
AX50 ... AX80	AX32 ... AX80	Horizontal	- 2	VE5-2	1SBN030210R1000	1	0.146
AX50 ... AX80	AX95 ... AX150	Horizontal	- 2	VE5-2 (2)	1SBN030210R1000	1	0.146
AX95 ... AX150	AX50 ... AX80	Horizontal	- 2	VE5-2 (2)	1SBN030210R1000	1	0.146
AX95 ... AX150	AX95 ... AX150	Horizontal	- 2	VE5-2	1SBN030210R1000	1	0.146

(1) Mechanical durability: VM5-1 = 5 millions cycles, VM300H = 1 million cycles.

VM19 = 0,5 million cycles, VM205/260 = 1 million cycles, VM300V = 1 million cycles.

(2) The combination of AX50 ... AX80 contactors interlocked with AX95 ... AX150 contactors cannot be mounted on symmetrical rail (75 mm, IEC/EN 60715).

Mechanical and electrical interlock units

Technical data

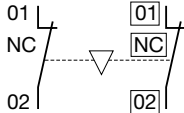
Contact utilization characteristics according to IEC

Types	VE5-1	VE5-2
Standards	IEC 60947-5-1 and EN 60947-5-1	
Rated insulation voltage U_i acc. to IEC 60947-5-1	690 V	
Rated operational voltage U_e max.	24...690 V	
Conventional thermal current I_{th} - $\theta \leq 40^\circ\text{C}$	16 A	
I_e / Rated operational current AC-15 acc. to IEC 60947-5-1	24-127 V 50/60 Hz	6 A
	220-240 V 50/60 Hz	4 A
	380-440 V 50/60 Hz	3 A
	500-690 V 50/60 Hz	2 A
Making capacity	10 x I_e AC-15 acc. to IEC 60947-5-1	
Breaking capacity	10 x I_e AC-15 acc. to IEC 60947-5-1	
I_e / Rated operational current DC-13 acc. to IEC 60947-5-1	24 V DC	6 A
	48 V DC	2.8 A
	72 V DC	1 A
	125 V DC	0.55 A
	250 V DC	0.3 A
Short-circuit protection device - gG type fuse	10 A	
Rated short-time withstand current I_{cw} $\theta = 40^\circ\text{C}$	for 1.0 s	100 A
	for 0.1 s	140 A
Power dissipation per pole at 6 A	0.15 W	
Mechanical durability	5 millions operating cycles	
Number of operating cycles	5 millions operating cycles	
Max. switching frequency	600 cycles/h	

Utilization characteristics according to UL / CSA

Standards	UL 508, CSA C22.2 N°14
Max. operational voltage	600 V

Connecting characteristics

Connection capacity (min. ... max.)		
 Rigid solid	1 x	1...4 mm ²
	2 x	1...4 mm ²
 Flexible with ferrule	1 x	0.75...2.5 mm ²
	2 x	0.75...2.5 mm ²
 Lugs	L <	8 mm
	I >	3.5 mm
Tightening torque	Recommended	1 Nm
	Max.	1.2 Nm
Degree of protection acc. to IEC 60947-1 / EN 60947-1 and IEC 60529 / EN 60529	IP20	
Screw terminals All terminals	Delivered in open position, screws of unused terminals must be tightened M3.5	
Screwdriver type	Flat Ø 5.5 / Pozidriv 2	
Terminal marking		

Technical note: when, during switching, the arc time is estimated to more than 40 ms, the closing signal of one of the two contactors must be delayed with respect to the opening signal of the other contactor in order to prevent a short-circuit.

Interface relays



RA5-1

RA5-1 interface relay is designed to receive 24 V DC signals delivered by PLC’s or other sources with a low output power and to restore them with sufficient power to operate the coils of the relevant AX50, AX65 and AX80 contactors.

RA5-1 interface relay is made up of a miniature electromechanical relay equipped with a N.O. contact and with a low consumption 24 V DC coil.

The interface relay coil is controlled by the PLC while the N.O. contact ensures switching of the power contactor.

Coil switching gives rise to overvoltages which have adverse effects on the electronic devices, insulators and, more generally, on component lifetime. The RA5-1 is equipped with surge suppressors:

- on the 24 V DC relay coil via a diode,
- on the power contactor coil via a varistor.

Furthermore, the RA5-1 is protected against relay pole reversal by a diode inserted between the E1 and E2 input terminals.

For contactors	Coil voltages	Rated control circuit voltage Uc	Type	Order code	Pkg qty	Weight (1 pce)
	V 50/60 Hz	V DC				kg
AX50 ... AX80	24...250	24	RA5-1	1SBN060300R1000	1	0.050
			RA5-1	1SBN060300T1000	10	0.050

Interface relays

Technical data

Type	RA5-1
Utilization characteristics according to IEC	
Standards	IEC 60255-5
Rated insulation voltage U_i acc. to IEC 60947-4-1	250 V AC
Ambient air temperature	
In free air operation	at $U_c = 24$ V DC (between E1 and E2) -25...+70 °C
from 0.85 to 1.1 x U_c	-25...+55 °C
Storage	-40...+70 °C
Climatic withstand	Complies with that of associated contactors
Maximum operating altitude	3000 m
Mounting positions	No limitation
Fixing	Using the contactor A1 and A2 terminal connecting parts

Connecting characteristics

Connection capacity (min. ... max.)	
 Rigid solid	1 x 1...4 mm ²
 Flexible with ferrule	2 x 1...4 mm ²
 Flexible with ferrule	1 x 0.75...2.5 mm ²
 Flexible with ferrule	2 x 0.75...2.5 mm ²
 Lugs	L < 8 mm
	l > 3.5 mm
Tightening torque	
Recommended	1 Nm
Max.	1.2 Nm
Degree of protection	Protection against direct contact in acc. with EN 50274
acc. to IEC 60947-1 / EN 60947-1 and IEC 60529 / EN 60529	RA5-1 wired and mounted on the associated contactor
Screw terminals	Delivered in open position, screws of unused terminals must be tightened
All terminals	M3.5
Screwdriver type	Flat Ø 5.5 / Pozidriv 2

Working data

Surge suppression	
For contactor coil	Varistor
For interface relay coil	Diode
Protection against polarity reversal between terminals E1 and E2	Diode
Interface relay operating time	Closing and drop-out ≤ 10 ms
Total operating time, interface relay + contactor	
Between energization and:	N.O. contact closing 20...37 ms
	N.C. contact opening 17...32 ms
Between de-energization and:	N.O. contact opening 17...25 ms
	N.C. contact closing 20...28 ms

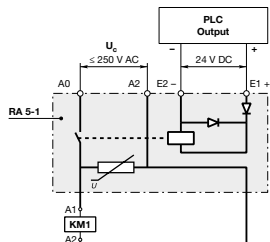
Electrical input data

Control voltage (E1 and E2 terminals) U_c	
Rated value	24 V DC
Max. range at ambient temperature 20 °C	19...30 V DC
Max. consumption for $U_c = 24$ V DC, $\theta = 20$ °C	0.3 W
"0" status (relay open)	for $U_c \leq 2.4$ V DC
	for $I_c < 1$ mA
"1" status (relay closed)	for $U_c \geq 19$ V DC
Max. short supply interruption immunity time	2 ms

Electrical output data

Switching voltage (A0 and A2 terminals)	≤ 250 V AC
Electrical durability	
Number of operating cycles	2 millions

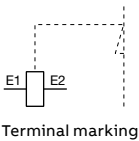
Connection

	The "E1+" and "E2-" input terminals must be connected, according to their polarity, to the PLC output. The RA5-1 is equipped with two terminal pads for connection to the A1 and the A2 terminals of the contactor coil. This coil is supplied between the A0 and the A2 terminals of the RA 5-1. Mounting: terminals pads clamped inside the contactor coil terminals.
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Mechanical latching units



WB75-A



Terminal marking

For converting standard contactors into latched contactors.
The WB75-A block contains a mechanical latching device with electromagnetic impulse unlatching (AC or DC) or manual unlatching.
Captive screw type connecting terminals, built-in cable clamps, M3.5 (+,-) pozidriv 2 screw with screwdriver guidance; delivered untightened and protected against accidental direct contact.

Operation

After closing, the contactor continues to be held in the closed position by the latching mechanism without supply voltage at the contactor coil terminals.

Contactor opening can be controlled:

- electrically by an impulse (AC or DC) on the WB75-A block coil.
(the coil is not designed to be permanently energized)
- manually by pressing the pushbutton on the front face of the WB75-A block.

Mounting

The WB75-A block is clipped onto the front face of the 1-stack contactor where it takes up two slots. The two other slots may accept CA5X single pole auxiliary contacts (1 block on each side of the mechanical latch).

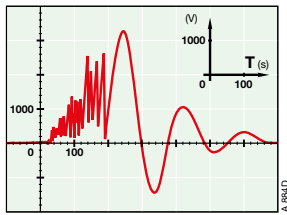
For contactors	Rated control circuit voltage Uc		Type	Order code	Pkg qty	Weight (1 pce) kg
	V 50 Hz or DC	V 60 Hz				
AX09 ... AX80	24	24...28	WB75-A	FPTN372726R1001	1	0.120
	42	42...48	WB75-A	FPTN372726R1002	1	0.120
	48	48...55	WB75-A	FPTN372726R1003	1	0.120
	110	110...127	WB75-A	FPTN372726R1004	1	0.120
	220...230	220...255	WB75-A	FPTN372726R1006	1	0.120
	230...240	230...277	WB75-A	FPTN372726R1005	1	0.120
	380...415	380...440	WB75-A	FPTN372726R1007	1	0.120
	415...440	440...480	WB75-A	FPTN372726R1008	1	0.120

Mechanical latching units

Technical data

Type		WB75-A	
Utilization characteristics according to IEC			
Rated insulation voltage Ui acc. to IEC 60947-1		690 V	
Max. electrical impulse time			
On AC coil (with load factor 5 %)		20 s	
On DC coil (with load factor 3 %)		8 s	
Min. electrical impulse time			
For latching (energizing of the contactor coil)		AC	50 ms
For pull-out (energizing of the WB block coil)		AC	30 ms
Coil operating limits		AC or DC supply	0.85...1.1 x Uc
AC control voltage 50/60 Hz			
Rated control circuit voltage Uc		24...480 V AC	
Coil consumption		Average pull-in value	90 VA
		Average holding value	60 VA
DC control voltage			
Rated control circuit voltage Uc		24...440 V DC	
Coil consumption		Average pull-in value	110 W
		Average holding value	110 W
Operating time			
On contactor closing (latching)			
Between coil energization and:		N.O. contact closing	No difference with the operation of a contactor without mechanical latching unit
		N.C. contact opening	No difference with the operation of a contactor without mechanical latching unit
On contactor opening (unlatching)			
Between WB coil energization and:		N.O. contact opening	5...25 ms
		N.C. contact closing	7...28 ms
Mechanical durability			
Number of operating cycles		1 million operating cycles	
Max. switching frequency		3600 cycles/h with on-load factor of 8 %	
Connecting characteristics			
Connection capacity (min. ... max.)			
 Rigid solid		1 x	1...4 mm²
		2 x	1...4 mm²
 Flexible with ferrule		1 x	0.75...2.5 mm²
		2 x	0.75...2.5 mm²
 Lugs		L <	8 mm
		I >	3.5 mm
Tightening torque		Recommended	1 Nm
		Max.	1.2 Nm
Screw terminals		Delivered in open position, screws of unused terminals must be tightened	
All terminals		M3.5	
Screwdriver type		Flat Ø 5.5 / Pozidriv 2	

Surge suppressors for contactor coils



The operation of inductive circuits causes overvoltages, in particular on opening of the contactor coil.

The electromagnetic energy stored in the coil during contactor closing is restored on opening in the form of surges, the slope and amplitude of which may rise to several kilovolts. A number of drawbacks are observed ranging from interference on the electronic devices to breakdown of insulators and even destruction of certain sensitive components.

The graph opposite reproduces the oscillogram showing voltage discharges at the terminals of a 42 V / 50 Hz coil without peak clipping. The coil was switched by 8 series-connected poles of a contactor relay.

Following a burst of discharges with a very steep slope a damped oscillation emerges with a peak value of 3500 V.

Overvoltage Factor

The overvoltage factor k is defined as the ratio of the maximum overvoltage peak value $\hat{U}_s$ to the peak value $\hat{U}_c$ of the coil rated control voltage U_c :

$$k = \frac{\hat{U}_s \text{ max.}}{\hat{U}_c} \quad \text{or in AC:} \quad k = \frac{\hat{U}_s \text{ max.}}{U_c \sqrt{2}}$$

For example the following is obtained for the above graph: $k = \frac{3500}{42 \sqrt{2}} \approx 60$



RV5/50

1SBC5740101F0301



RC5-1/50

1SBC573891F0301

To reduce the harmful effects of these overvoltages, ABB has developed a range of surge suppressors designed to reduce the k factor defined above and to limit or even completely eliminate the high pre-damping voltage frequencies.

Each case is different, but the technical data tolerances and the generous sizing of parts have enabled us to reduce the number of variants.

We have chosen the following solutions: varistors and RC blocks.

Note: A varistor is a resistor whose value decreases to a very large extent when a certain voltage is applied at its terminals.

For contactors	Rated control circuit voltage U_c V AC	Type	Order code	Pkg qty	Weight (1 pce) kg
AX09 ... AX150	24...50	RV5/50	1SBN050010R1000	2	0.015
	50...133	RV5/133	1SBN050010R1001	2	0.015
	110...250	RV5/250	1SBN050010R1002	2	0.015
	250...440	RV5/440	1SBN050010R1003	2	0.015
AX09 ... AX40	24...50	RC5-1/50	1SBN050100R1000	2	0.012
	50...133	RC5-1/133	1SBN050100R1001	2	0.012
	110...250	RC5-1/250	1SBN050100R1002	2	0.012
	250...440	RC5-1/440	1SBN050100R1003	2	0.012
AX50 ... AX150	24...50	RC5-2/50	1SBN050200R1000	2	0.015
	50...133	RC5-2/133	1SBN050200R1001	2	0.015
	110...250	RC5-2/250	1SBN050200R1002	2	0.015
	250...440	RC5-2/440	1SBN050200R1003	2	0.015
AX185 ... AX205	250...440	RC5-3/440	1SBN050300R1003		

Surge suppressors for contactor coils

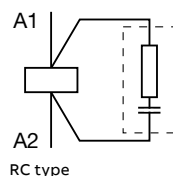
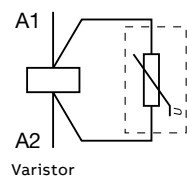
Technical data

Varistor	RV5/50	RV5/133	RV5/250	RV5/440
Rated control circuit voltage U_c	24...50 V AC	50...133 V AC	110...250 V AC	250...440 V AC
Residual overvoltage (clipping voltage)	132 V AC	270 V AC	480 V AC	825 V AC
Opening time growth factor	1.1...1.5			
Operating temperature	-20...+70 °C			
Connection to the coil terminals (parallel mounting)	Clip-on for both fixing and connection.			
Fixing	Clipped onto the top part of the contactor base without change in contactor overall dimensions.			
Advantages	High energy absorption: good damping - Unpolarized system.			
Drawback	Clipping as from Uvdr (1), thus voltage front up to this point.			

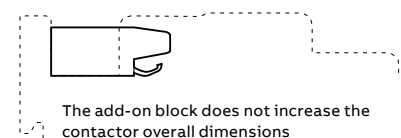
(1) Uvdr = Varistor operating voltage (voltage dependent resistor), tolerance $\pm 10\%$.

RC type	RC5-1/50	RC5-1/133	RC5-1/250	RC5-1/440
	RC5-2/50	RC5-2/133	RC5-2/250	RC5-2/440
	–	–	–	RC5-3/440
Rated control circuit voltage U_c	24...50 V AC	50...133 V AC	110...250 V AC	250...440 V AC
Residual overvoltage (clipping voltage)	2 to 3 x U_c max.			
Opening time growth factor	1.2...1.3			
Operating temperature	-20...+70 °C			
Connection to the coil terminals (parallel mounting)	Clip-on for both fixing and connection.			
Fixing	Clipped onto the top part of the contactor base without change in contactor overall dimensions.			
Advantages	Very fast clipping - Attenuation of steep fronts and thus of high frequencies. No operating delays.			

Wiring diagrams



Dimensions



RV5, RC5

Additional terminal blocks



LD75

1SBC580745FC0301



LD110

1SBC580723FC0301

The LD terminal block is designed to increase the connecting capacity of the contactor on which it is fitted and for preparation of the wiring before final connection on the contactor.

The LD blocks are 3-pole terminal blocks with tunnel terminals. The available range can be used on AX50 to AX150 contactors.

The LD75 and LD110 terminal blocks are fixed in the 3 independent slots located above the built-in connectors.

For contactors	Type	Order code	Pkg qty	Weight (1 pce) kg
AX50 ... AX80	LD75	1SBN073508R1000	1	0.115
AX95 ... AX150	LD110 (1)	1SFN074308R1000	1	0.150

(1) up to 160 AAC-1.

Technical data

Types	LD75	LD110
Rated insulation voltage U_i acc. to IEC 60947-4-1 acc. to UL / CSA		
Main terminals	 Screw terminals with single connector 10 x 11 mm	 Screw terminals with single connector 12 x 12 mm
Connection capacity (min. ... max.)		
 Rigid Solid ($\leq 4 \text{ mm}^2$)	1 x 6...50 mm ²	10...70 mm ²
 Stranded ($\geq 6 \text{ mm}^2$)	2 x 6...25 mm ²	10...35 mm ²
 Flexible with ferrule	1 x 6...35 mm ²	10...50 mm ²
	2 x 6...16 mm ²	10...25 mm ²
Bars	10 mm	12 mm
Tightening torque	4 Nm	6 Nm
Degree of protection acc. to IEC 60947-1 / EN 60947-1 and IEC 60529 / EN 60529		
Screw terminals	M6	M8
Screwdriver type	Pozidriv 2	Hexagon socket (s = 4 mm)

Note: The utilization of LD additional terminal blocks keeps the possibility to connect the following cables directly in the contactor main terminals but the BED and BEM connecting sets can no longer be used.

	LD75	LD110
Possible cross section of rigid cable in the contactor terminals	50 mm ²	95 mm ²

Function markers

Terminals for control lead connections



BA5-50

1SBC575874F0301



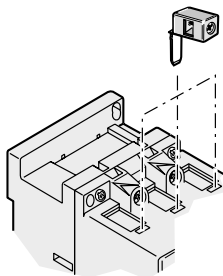
BA4

1SNC160101F0014



LK75-F

1SBC575779F0301



LK positioning

Function markers AX09 ... AX150

Set of 50 function markers designed to be clipped onto the front face of devices. Details can be added to these markers using a ball point pen, indelible felt-tip pen or pentel white.

Self-adhesive labels (not supplied) can also be added to them.

Marker dimensions: 7 x 19 mm (0.276" x 0.748").

For contactors	Type	Order code	Pkg qty	Weight (1 pce) kg
AX09 ... AX150 and accessories	BA5-50	1SBN110000R1000	1	0.017

Function markers AX260 ... AX370

Box of 16 blank cards (16 markers by card) printable on HTP500 thermal transfer printer and AMS 500 marking table to identify your contactors, overload relays or manual motor starters.

Marker dimensions: 7 x 20 mm (.276" x .787").

For contactors	Type	Order code	Pkg qty	Weight (1 pce) kg
AX260 ... AX370 contactors, EF electronic overload relays and MS116 manual motor starters	BA5-50	1SNA235256R2700	16	0.011
AMS 500 support plate for 8 BA4	SPRC 1	1SNA360010R1500	1	0.220
HTP500 support plate	HTP500-BA4	1SNA235712R2400	1	0.290

Terminals for control lead connections

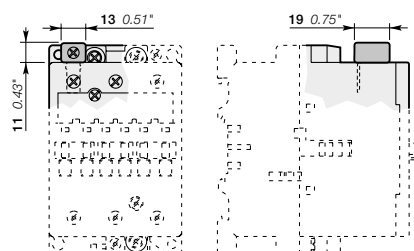
Terminals designed to connect the control conductors to the main poles of the AX50 ... AX80 contactors.

Accessories clipped into the slots placed above each power terminal connector.

The LK75 are fitted with a pin designed to hold them in place until the connector has been fully clamped with its power cable.

- Degree of protection IP20
- Connecting terminal delivered in open position: cable clamp and M3.5 (+,-) pozidriv 2 screw.
- Cable cross-sectional area:
 - 1 or 2 rigid conductors 1...4 mm²
 - 1 or 2 flexible conductors with cable end 0.75...2.5 mm²
- Tightening torque for the LK screw:
 - recommended 1.00 Nm
 - maxi. 1.20 Nm

For contactors	Type	Order code	Pkg qty	Weight (1 pce) kg
AX50 ... AX80	LK75-F	1SBN073552R1002	2	0.006



LK75-F

Main dimensions mm, inches

Terminal shrouds

Terminals enlargements and extension



LT ... AC

1SFT98099-019C3



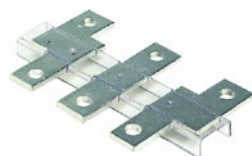
LT ... AL

1SFT98099-125



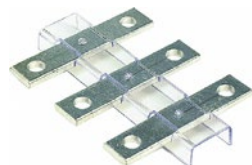
LT ... AY

1SFT98000-014



LW

1SFT98000-011C3



LX

1SFT98000_012C3

Terminal shrouds

Main terminal protection for AX185 ... AF370 contactors.

The auxiliary contact blocks and coils are designed to provide an IP 20 degree of protection.

The main terminals, equipped with compression lugs or cable clamps, can be protected against accidental direct contact after wiring (EN 50274) by the addition of terminal shrouds (see table below).

For contactors	Type	Order code	Pkg qty	Weight (1 pce) kg
AX185 ... AX205 with connectors	LT185-AC	1SFN124701R1000	2	0.050
AX185 ... AX205 with lugs	LT185-AL	1SFN124703R1000	2	0.220
AX185 ... AX205 with shorting bar or between contactor and TOL/EOL in DOL starters	LT185-AY	1SFN124704R1000	2	0.050
AX260 ... AX370, with cable clamps	LT370-30C	1SFN125401R1000	2	0.035
AX260 ... AX370, with compression lugs	LT370-30L	1SFN125403R1000	2	0.280
AX260 ... AX370, with shorting bar or between contactor and TOL/EOL in DOL starters	LT370-30Y	1SFN125404R1000	1	0.075

Terminal enlargements

Enlargement pieces designed to increase the width of the contactor terminal pads in order to allow larger connections to be mounted.

Sets containing 3 tin plated copper bars fixed by an isolating spacer.

For contactors	Dimension	Bar	Type	Order code	Pkg qty	Weight (1 pce) kg
	hole Ø mm	mm				
AX185 ... AX205	10.5	20 x 5	LW185	1SFN074707R1000	1	0.250
AX260 ... AX370	10.5	20 x 5	LW370	1SFN075407R1000	1	0.340

Terminal extension

Extension pieces designed to extend the main terminals of contactors for combined mounting of contactors and connection sets.

Sets containing 3 tin plated copper bars fixed by an isolating spacer.

For contactors	Dimension	Bar	Type	Order code	Pkg qty	Weight (1 pce) kg
	hole Ø mm	mm				
AX185 ... AX205	8.5	17.5 x 5	LX185	1SFN074710R1000	1	0.250
AX260 ... AX370	10.5	20 x 5	LX370	1SFN075410R1000	1	0.234

Terminal connecting strips and shorting bars



LP185

1SFT98000-010C3



LY185

1SFC100088V0001

Parallel and series connection of 3-pole contactors:

- To obtain a star point (3 parallel-connected poles): LY allows 3 phases to be short-circuited.
- To connect poles in parallel and thus increase the AC load passing through the flow path made up of the parallel-connected poles: LP (2 poles); LY (3 poles).
- For the maximum permissible current values with parallel-connected poles see "Parallel connection of main poles".

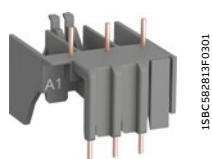
The relevant cable cross-sectional area may limit the maximum permissible current. Consult information in table below.

- To connect poles in series and thus increase the DC load controlled by the poles: LP.

Types	for connection of "n" poles	with terminal	insulated
LP	n = 2	no	yes
LY	n = 3	no	yes

For contactors	max. nominal continuous current with "n" poles A	continuous mm ²	Type	Order code	Pkg qty	Weight (1 pce) kg
AX09	30	6	LP16	FPEP407000R0001	10	0.002
AX12	32	6				
AX18	34	6				
NX	-	6				
AX185, AX205	300	-	LP185	1SFN074712R1000	2	0.300
AX260, AX300, AX370	475	-	LP300	1SFN075112R1000	2	0.400
AX09	33	6	LY16	FPEP407002R0001	10	0.005
AX12	36	6				
AX18	39	6				
AX185, AX205	400	-	LY185	1SFN074703R1000	1	0.200
AX260, AX300, AX370	670	-	LY300	1SFN075103R1000	1	0.300

Connection accessories for starting solutions



BEA

1SBC582813F0301



BEA 300

1SFT98001-005C3

Connection links between contactors and manual motor starters

The BEA connecting links are used to connect a contactor to associated manual motor starters.

These are then used together as DOL or reversing starters in type 1 or type 2 coordination, complying with IEC 60947-4-1 and EN 60947-4-1.

The BEA insulated 3-pole connecting link (touch safe) ensures the electrical linking between the contactor and the corresponding manual motor starter.

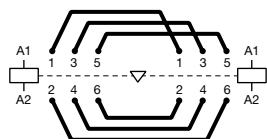
For contactors	Manual motor starter	Type	Order code	Pkg qty	Weight (1 pce) kg
AX09 ... AX18	MS116-0.16 ... MS116-16	BEA16/116	1SBN081406R1000	1	0.020
AX25	MS116-0.16 ... MS116-16	BEA25/116	1SBN089306T1000	1	0.020
AX25	MS116-20 ... MS116-32	BEA25/132	1SBN089306T1001	1	0.020

(1) up to 160 A AC-1

Connection bars between contactors and MCCB

Connection between contactors/starters and moulded case circuit breakers. These connection sets are solid copper bars.

For contactors	MCCB	Type	Order code	Pkg qty	Weight (1 pce) kg
AX185, AX205	T3	BEA185/T3	1SFN084706R1003	1	0.150
AX260, AX300, AX370	T5	BEA370/T5	1SFN085406R1000	1	0.350



BEM ... connections

Connection sets for reversing contactors

Connections between the main poles of two 3-pole contactors mounted side by side as reversing contactors with mechanical or electrical interlock.

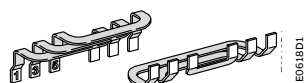
The sets are made up of three upstream connections and three downstream connections.

- BER16V ... BER40V: Insulated, stranded, rigid copper wires
- BEM75-30 ... BEM185-30 and BER370-4: Insulated, solid copper bars

On the AX contactors, the power supply by bars or cables equipped with lugs is directly connected to the terminal pads of the main poles. For flange connectors, LX terminal extension pieces should be used.

For contactors	Type	Order code	Pkg qty	Weight (1 pce) kg
AX09 ... AX18	BER16V	1SBN081411R1000	1	0.045
AX32, AX40	BER40V	1SBN082411R1000	1	0.085
AX50 ... AX80	BEM75-30	1SBN083501R1000	1	0.243
AX95 ... AX150	BEM110-30 (1)	1SFN084301R1000	1	0.450
AX185, AX205	BEM185-30	1SFN084701R1000	1	0.900
AX260 ... AX370	BER370-4	1SFN085411R1000	1	2.140

(1) up to 160 A AC-1

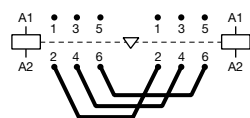


BEM 75-30

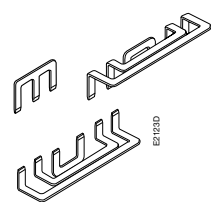
E061801

Phase to phase connections

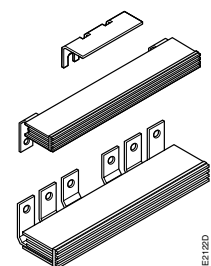
Connection sets for star-delta starters



BEP, BES



BED 110



BED 185

Phase to phase connections

Connections between the main poles of two 3-pole contactors horizontal mounted.

This set is made up of three downstream or upstream connections.

For contactors	Type	Order code	Pkg qty	Weight (1 pce) kg
AX50 ... AX80	BES75-30	1SBN083504R1000	1	0.130
AX95 ... AX150	BES110 (1)	1SFN084304R1000	1	0.250
AX185, AX205	BES185	1SFN084704R1000	1	0.500
AX260 ... AX370	BEP370-30	1SFN085414R1000	1	0.926

(1) up to 160 A AC-1.

Connections sets for star-delta starters

Connections between the main poles of a star-delta starter.

These sets are made up of:

- Three line contactor / delta contactor connections, upstream side
- Three connections for star and delta contactors, downstream side
- The necessary elements to create the star point upstream of the star contactor.
- Insulated, solid copper bars.

BED are designed for both star and delta contactors with or without mechanical interlock unit.

For line and delta contactors	For star contactors	Interlock unit between delta and star contactors	Type	Order code	Pkg qty	Weight (1 pce) kg
AX150	AX95	VE5-2	BED110 (1)	1SFN084503R1000	1	0.500
AX185	AX115	VM300H	BED145A	1SFN084703R1000	1	1.300
AX205	AX150	VM300H	BED185	1SFN084903R1000	1	1.100
AX370	AX260	VM19	BEY370-4	1SFN085813R1000	1	2.020

(1) up to 160 A AC-1.

Voltage code table

The below tables indicate the available coil voltages and corresponding digits for order codes. When placing an order, please give the order code. Select a standard contactor from ordering detail pages. Change the coil voltage code in the order code according to the table below. Example: for contactor AX40-30-10 and coil 200 V 50 Hz, the order code is 1SBL321074R.

AX contactors

Auxiliary contact
N.O. N.C.

Order code

AX 40 - 30 - 10 - 80

1SBL321074R

80

10

N.O. N.C.
Main contacts

AC coil code
Contactors: AX

	50 Hz	60 Hz
81	24 V	24 V
84	110 V	110...120 V
80	220...230 V	230...240 V
88	230...240 V	240...460 V
85	380...400 V	400...415 V
86	400...415 V	410...440 V

Codes in bold for dual frequency coils.

AC-3 Current rating

Contactor type
AC operated

NX contactor relays

N.O. N.C.
Number contacts

Order code

NX 40 E - 80

1SBH901074R

80

40

AC coil code
Contactors: NX

	50 Hz	60 Hz
81	24 V	24 V
84	110 V	110...120 V
80	220...230 V	230...240 V
88	230...240 V	240...460 V
85	380...400 V	400...415 V
86	400...415 V	410...440 V

Codes in bold for dual frequency coils.

Contactor relay type
AC operated



Manual motor starters

	Overview
84	Overview
	Ordering details – 0.10 to 65 A with thermal and electromagnetic protection
85	MS116 manual motor starters
86	Technical data
90	Main accessories
97	General accessories

Manual motor starters

Overview



Type	MS116
Thermal and electromagnetic protection	Yes
Electromagnetic protection	-
Phase loss sensitivity	Yes
Switch position	ON/OFF
Magnetic trip indication	-
Lockable handle without accessories	-
Disconnecting feature	Yes
Width	45 mm
Rated operational current I _e	0.16 ... 32 A
Setting range	0.1 ... 32 A
Ambient air temperature	-25 ... +55°C (1)

(1) Compensated

Main accessories

Auxiliary contact blocks		HKF1, HK1
Signalling contact	for tripped alarm	SK1
	for short circuit alarm	-
Shunt trip		AA1
Undervoltage release		UA1

Table for short circuit ratings for 400/415 V

Standard range MS116				
Selection parameters				
Rated operational power	Setting range for thermal release	Type	Short-circuit breaking capacity	
			I _{cu}	I _{cs}
-	0.1 ... 0.16 A	MS116-0.16	50 kA	50 kA
0.06 kW	0.16 ... 0.25 A	MS116-0.25	50 kA	50 kA
0.09 kW	0.25 ... 0.4 A	MS116-0.4	50 kA	50 kA
0.18 kW	0.4 ... 0.63 A	MS116-0.63	50 kA	50 kA
0.25 kW	0.63 ... 1.0 A	MS116-1.0	50 kA	50 kA
0.55 kW	1.0...1.6 A	MS116-1.6	50 kA	50 kA
0.75 kW	1.6...2.5 A	MS116-2.5	50 kA	50 kA
1.5 kW	2.5...4.0 A	MS116-4.0	50 kA	50 kA
2.2 kW	4.0...6.3 A	MS116-6.3	50 kA	50 kA
4.0 kW	6.3...10 A	MS116-10	50 kA	50 kA
5.5 kW	8...12 A	MS116-12	25 kA	25 kA
7.5 kW	10...16 A	MS116-16	16 kA	16 kA
7.5 kW	16 ... 20 A	MS116-20	15 kA	10 kA
11 kW	20 ... 25 A	MS116-25	15 kA	10 kA
15 kW	25 ... 32 A	MS116-32	10 kA	10 kA

MS116 manual motor starters

0.10 to 32 A - with thermal and electromagnetic protection



2CDC241010F0011

MS116-16



2CDC241001F0011

MS116-25



2CDC241010F0011

MS116-0.16-HKF1-11



2CDC241012F0011

MS116-32-HKF1-11

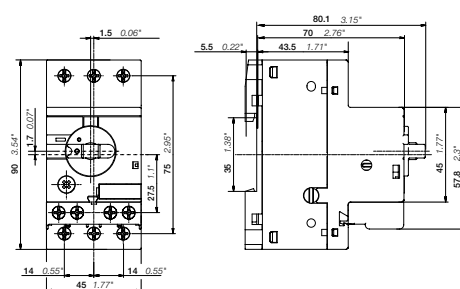
MS116 is a compact and economic range for motor protection up to 15 kW (400 V) / 32 A in width of 45 mm. Further features are the build-in disconnect function, temperature compensation, trip-free mechanism and a rotary handle with a clear switch position indication. The manual motor starter is suitable for three- and single phase applications. Auxiliary contacts, signaling contacts, undervoltage releases, shunt trips, power in-feed blocks and locking devices for protection against unauthorized changes are available as accessory. These are suitable throughout the MS116 range.

Rated operational power 400 V	Setting range	Short-circuit breaking capacity Ics at 400 V AC	Rated instantaneous short-circuit current setting Ii	Type	Order code	Weight (1 pce)
AC-3 kW	A	kA	A			
0.03(2)	0.10 ... 0.16	50	2.00 (1)	MS116-0.16	1SAM250000R1001	0.225
0.06	0.16 ... 0.25	50	3.10 (1)	MS116-0.25	1SAM250000R1002	0.225
0.09	0.25 ... 0.40	50	5.00 (1)	MS116-0.4	1SAM250000R1003	0.225
0.18	0.40 ... 0.63	50	7.90 (1)	MS116-0.63	1SAM250000R1004	0.225
0.25	0.63 ... 1.00	50	12.5 (1)	MS116-1.0	1SAM250000R1005	0.225
0.55	1.00 ... 1.60	50	20.0 (1)	MS116-1.6	1SAM250000R1006	0.265
0.75	1.60 ... 2.50	50	31.3 (1)	MS116-2.5	1SAM250000R1007	0.265
1.50	2.50 ... 4.00	50	50.0	MS116-4.0	1SAM250000R1008	0.265
2.20	4.00 ... 6.30	50	78.8	MS116-6.3	1SAM250000R1009	0.265
4.00	6.30 ... 10.0	50	150	MS116-10	1SAM250000R1010	0.265
5.50	8.00 ... 12.0	25	180	MS116-12	1SAM250000R1012	0.265
7.50	10.0 ... 16.0	16	240	MS116-16	1SAM250000R1011	0.265
7.50	16.0 ... 20.0	10	300	MS116-20	1SAM250000R1013	0.310
11.0	20.0 ... 25.0	10	375	MS116-25	1SAM250000R1014	0.310
15.0	25.0 ... 32.0	10	480	MS116-32	1SAM250000R1015	0.310
0.03(2)	0.10 ... 0.16	50	2.00 (1)	MS116-0.16-HKF1-11	1SAM250005R1001	0.240
0.06	0.16 ... 0.25	50	3.10 (1)	MS116-0.25-HKF1-11	1SAM250005R1002	0.240
0.09	0.25 ... 0.40	50	5.00 (1)	MS116-0.4-HKF1-11	1SAM250005R1003	0.240
0.18	0.40 ... 0.63	50	7.90 (1)	MS116-0.63-HKF1-11	1SAM250005R1004	0.240
0.25	0.63 ... 1.00	50	12.5 (1)	MS116-1.0-HKF1-11	1SAM250005R1005	0.240
0.55	1.00 ... 1.60	50	20.0 (1)	MS116-1.6-HKF1-11	1SAM250005R1006	0.280
0.75	1.60 ... 2.50	50	31.3 (1)	MS116-2.5-HKF1-11	1SAM250005R1007	0.280
1.50	2.50 ... 4.00	50	50.0	MS116-4.0-HKF1-11	1SAM250005R1008	0.280
2.20	4.00 ... 6.30	50	78.8	MS116-6.3-HKF1-11	1SAM250005R1009	0.280
4.00	6.30 ... 10.0	50	150	MS116-10.0-HKF1-11	1SAM250005R1010	0.280
5.50	8.00 ... 12.0	25	180	MS116-12.0-HKF1-11	1SAM250005R1012	0.280
7.50	10.0 ... 16.0	16	240	MS116-16.0-HKF1-11	1SAM250005R1011	0.280
7.50	16.0 ... 20.0	10	300	MS116-20-HKF1-11	1SAM250005R1013	0.326
11.0	20.0 ... 25.0	10	375	MS116-25-HKF1-11	1SAM250005R1014	0.326
15.0	25.0 ... 32.0	10	480	MS116-32-HKF1-11	1SAM250005R1015	0.326

Note: Manual motor starters should always be selected so that the actual motor current is within the setting range.

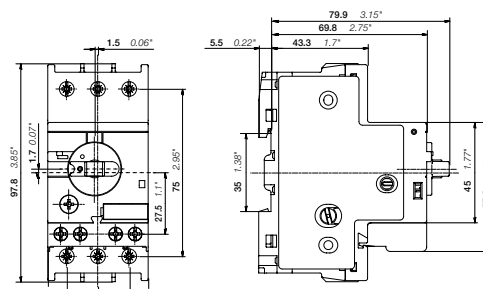
(1) For other voltage version see voltage code table.

(2) 690 V



MS116 ≤ 16 A & MS116-HK1-11 ≤ 16 A

Main dimensions mm, inches



MS116 ≤ 20 A & MS116-HK1-11 ≤ 20 A

2CDC131062C0201-revA

MS116

Technical data

Main circuit - Utilization characteristics according to IEC/EN

Type	MS116	
Standards	IEC/EN 60947-2, IEC/EN 60947-4-1, IEC/EN 60947-1	
Rated operation voltage Ue	690 V AC	
Rated frequency	50/60 Hz	
Operational frequency	50/60 Hz	
Trip class	10A	
Number of poles	3	
Duty time	100%	
Mechanical durability	100000 cycles	
Electrical durability	up to 16 A	100000 cycles
	20 ... 65 A	50000 cycles
Rated impulse withstand voltage Uimp	6 kV	
Rated insulation voltage Ui	690 V	
Rated operational current Ie	See ordering details	
Rated operational current DC-5 Ie 3 conducting paths in series up to 250 V	-	
Rated instantaneous short-circuit current setting Ii	See ordering details	
Rated service short-circuit breaking capacity Ics	See table "Short-circuit breaking capacity and back-up fuses"	
Rated ultimate short-circuit breaking capacity Icu	See table "Short-circuit breaking capacity and back-up fuses"	
Rated service short-circuit breaking capacity DC Ics 3 conducting paths in series up to 250 V	-	

Short-circuit breaking capacity and back-up fuses

ICS Rated service short-circuit breaking capacity

ICU Rated ultimate short-circuit breaking capacity

ICC Prospective short-circuit current at installation location

Note: Maximum rated current of the back-up fuses if $I_{cc} > I_{cs}$

Type	230 V AC			400 V AC			440 V AC			500 V AC			690 V AC		
	I _{cs} kA	I _{cu} kA	gG, aM A	I _{cs} kA	I _{cu} kA	gG, aM A	I _{cs} kA	I _{cu} kA	gG, aM A	I _{cs} kA	I _{cu} kA	gG, aM A	I _{cs} kA	I _{cu} kA	gG, aM A
MS116-0.16	50	50	-(1)	50	50	-(1)	30	30	-(1)	30	30	-(1)	30	30	-(1)
MS116-0.25	50	50	-(1)	50	50	-(1)	30	30	-(1)	30	30	-(1)	30	30	-(1)
MS116-0.4	50	50	-(1)	50	50	-(1)	30	30	-(1)	30	30	-(1)	30	30	-(1)
MS116-0.63	50	50	-(1)	50	50	-(1)	30	30	-(1)	30	30	-(1)	30	30	-(1)
MS116-1.0	50	50	-(1)	50	50	-(1)	30	30	-(1)	30	30	-(1)	30	30	-(1)
MS116-1.6	50	50	-(1)	50	50	-(1)	30	30	-(1)	30	30	-(1)	30	30	-(1)
MS116-2.5	50	50	-(1)	50	50	-(1)	10	10	25(2)	10	10	25(2)	5	5	25(2)
MS116-4.0	50	50	-(1)	50	50	-(1)	6	6	25(2)	6	6	25(2)	2	2	25(2)
MS116-6.3	50	50	-(1)	50	50	-(1)	6	6	63(2)	6	6	63(2)	2	2	40(2)
MS116-10	50	50	-(1)	50	50	-(1)	6	6	63(2)	6	6	63(2)	2	2	50(2)
MS116-12	25	25	80(2)	25	25	80(2)	6	6	63(2)	6	6	63(2)	2	2	50(2)
MS116-16	16	16	80(2)	16	16	80(2)	4	4	63(2)	4	4	63(2)	2	2	63(2)
MS116-20	10	15	125(2)	10	15	125(2)	3	6	125(2)	3	4	125(2)	2	2	80(2)
MS116-25	10	15	125(2)	10	15	125(2)	3	6	125(2)	3	4	125(2)	2	2	100(2)
MS116-32	10	10	125(2)	10	10	125(2)	3	6	125(2)	3	4	125(2)	2	2	100(2)

(1) No back-up fuse required, because short-circuit proof up to 50 kA

(2) Rated back-up fuse for short-circuit up to 50 kA

MS116

Technical data

Main circuit - Utilization characteristics according to UL/CSA

Type	MS116	
Standards	UL 60947-1, UL 60947-4-1 (UL 508), CSA C22.2 No.60947-4-1 (CSA C22.2 No.14)	
Rated operation voltage U _e acc. to UL/CSA	600 V AC	
Trip class	10 A	
Motor ratings (1)	Horsepower	See table "Motor ratings, three phase"
	Full Load Amps (FLA)	See table "Motor ratings, three phase"
	Locked Rotor Amps (LRA)	See table "Motor ratings, three phase"

(1) See product data sheets for UL/CSA single phase motor and general use (AC-1) ratings.

UL/CSA ratings overview

Type	MS116
Manual Motor Controller	X
Manual Motor Controller, Suitable as Motor Disconnect	X
Manual Motor Controller, Suitable for use in Group Installations	X
Manual Motor Controller, Suitable for Tap Conductor Protection in Group Installations	-
Manual self-protected Combination Motor Controller (Type E)	-
Combination Motor Controller (Type F)	-

UL/CSA Motor ratings, three phase – MS116

Type	200 V AC			208 V AC			220 ... 240 V AC			440 ... 480 V AC			550 ... 600 V AC		
	hp	FLA	LRA	hp	FLA	LRA	hp	FLA	LRA	hp	FLA	LRA	hp	FLA	LRA
MS116-0.16	-	0.16	0.96	-	0.16	0.96	-	0.16	0.96	-	0.16	0.96	-	0.16	0.96
MS116-0.25	-	0.25	1.5	-	0.25	1.5	-	0.25	1.5	-	0.25	1.5	-	0.25	1.5
MS116-0.40	-	0.4	2.4	-	0.4	2.4	-	0.4	2.4	-	0.4	2.4	-	0.4	2.4
MS116-0.63	-	0.63	3.78	-	0.63	3.78	-	0.63	3.78	-	0.63	3.78	-	0.63	3.78
MS116-1.0	-	1	6	-	1	6	-	1	6	-	1	6	1/2	1	6
MS116-1.6	-	1.6	9.6	-	1.6	9.6	-	1.6	9.6	3/4	1.6	9.6	3/4	1.6	9.6
MS116-2.5	1/2	2.5	15	1/2	2.5	15	1/2	2.5	15	1	2.5	15	1 1/2	2.5	15
MS116-4.0	3/4	4	24	3/4	4	24	1	4	24	2	4	24	3	3.9	25.6
MS116-6.3	1	6.3	37.8	1	6.3	37.8	1 1/2	6.3	37.8	3	4.8	32	5	6.1	36.8
MS116-10	2	7.8	57.5	2	7.5	55	3	9.6	64	5	7.6	46	7 1/2	9	50.8
MS116-12	3	11	73.6	3	10.6	71	3	9.6	64	7 1/2	11	63.5	10	11	64.8
MS116-16	3	11	73.6	3	10.6	71	5	15.2	92	10	14	81	10	11	64.8
MS116-20	5	17.5	105.8	5	16.7	102	5	15.2	92	10	14	81	15	17	93
MS116-25	5	17.5	105.8	7 1/2	24.2	140	7 1/2	22	127	15	21	116	20	22	116
MS116-32	7 1/2	25.3	146	10	30.8	179	10	28	162	20	27	145	25	27	146

MS116

Technical data

UL/CSA Maximum short-circuit current ratings – MS116

Type	Manual Motor Controllers					
	Branch circuit protection, max. size per NEC/CEC(1)		for motor disconnect(2)		for group installations	
	Fuses	Circuit breaker	480 V	600 V	480 V	600 V
	A	A	kA	kA	kA	kA
MS116-0.16	100	-	30	5	30	5
MS116-0.25	100	-	30	5	30	5
MS116-0.40	100	-	30	5	30	5
MS116-0.63	100	-	30	5	30	5
MS116-1.0	100	-	30	5	30	5
MS116-1.6	100	-	30	5	30	5
MS116-2.5	100	-	30	5	30	5
MS116-4.0	100	-	18	5	18	5
MS116-6.3	100	-	18	5	18	5
MS116-10	100	-	18	5	18	5
MS116-12	100	-	18	5	18	5
MS116-16	100	-	18	5	18	5
MS116-20	100	-	18	5	18	5
MS116-25	100	-	18	5	18	5
MS116-32	100	-	18	5	18	5

(1) NEC: NFPA®70 National Electrical Code®; CEC: CSA C22.1 Canadian Electrical Code.

(2) Suitable as motor disconnect only when provide with padlock adaptor SA1 or SA3.

MS116

Technical data

General technical data

Type		MS116
Pollution degree		3
Phase loss sensitivity		Yes
Disconnect function acc. to IEC/EN 60947-2		Yes
Ambient air temperature		
Operation	Open - compensated	-25 ... +55 °C
	Open	-25 ... +70 °C
	Enclosed (IB132 ⁽¹⁾)	0 ... +40 °C
Storage		-50 ... +80 °C
Ambient air temperature compensation		Acc. to IEC/EN60947-4-1
Maximum operating altitude permissible		2000 m
Resistance to shock acc. to IEC 60068-2-27		25g / 11 ms
Resistance to vibrations acc. to IEC 60068-2-6		5g / 3 ... 150 Hz
Mounting position		Position 1-6 (optional for single mounting)
Mounting		DIN-rail (EN 60715)
Group mounting		On request
Recommended screw for mounting plate		-
Screw torque for mounting plate		-
Minimum distance to other units same type	Horizontal	0 mm
	Vertical	150 mm
Minimum distance to electrical conductive board	Horizontal, up to 400 V	0 mm
	Horizontal, up to 690 V	> 1.5 mm
	Vertical	75 mm
Degree of protection	Housing	IP20
	Main circuit terminals	IP10

(1) not suitable for MS165 and MO165

Connecting characteristics - Main circuit

Type		MS116 ≤ 16 A	MS116 ≥ 20 A
Connecting capacity			
	Rigid	1 or 2 x 1 ... 4 mm ²	2.5 ... 6 mm ²
	Flexible with ferrule	1 or 2 x 0.75 ... 2.5 mm ²	1 ... 6 mm ²
	Flexible with insulated ferrule	1 or 2 x 0.75 ... 2.5 mm ²	1 ... 6 mm ²
	Flexible	1 or 2 x 0.75 ... 2.5 mm ²	1 ... 6 mm ²
	Stranded acc. to UL/CSA	1 or 2 x AWG 16-12	AWG 16-8
Stripping length		9 mm	10 mm
Tightening torque		0.8 ... 1.2 Nm / 10 ... 12 lb.in	2.0 Nm / 18 lb.in
Recommended screw driver		Pozidriv 2	Pozidriv 2

Main accessories

MS116



HKF1-11



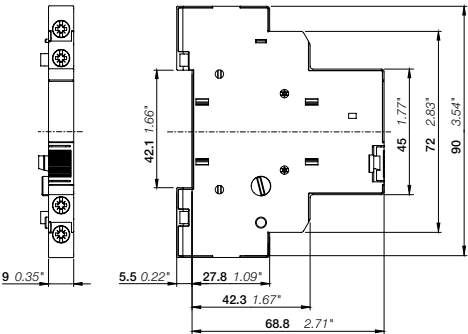
HK1-11



SK1-11

MS116 can be equipped with auxiliary contacts for lateral/front mounting, signaling contacts for lateral mounting, undervoltage releases and shunt trips. Two different signaling contacts are available. The accessories can be fitted wiring free and without tools. A variety of combinations is possible as required for the application. The auxiliary contacts change position with the main contacts. The signaling contact SK signals tripping regardless if it was caused by short-circuit or overload. The signaling contact CK signals tripping in case it was caused by short-circuit. Undervoltage releases are used for remote tripping of the manual motor starters especially for emergency stop circuits. Shunt trips release the MMS used for remote tripping. These main accessories are suitable throughout the MS116 range.

Suitable for	Auxiliary contacts N.O.	Auxiliary contacts N.C.	Description	Type	Order code		Pkg qty	Weight (1 pce)
Auxiliary contacts – mountable on the front								
MS116	1	1		HKF1-11	1SAM201901R1001		10	0.015
	1	0		HKF1-10	1SAM201901R1003		10	0.013
	0	1		HKF1-01	1SAM201901R1004		10	0.013
	2	0		HKF1-20	1SAM201901R1002		10	0.015
Auxiliary contacts – mountable on the right								
MS116	1	1	max. 2 pieces	HK1-11	1SAM201902R1001		2	0.035
	2	0	max. 2 pieces	HK1-20	1SAM201902R1002		2	0.035
	0	2	max. 2 pieces	HK1-02	1SAM201902R1003		2	0.035
	2	0	with lead contacts	HK1-20L	1SAM201902R1004		2	0.035
Signaling contacts – mountable on the right								
MS116	1	1	for tripped alarm, max. 2 pieces	SK1-11	1SAM201903R1001		2	0.035
	2	0	for tripped alarm, max. 2 pieces	SK1-20	1SAM201903R1002		2	0.035
	0	2	for tripped alarm, max. 2 pieces	SK1-02	1SAM201903R1003		2	0.035



HK1

Main dimensions mm, inches

Main accessories

MS116

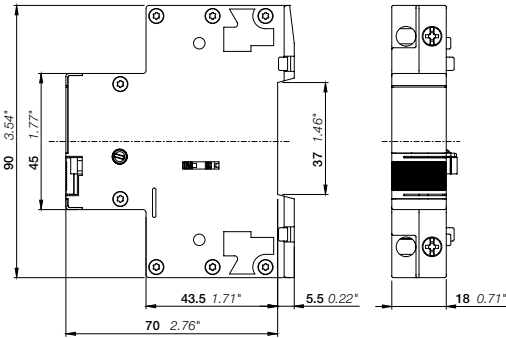


AA1-24



UA1-24

Suitable for	Rated control supply voltage		Type	Order code	Pkg qty	Weight (1 pce)
	50 Hz V AC	60 Hz V AC				kg
Shunt trips - mountable on the left						
MS116	20 ... 24	20 ... 24	AA1-24	1SAM201910R1001	1	0.100
	110	110	AA1-110	1SAM201910R1002	1	0.100
	200 ... 240	200 ... 240	AA1-230	1SAM201910R1003	1	0.100
	350 ... 415	350 ... 415	AA1-400	1SAM201910R1004	1	0.100
Undervoltage releases - mountable on the left						
MS116	20	24	UA1-20	1SAM201904R1010	1	0.100
	24	-	UA1-24	1SAM201904R1001	1	0.100
	48	-	UA1-48	1SAM201904R1002	1	0.100
	60	-	UA1-60	1SAM201904R1003	1	0.100
	110	120	UA1-110	1SAM201904R1004	1	0.100
	-	208	UA1-208	1SAM201904R1008	1	0.100
	230	240	UA1-230	1SAM201904R1005	1	0.100
	400	-	UA1-400	1SAM201904R1006	1	0.100
	415	480	UA1-415	1SAM201904R1007	1	0.100
	-	575	UA1-575	1SAM201904R1009	1	0.100
	-	-	-	-	-	-



AA1, UA1

Main dimensions mm, inches

Main accessories

MS116

General technical data

Type		HK1, SK1, CK1	HKF1
Standards		IEC/EN 60947-2, IEC/EN 60947-4-1, IEC/EN 60947-1	
Rated operational voltage Ue		690 V AC / 600 DC	
Conventional free-air thermal current Ith		6 A	250 V AC / 250 V DC 5 A
Rated frequency		50/60 Hz	
Rated impulse withstand voltage Uimp		6 kV	
Rated insulation voltage Ui		690 V AC	250 V AC
Pollution degree		3	
Ambient air temperature	Operation	-25 ... +70 °C	
	Storage	-50 ... +80 °C	
Resistance to shock acc. to IEC 60068-2-27		25g / 11 ms	
Resistance to vibrations acc. to IEC 60068-2-6		5g / 3 ... 150 Hz	
Ie / Rated operational current AC-15 acc. to IEC/EN 60947-5-1 for utilization category			
	24 V, 120 V	6 A	3 A
	240 V	4 A	1.5 A
	400 V	3 A	-
	440 V, 690 V	1 A	-
Ie / Rated operational current DC-13 acc. to IEC/EN 60947-5-1 for utilization category			
	24 V	2 A	1 A
	125 V	0.55 A	0.27 A
	250 V	0.27 A	0.11 A
	440 V, 600 V	0.15 A	-
Minimum switching capacity		17 V / 5 mA	
Short-circuit protective device	N.C., 95-96	10 A Type gG	
	N.O., 97-98	10 A Type gG	
Duty time		100 %	
Mounting		Right side of MMS	Front of MMS
Mounting positions		1-6	
Mechanical durability		50000 cycles	-
Electrical durability		50000 cycles	-

Contact utilization characteristics according to UL/CSA

Type	HK1, SK1, CK1		HKF1
Standards	UL 60947-1, UL 60947-4-1 (UL 508), CSA C22.2 No.60947-4-1 (CSA C22.2 No.14)		
Rated operational voltage Ue acc. to UL/CSA	600 V AC / 600 V DC		250 V AC / 250 V DC
Pilot duty	A600, Q600		B300, Q300
AC thermal rated current	10 A		5 A
AC maximum volt-ampere making	7200 VA		3600 VA
AC maximum volt-ampere breaking	720 VA		360 VA
DC thermal rated current	2.5 A		2.5 A
DC maximum volt-ampere making-breaking	69 VA		69 VA

Connecting characteristics - Auxiliary circuit

Type	HK1, SK1, CK1		HKF1
Connecting capacity			
 Rigid	1 or 2 x	1 ... 1.5 mm ²	1 ... 2.5 mm ²
 Flexible with ferrule	1 or 2 x	0.75 ... 1.5 mm ²	
 Flexible with insulated ferrule	1 or 2 x	0.75 ... 1.5 mm ²	
 Flexible	1 or 2 x	0.75 ... 1.5 mm ²	
Stranded acc. to UL/CSA	1 or 2 x	AWG 16-14	
Stripping length	8 mm		
Tightening torque	0.8 ... 1.2 Nm / 7 lb.in		
Recommended screw driver	Pozidriv 2		

Main accessories

MS116

General technical data

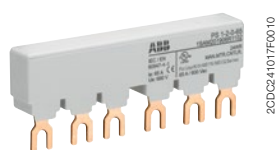
Type	UA1		AA1
Standards	IEC/EN 60947-1, IEC/EN 60947-4-1, UL 60947-1, UL 60947-4-1 (UL 508), CSA C22.2 No.60947-4-1 (CSA C22.2 No.14)		
Rated control supply voltage	see ordering details		AA1-24: 20-24 V 50/60 Hz; 20-70 V 50/60 Hz KB = 5 s, 20-70 V DC KB = 5 s AA1-100: 110 V 50/60 Hz; 110-200 V 50/60 Hz KB = 5 s, 110-200 V DC KB = 5 s AA1-230: 200-240 V 50/60 Hz, 200-350 V 50/60 Hz KB = 5 s, 200-350 V DC KB = 5 s AA1-400: 350-415 V 50/60 Hz, 350-500 V 50/60 Hz KB = 5 s, 350-500 V DC KB = 5 s
Rated frequency	see ordering details		50/60 Hz, DC
Operating voltage	Tripping	0.35 ... 0.7 x Us	0.7 ... 1.1 x Us
	Coil operating voltage	0.85 ... 1.1 x Us	-
Power consumption	Pull-in	AC	on request
		DC	on request
	Holding	AC	-
		DC	-
Rated impulse withstand voltage Uimp	6 kV		6 kV
Rated insulation voltage Ui	690 V		690 V
Pollution degree	3		3
Ambient air temperature	Operation	-25 ... +60 °C	-25 ... +60 °C
	Storage	-50 ... +80 °C	-50 ... +80 °C
Resistance to shock acc. to IEC 60068-2-27	25g / 11 ms		25g / 11 ms
Resistance to vibrations acc. to IEC 60068-2-6	5g / 3 ... 150 Hz		5g / 3 ... 150 Hz
Mounting	left side of MMS		left side of MMS
Mounting positions	-		-

Connecting characteristics - Auxiliary circuit

Type	UA1		AA1
Connecting capacity			
 Rigid	1 or 2 x	1 ... 4 mm ²	
 Flexible with ferrule	1 or 2 x	0.75 ... 2.5 mm ²	
 Flexible with insulated ferrule	1x	0.75 ... 2.5 mm ²	
	2 x	0.75 ... 1.5 mm ²	
 Flexible	1 or 2 x	0.75 ... 2.5 mm ²	
	Stranded acc. to UL/CSA	1 or 2 x	AWG 16-12
Stripping length	10 mm		
Tightening torque	0.8 ... 1.2 Nm / 7 lb.in		
Recommended screw driver	Pozidriv 2		

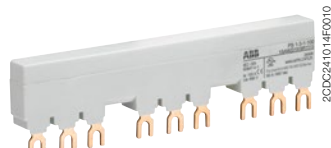
Main accessories

MS116



PS1-2-0-65

2DCD24107FD010



PS1-3-1-100

2DCD241014FD010



S1-M1-25

1SBC101226FD014



S1-M2-25

1SBC101226FD014



SA2



2DCD241023FD013



SA1

SK0108B91



PB1-1-32

2DCD241004FD014



S1-PB1-25

Three-phase busbars ensure a quick and safe connection and are therefore a cost effective solution. A variety of different three-phase busbars up to 100 A are in the assortment. Between 2 and 5 manual motor starters with none, one or two lateral auxiliary contacts can be connected. Different three-phase feeder terminals are available according to the application. Phase connecting links and phase power infeed blocks are also available for single-phase applications.

Suitable for	Rated operational current A	Number of MMS	Number of lateral aux.	Type	Order code		Pkg qty	Weight (1 pce) kg
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Three-phase busbars

MS116	65	2	0	PS1-2-0-65	1SAM201906R1102		10	0.034
	65	3	0	PS1-3-0-65	1SAM201906R1103		10	0.055
	65	4	0	PS1-4-0-65	1SAM201906R1104		10	0.077
	65	5	0	PS1-5-0-65	1SAM201906R1105		10	0.098
	65	2	1	PS1-2-1-65	1SAM201906R1112		10	0.036
	65	3	1	PS1-3-1-65	1SAM201906R1113		10	0.060
	65	4	1	PS1-4-1-65	1SAM201906R1114		10	0.087
	65	5	1	PS1-5-1-65	1SAM201906R1115		10	0.108
	65	2	2	PS1-2-2-65	1SAM201906R1122		10	0.040
	65	3	2	PS1-3-2-65	1SAM201906R1123		10	0.067
	65	4	2	PS1-4-2-65	1SAM201906R1124		10	0.095
	65	5	2	PS1-5-2-65	1SAM201906R1125		10	0.122
MS116	100	3	0	PS1-3-0-100	1SAM201916R1103		10	0.084
	100	4	0	PS1-4-0-100	1SAM201916R1104		10	0.117
	100	5	0	PS1-5-0-100	1SAM201916R1105		10	0.154
	100	3	1	PS1-3-1-100	1SAM201916R1113		10	0.094
	100	4	1	PS1-4-1-100	1SAM201916R1114		10	0.134
	100	5	1	PS1-5-1-100	1SAM201916R1115		10	0.172
	100	3	2	PS1-3-2-100	1SAM201916R1123		10	0.105

Suitable for	Rated operational current A	Number of MMS mm ²	Number of lateral aux.	Type	Order code		Pkg qty	Weight (1 pce) kg
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Three-phase terminals

MS116	65	25	Flat	S1-M1-25	1SAM201907R1101		10	0.038
	65	25	High	S1-M2-25	1SAM201907R1102		10	0.051
	65	25	UL/CSA Type E/F and IEC	S1-M3-25	1SAM201907R1103		10	0.042
	100	35	UL/CSA Type E/F and IEC	S1-M3-35	1SAM201913R1103		10	0.060

Suitable for	Description	Type	Order code		Pkg qty	Weight (1 pce) kg
MS116	Protection cover for busbars	BS1-3	1SAM201908R1001		50	0.003
MS116	Screw fixing kit	FS116	1SAM201909R1001		1	0.020
	Padlock + two keys	SA2	GJF1101903R0002		10	0.020
MS116	Lock handle	SA1	GJF1101903R0001		10	0.003
	Lock handle box SA1/SA2	SA3	GJF1101903R0003		10	0.050

Accessories for single-phase connection (IEC only)

MS116	Phase connecting link	PB1-1-32	1SAM201914R1001		1	0.009
	Phase power infeed block	S1-PB1-25	1SAM201914R1002		1	0.013

Main accessories

MS116

General technical data

Type	PS1-xxx-65		PS1-xxx-100		S1-Mx-25		S1-Mx-35	
Standards	IEC/EN 60947-4-1, IEC/EN 60947-1, UL 60947-1, UL 60947-4-1 (UL 508), CSA C22.2 No.60947-4-1 (CSA C22.2 No.14)							
Rated operational voltage Ue	690 V							
Rated operational voltage Ue acc. to UL/CSA	600 V AC							
Rated operational current Ie	65 A		100 A		65 A		100 A	
Rated operational current Ie acc. to UL/CSA	65 A		92 A		65 A		92 A	
Rated frequency	50/60 Hz							
Rated impulse withstand voltage Uimp	6 kV							
Rated insulation voltage Ui	690 V AC							
Pollution degree	3							
Cross-section	10 mm²		16 mm²		25 mm²		35 mm²	
Ambient air temperature	Operation	-25 ... +70 °C						
	Storage	-50 ... +80 °C						
Resistance to shock acc. to IEC 60068-2-27	25g / 11 ms							
Resistance to vibrations acc. to IEC 60068-2-6	5g / 3 ... 150 Hz							

Electrical connection

Type	S1-Mx-25	S1-Mx-35
Connecting capacity		
 Rigid	1 x 6 ... 25 mm ²	10 ... 35 mm ²
 Flexible with ferrule	1 x 6 ... 16 mm ²	10 ... 35 mm ²
 Flexible with insulated ferrule	1 x 6 ... 16 mm ²	10 ... 35 mm ²
 Flexible	1 x 6 ... 16 mm ²	10 ... 35 mm ²
Stranded acc. to UL/CSA	1 x AWG 10-4	AWG 8-2
Stripping length	10 mm	12 mm
Tightening torque	2.5 Nm / 22 lb.in	4.5 Nm / 40 lb.in
Recommended screw driver	Pozidriv 2	Hexagon SW4

Technical data for PS2-xxx on request.

Main accessories

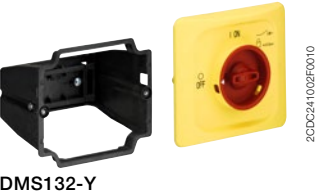
MS116



IB132-Y



IB132-G



DMS132-Y



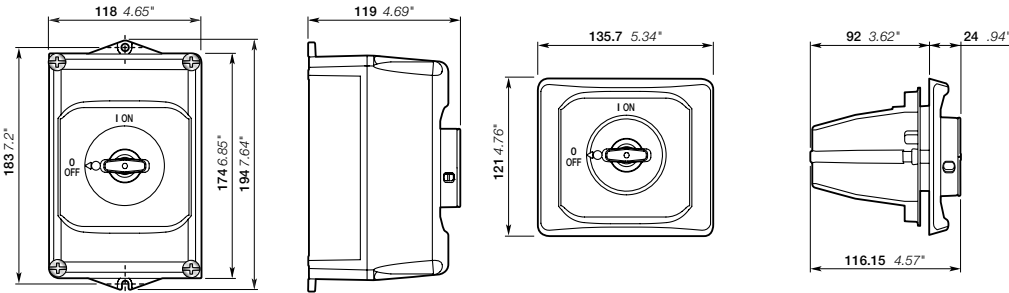
DMS132-G

IB132 are IP65 (UL/CSA Type 12) enclosures for single MMS installation. Additional mounting of auxiliary and signaling contacts, shunt trips and undervoltage release is possible. The handle is lockable in OFF position. For detailed specification see installation instruction.

DMS132 are IP65 (UL/CSA Type 12) door mounting kits for MMS installation in any enclosure. Additional mounting of auxiliary, signaling, shunt trips and undervoltage release is possible. The handle is lockable in OFF position. For detailed specification see installation instruction.

Suitable for	Description	Color	Type	Order code	Pkg qty	Weight (1 pce) kg
IP65 enclosures (UL/CSA Type 12)						
MS116	Padlockable max. 3 padlocks with bail diameter 4 ... 6.5 mm	Yellow/red	IB132-Y	1SAM201911R1011	1	0.370
		Grey/black	IB132-G	1SAM201911R1010	1	0.370
IP65 door mounting kits (UL/CSA Type 12)						
MS116	Padlockable max. 3 padlocks with bail diameter 4 ... 6.5 mm	Yellow/red	DMS132-Y	1SAM201912R1011	1	0.170
		Grey/black	DMS132-G	1SAM201912R1010	1	0.170

Indication I-O-T and ON-OFF-T
Please check for further equipment chapter General accessories.



IB132

DMS132

Main dimensions mm, inches

General accessories

MS116



MSHD-LB

2CDC241003F0011



MSHD-LY

2CDC241002S0011



MSMN

2CDC241004F0011



MSH-AR

2CDC241001F0012



MSAH1

2CDC241017V0013

With this solution of door coupling rotary mechanism it is possible to operate a manual motor starter in the back of a switch cabinet from outside. The door coupling mechanism prevents opening of the door of a switch cabinet with the manual motor starter in ON position.

The complete mechanism includes handle, shaft, driver, shaft alignment ring and shaft supporter. Most accessories fit for 6 mm shafts with a maximum length of 180 mm. The degree of protection for handles MSHD is IP64 (UL/CSA Type 1, 3R, 12).

Suitable for	Description	Shaft length mm	Color	Type	Order code	Pkg qty	Weight (1 pce) kg
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Shafts

MS116	For MSHD handles. Shaft diameter 6 mm. Shaft extension for door coupling driver.	85		OX56X85	1SCA101647R1001	1	0.020
		105		OX56X105	1SCA108043R1001	1	0.020
		130		OX56X130	1SCA101655R1001	1	0.030
		180		OX56X180	1SCA101659R1001	1	0.040

IP64 handles (UL/CSA Type 1, 3R, 12)

MS116	Padlockable max. 3 padlocks with bail diameter 5 ... 8 mm, door interlock in ON position defeatable, for use with 6 mm OXS6... types up to 180 mm or driver shafts MSOX.		Black	MSHD-LB (1)	1SAM201920R1001	1	0.065
			Yellow	MSHD-LY (1)	1SAM201920R1002	1	0.065
			Black	MSHD-LTB (2)	1SAM201920R1011	1	0.065
			Yellow	MSHD-LTY (2)	1SAM201920R1012	1	0.065

Driver

MS116	Coupling driver for use with 6 mm OXS6... types up to 180 mm.			MSMN (3)	1SAM101923R0002	1	0.002
				MSMNO (4)	1SAM101923R0012	1	0.002

Shaft alignment ring

MS116	The MSH-AR supports the long shafts for alignment to the handle inlet. It makes closing panel doors more easy. Use for OXS6X > 105 mm.			MSH-AR	1SAM201920R1000	1	0.010
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Shaft supporter

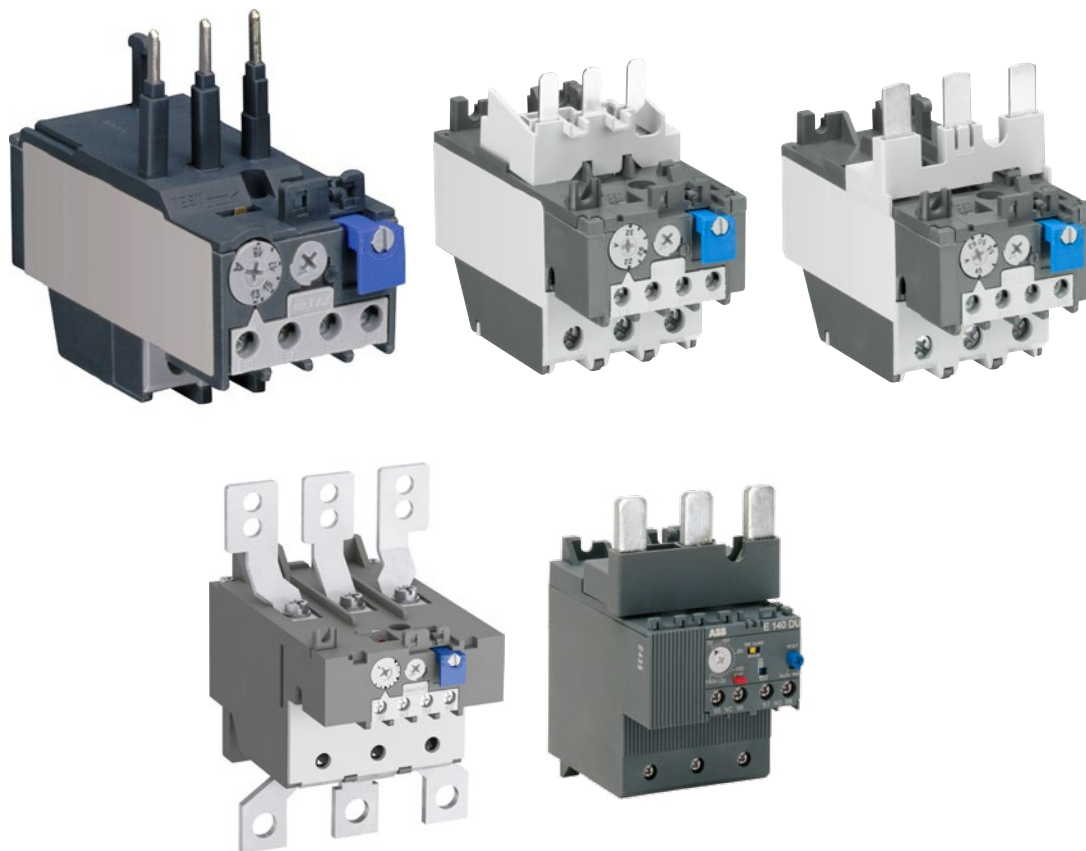
MS116	With the MSAH1 it is possible to support the shaft in the extension of handle (MSHD). It is mandatory for the usage of shafts >130 mm.			MSAH1	1SAM201909R1021	1	0.035
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(1) Indication I-O and ON-OFF (recommended for MS116)

(2) Indication I-O and ON-OFF + Trip indication

(3) Coded - Positioning of ON indication dependent from mounting orientation of the MMS

(4) Uncoded - Positioning of ON indication independent from mounting orientation of the MMS

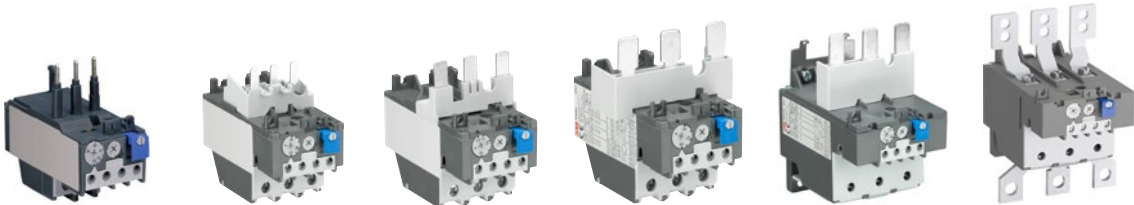


Overload relays

100	Overview
	Thermal overload relays
	TA25DU-M / TA42DU-M / TA75DU-M (0.10 ... 80 A)
101	Ordering details
102	Technical data
105	Accessories
	TA80DU-M / TA110DU-M / TA200DU (29...200 A)
106	Ordering details
107	Technical data
110	Dimensions
	Electronic overload relays
	EF205, EF370 (63 ... 380 A)
112	Ordering details
113	Technical data
116	Dimensions

Overload relays

Thermal overload relay



Type	TA25DU-M	TA42DU-M	TA75DU-M	TA80DU-M	TA110DU-M	TA200DU
Current range	0.10 ... 32 A	18 ... 42 A	18 ... 80 A	29 ... 80 A	66 ... 110 A	66 ... 200 A
Trip class	10A	10A	10A	10A	10A	10A
Single mounting kit	DB25	DB80	DB80	DB80	DB200	DB200
For contactors	AX09 ... AX40	AX32 ... AX40	AX50 ... AX80	AX95 ... AX150	AX95 ... AX150	AX185 ... AX205

Electronic overload relay



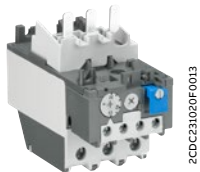
Type	EF205	EF370
Current range	63 ... 210 A	115 ... 380 A
Trip class	10E, 20E, 30E selectable	
Single mounting kit	-	-
For contactors	A185 ... AX205	AX260 ... AX370

Thermal overload relays TA25DU-M / TA42DU-M / TA75DU-M

0.10 to 80 A



TA25DU-M



TA42DU-M



TA75DU-M

The TA25DU-M / TA42DU-M and TA75DU-M thermal overload relays are economic electromechanical protection devices for the main circuit. They offer reliable protection for motors in the event of overload or phase failure. The devices have trip class 10A.

The thermal overload relays are three pole relays with bimetal tripping elements. The motor current flows through the bimetal tripping elements and heats them directly and indirectly. In case of an overload (over current), the bimetal elements bent as a result of the heating. This leads to a release of the relay and a change of the contacts switching position (95-96 / 97-98).

- Manual or automatic reset selectable
- Phase loss sensitive acc. to IEC/EN 60947-4-1
- Two electrically isolated auxiliary contacts – 1 N.O. + 1 N.C.
- TEST and STOP function – Trip indication on the front
- Temperature compensation
- Suitable for three- and single-phase applications

Setting range	Short-circuit protective device	Trip class	Type	Order code	Weight (1 pce) kg
A					
TA25DU-M					
0.10 ... 0.16	0.50 A, Fuse type F	10A	TA25DU-0.16M	1SAZ211201R2005	0.150
0.16 ... 0.25	0.63 A, Fuse type F	10A	TA25DU-0.25M	1SAZ211201R2009	0.150
0.25 ... 0.40	1.25 A, Fuse type F	10A	TA25DU-0.4M	1SAZ211201R2013	0.150
0.40 ... 0.63	2 A, Fuse type gG / -	10A	TA25DU-0.63M	1SAZ211201R2017	0.150
0.63 ... 1.00	4 A, Fuse type gG / 2 A aM	10A	TA25DU-1.0M	1SAZ211201R2021	0.150
1.00 ... 1.40	6 A, Fuse type gG / 2 A aM	10A	TA25DU-1.4M	1SAZ211201R2023	0.150
1.30 ... 1.80	6 A, Fuse type gG / 4 A aM	10A	TA25DU-1.8M	1SAZ211201R2025	0.150
1.70 ... 2.40	6 A, Fuse type gG / 4 A aM	10A	TA25DU-2.4M	1SAZ211201R2028	0.150
2.20 ... 3.10	10 A, Fuse type gG / 6 A aM	10A	TA25DU-3.1M	1SAZ211201R2031	0.150
2.80 ... 4.00	10 A, Fuse type gG / 6 A aM	10A	TA25DU-4.0M	1SAZ211201R2033	0.150
3.50 ... 5.00	16 A, Fuse type gG / 10 A aM	10A	TA25DU-5.0M	1SAZ211201R2035	0.150
4.50 ... 6.50	20 A, Fuse type gG / 16 A aM	10A	TA25DU-6.5M	1SAZ211201R2038	0.150
6.00 ... 8.50	20 A, Fuse type gG / 20 A aM	10A	TA25DU-8.5M	1SAZ211201R2040	0.150
7.50 ... 11.00	35 A, Fuse type gG / 25 A aM	10A	TA25DU-11M	1SAZ211201R2043	0.150
10.00 ... 14.00	35 A, Fuse type gG / 25 A aM	10A	TA25DU-14M	1SAZ211201R2045	0.150
13.00 ... 19.00	50 A, Fuse type gG / 35 A aM	10A	TA25DU-19M	1SAZ211201R2047	0.170
18.00 ... 25.00	63 A, Fuse type gG / 50 A aM	10A	TA25DU-25M	1SAZ211201R2051	0.170
24.00 ... 32.00	80 A, Fuse type gG / 63 A aM	10A	TA25DU-32M	1SAZ211201R2053	0.200
TA42DU-M					
18 ... 25	63 A, Fuse type gG / 50 A aM	10A	TA42DU-25M	1SAZ311201R2001	0.335
22 ... 32	80 A, Fuse type gG / 63 A aM	10A	TA42DU-32M	1SAZ311201R2002	0.335
29 ... 42	100 A, Fuse type gG / 80 A aM	10A	TA42DU-42M	1SAZ311201R2003	0.335
TA75DU-M					
18 ... 25	63 A, Fuse type gG / 50 A aM	10A	TA75DU-25M	1SAZ321201R2001	0.335
22 ... 32	80 A, Fuse type gG / 63 A aM	10A	TA75DU-32M	1SAZ321201R2002	0.335
29 ... 42	100 A, Fuse type gG / 80 A aM	10A	TA75DU-42M	1SAZ321201R2003	0.335
36 ... 52	125 A, Fuse type gG / 100 A aM	10A	TA75DU-52M	1SAZ321201R2004	0.335
45 ... 63	160 A, Fuse type gG / 125 A aM	10A	TA75DU-63M	1SAZ321201R2005	0.335
60 ... 80	200 A, Fuse type gG / 160 A aM	10A	TA75DU-80M	1SAZ321201R2006	0.370

Thermal overload relays TA25DU-M / TA42DU-M / TA75DU-M

Technical data

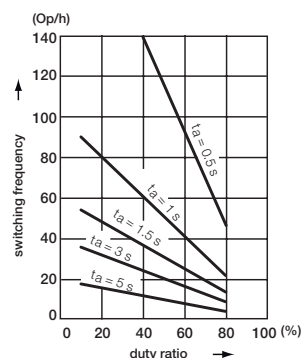
Main circuit – Utilization characteristics according to IEC/EN

Type	TA25DU-M	TA42DU-M	TA75DU-M
Standards	IEC/EN 60947-4-1, IEC/EN 60947-5-1, IEC/EN 60947-1		
Rated operational voltage U _e	690 V AC		
Rated frequency	DC, 50/60 Hz		
Frequency range	0 ... 400 Hz		
Trip class	10A		
Number of poles	3		
Duty time	100 %		
Operating frequency without early tripping	Up to 15 operations/h, see "Technical diagram – Intermittent periodic duty"		
Rated impulse withstand voltage U _{imp}	6 kV		
Rated insulation voltage U _i	690 V AC		

Auxiliary circuit according to IEC/EN

Type	TA25DU-M		TA42DU-M	TA75DU-M
Rated operational voltage Ue	500 V AC, 440 V DC			
Conventional free air thermal current Ith	N.C., 95-96	10 A		
	N.O., 97-98	6 A		
Rated frequency	DC, 50/60 Hz			
Number of poles	1 N.O. + 1 N.C.			
Ie / Rated operational current AC-15 acc. to IEC/EN 60947-5-1 for utilization category				
110-120 V	N.C., 95-96	3.00 A		
	N.O., 97-98	1.50 A		
220-230-240 V	N.C., 95-96	3.00 A		
	N.O., 97-98	1.20 A		
440 V	N.C., 95-96	0.75 A		
	N.O., 97-98	0.37 A		
480-500 V	N.C., 95-96	0.30 A		
	N.O., 97-98	0.25 A		
Ie / Rated operational current DC-13 acc. to IEC/EN 60947-5-1 for utilization category				
24 V	N.C., 95-96	1.25 A		
	N.O., 97-98	1.25 A		
60 V	N.C., 95-96	0.25 A		
	N.O., 97-98	0.25 A		
110-120-125 V	N.C., 95-96	0.25 A		
	N.O., 97-98	0.25 A		
250 V	N.C., 95-96	0.12 A		
	N.O., 97-98	0.04 A		
Minimum switching capacity	17 V / 3 mA			
Short-circuit protective device	N.C., 95-96	10 A, Fuse type gG		
	N.O., 97-98	6 A, Fuse type gG		
Rated impulse withstand voltage Uimp	6 kV			
Rated insulation voltage Ui	690 V			

Technical diagram – Intermittent periodic duty



t_a: Motor starting time - TA25DU-M,
TA42DU-M, TA75DU-M

Thermal overload relays TA25DU-M / TA42DU-M / TA75DU-M

Technical data

Main circuit – Utilization characteristics according to UL/CSA

Type	TA25DU-M / TA42DU-M / TA75DU-M
Standards	UL 508, CSA 22.2 No. 14
Maximum operational voltage	600 V AC/DC
Trip rating	125 % of FLA
Full load amps (FLA)	See table "Full load amps and short-circuit protective device"
Short-circuit rating RMS symmetrical	See table "Full load amps and short-circuit protective device"
Short-circuit protective device	See table "Full load amps and short-circuit protective device"

Auxiliary circuit according to UL/CSA

Type	TA25DU-M / TA42DU-M / TA75DU-M
Contact rating	N.C., 95-96 B600 N.O., 97-98 C600
Conventional free-air thermal current	5 A

Full load amps and short-circuit protective device

Type	Full load amps (FLA)	Short-circuit protective device			
		480 / 600 V AC		480 / 600 V AC	
		Short circuit rating RMS symmetrical	Fuse K5 / RK5	Short circuit rating RMS symmetrical	Fuse J
TA25DU-0.16M	0.16 A	5000 A	1 A	50000 A	30 A
TA25DU-0.25M	0.25 A	5000 A	1 A	50000 A	30 A
TA25DU-0.4M	0.40 A	5000 A	3 A	50000 A	30 A
TA25DU-0.63M	0.63 A	5000 A	3 A	50000 A	30 A
TA25DU-1.0M	1.0 A	5000 A	6 A	50000 A	30 A
TA25DU-1.4M	1.4 A	5000 A	6 A	50000 A	30 A
TA25DU-1.8M	1.8 A	5000 A	6 A	50000 A	30 A
TA25DU-2.4M	2.4 A	5000 A	10 A	50000 A	30 A
TA25DU-3.1M	3.1 A	5000 A	10 A	50000 A	30 A
TA25DU-4.0M	4.0 A	5000 A	15 A	50000 A	30 A
TA25DU-5.0M	5.0 A	5000 A	20 A	50000 A	30 A
TA25DU-6.5M	6.5 A	5000 A	25 A	50000 A	30 A
TA25DU-8.5M	8.5 A	5000 A	35 A	50000 A	30 A
TA25DU-11M	11 A	5000 A	45 A	50000 A	35 A
TA25DU-14M	14 A	5000 A	60 A	50000 A	60 A
TA25DU-19M	19 A	5000 A	60 A	50000 A	60 A
TA25DU-25M	25 A	5000 A	70 A	50000 A	100 A
TA25DU-32M	32 A	5000 A	100 A	50000 A	100 A
TA42DU-25M	25 A	5000 A	80 A	50000 A	100 A
TA42DU-32M	32 A	5000 A	100 A	50000 A	100 A
TA42DU-42M	42 A	5000 A	150 A	50000 A	200 A
TA75DU-25M	25 A	5000 A	80 A	50000 A	100 A
TA75DU-32M	32 A	5000 A	100 A	50000 A	100 A
TA75DU-42M	42 A	5000 A	150 A	50000 A	200 A
TA75DU-52M	52 A	5000 A	175 A	50000 A	200 A
TA75DU-63M	63 A	10000 A	200 A	50000 A	200 A
TA75DU-80M	80 A	10000 A	250 A	50000 A	200 A

Thermal overload relays TA25DU-M / TA42DU-M / TA75DU-M

Technical data

General technical data

Type	TA25DU-M	TA42DU-M	TA75DU-M
Pollution degree	3		
Phase loss sensitive	Yes		
Ambient air temperature			
Operation	Open - compensated	-25 ... +55 °C	
	Open	-25 ... +55 °C	
Storage		-40 ... +70 °C	
Ambient air temperature compensation	Acc. to IEC/EN60947-4-1		
Maximum operating altitude permissible	2000 m		
Resistance to shock acc. to IEC 60068-2-27	12g / 15 ms		
Mounting position	Position 1-6		
Mounting	Mount on the contactor and tighten the screws of the main circuit terminals or with single mounting kit on DIN rail (35 mm)		
Degree of protection	Housing	IP20	
	Main circuit terminals	IP10	

Electrical connection - main circuit

Type	TA25DU-M (0.16 ... 11 A)	TA25DU-M (14 ... 25 A)	TA25DU-M (32 A)
Connecting capacity			
 Rigid	1 x 0.75 ... 4 mm ² 2 x 0.75 ... 4 mm ²	1.5 ... 6 mm ² 1.5 ... 6 mm ²	1.5 ... 10 mm ² -
 Flexible with ferrule	1 x or 2 x 0.75 ... 4 mm ²	1.5 ... 4 mm ²	1.5 ... 6 mm ²
 Flexible	1 x or 2 x 0.75 ... 4 mm ²	1.5 ... 4 mm ²	1.5 ... 6 mm ²
Stranded acc. to UL/CSA	1 x or 2 x AWG 16-8	AWG 16-8	AWG 10-8
Flexible acc. to UL/CSA	1 x or 2 x AWG 16-8	AWG 16-8	AWG 10-8
Stripping length	12 mm	12 mm	15 mm
Tightening torque	1.5 ... 1.9 Nm / 12 in-lb	1.5 ... 1.9 Nm / 12 in-lb	2.5 ... 3.2 Nm / 20 in-lb
Recommended screw driver	M4 (Pozidriv 2)	M4 (Pozidriv 2)	M5 (Pozidriv 2)

Type	TA42DU-M	TA75DU-M
Connecting capacity		
 Rigid	1 x 2.5 ... 25 mm ² 2 x 2.5 ... 16 mm ²	
 Flexible with ferrule	1 x 2.5 ... 25 mm ² 2 x 2.5 ... 10 mm ²	
Stranded acc. to UL/CSA	1 x or 2 x AWG 8-1	
Flexible acc. to UL/CSA	1 x or 2 x AWG 8-1	
Stripping length	14 mm	
Tightening torque	4.5 Nm / 40 in-lb	
Recommended screw driver	M6 (Pozidriv 2)	

Auxiliary circuit

Type	TA25DU-M	TA42DU-M	TA75DU-M
Connecting capacity			
 Rigid	1 x or 2 x 0.75 ... 4 mm ²		
 Flexible	1 x or 2 x 0.75 ... 2.5 mm ²		
Stranded acc. to UL/CSA	1 x or 2 x AWG 18-14		
Flexible acc. to UL/CSA	1 x or 2 x AWG 18-14		
Stripping length	9 mm		
Tightening torque	1 ... 1.3 Nm / 12 in-lb		
Recommended screw driver	M3.5 (Pozidriv 2)		

Thermal overload relays TA25DU-M / TA42DU-M / TA75DU-M

Accessories



DX25



DB25/25A



KPR-101L



DB80

The single mounting kits offer the possibility to mount the overload relays separately from the contactor.

For thermal overload relays	Description	Type	Order code	Weight (1 pce) kg
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Terminal block and mounting kits

TA25DU-0.16M; ... 25M / DB25/25 A	Terminal block 10 mm ²	DX25	1SAZ201307R0002	0.030
TA25DU-0.16M ... 25M	Single mounting kit	DB25/25A	1SAZ201108R0001	0.055
TA25DU-32M	Single mounting kit	DB25/32A	1SAZ201108R0002	0.080
TA42DU-M / TA75DU-M	Single mounting kit	DB80	1SAZ301110R0001	0.155

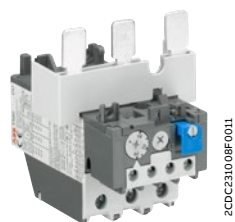
Reset push button

TA25DU-M / TA42DU -M / TA75DU -M	Reset push button*	KPR-101L	1SFA616162R1014	0.027
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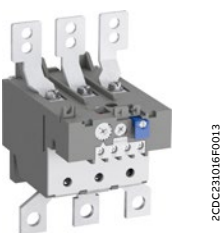
* The remote reset coil is to be connected to auxiliary contact 97-98 of TA25DU-M.
The coil is not suitable for continuous operation. Impulse duration: maximum 0.2 seconds.

Thermal overload relays TA80DU-M / TA110DU-M / TA200DU

29 to 200 A



TA80DU-M



TA200DU



DB80



KPR-101L

The TA80DU-M / TA110DU-M and TA200DU thermal overload relays are economic electromechanical protection devices for the main circuit. They offer reliable protection for motors in the event of overload or phase failure. The devices have trip class 10A.

The thermal overload relays are three pole relays with bimetal tripping elements. The motor current flows through the bimetal tripping elements and heats them directly and indirectly. In case of an overload (over current), the bimetal elements bent as a result of the heating. This leads to a release of the relay and a change of the contacts switching position (95-96 / 97-98).

- Manual or automatic reset selectable
- Phase loss sensitive acc. to IEC/EN 60947-4-1
- Two electrically isolated auxiliary contacts – 1 N.O. + 1 N.C.
- TEST and STOP function – Trip indication on the front
- Temperature compensation
- Suitable for three- and single-phase applications

Setting range	Short-circuit protective device	Trip class	Type	Order code	Weight (1 pce) kg
A					
TA80DU-M					
29 ... 42	100 A, Fuse type gG / 80 A aM	10A	TA80DU-42M	1SAZ331201R2003	0.360
36 ... 52	125 A, Fuse type gG / 100 A aM	10A	TA80DU-52M	1SAZ331201R2004	0.365
45 ... 63	160 A, Fuse type gG / 125 A aM	10A	TA80DU-63M	1SAZ331201R2005	0.365
60 ... 80	200 A, Fuse type gG / 160 A aM	10A	TA80DU-80M	1SAZ331201R2006	0.375
TA110DU-M					
66 ... 90	200 A, Fuse type gG / 160 A aM	10A	TA110DU-90M	1SAZ411201R2001	0.750
80 ... 110	224 A, Fuse type gG / 200 A aM	10A	TA110DU-110M	1SAZ411201R2002	0.755
TA200DU					
66 ... 90	200 A, Fuse type gG / 125 A aM	10A	TA200DU-90	1SAZ421201R1001	0.755
80 ... 110	224 A, Fuse type gG / 160 A aM	10A	TA200DU-110	1SAZ421201R1002	0.760
100 ... 135	224 A, Fuse type gG / 200 A aM	10A	TA200DU-135	1SAZ421201R1003	0.760
110 ... 150	250 A, Fuse type gG / 200 A aM	10A	TA200DU-150	1SAZ421201R1004	0.760
130 ... 175	315 A, Fuse type gG / 250 A aM	10A	TA200DU-175	1SAZ421201R1005	0.770
150 ... 200	315 A, Fuse type gG / 250 A aM	10A	TA200DU-200	1SAZ421201R1006	0.785

Ordering details accessories

For thermal overload relays	Description	Type	Order code	Weight (1 pce) kg
TA80DU-M	Single mounting kit	DB80	1SAZ301110R0001	0.155
TA200DU	Terminal shroud	LT200/A	1SAZ401901R1001	0.090
TA110DU-M / TA200DU	Single mounting kit	DB200	1SAZ401110R0001	0.225
TA80DU-M / TA110DU-M / TA200DU	Reset push button	KPR-101L	1SFA616162R1014	0.027

Thermal overload relays TA80DU-M / TA110DU-M / TA200DU

Technical data

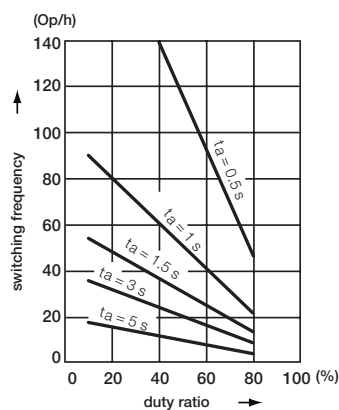
Main circuit – Utilization characteristics according to IEC/EN

Type	TA80DU-M	TA110DU-M	TA200DU
Standards	IEC 60947-1 / 60947-4-1 and EN 60947-1 / 60947-4-1		
Rated operational voltage Ue	690 V AC		
Rated frequency	DC, 50/60 Hz		
Frequency range	0 ... 400 Hz		
Trip class	10A		
Number of poles	3		
Duty time	100 %		
Operating frequency without early tripping	Up to 15 operations/h, see "Technical diagram – Intermittent periodic duty"		
Rated impulse withstand voltage Uimp	6 kV		
Rated insulation voltage Ui	690 V AC		

Auxiliary circuit according to IEC/EN

Type		TA80DU-M	TA110DU-M	TA200DU
Rated operational voltage Ue		500 V AC, 440 V DC		
Conventional free air thermal current Ith	N.C., 95-96	10 A		
	N.O., 97-98	6 A		
Rated frequency		DC, 50/60 Hz		
Number of poles		1 N.O. + 1 N.C.		
Ie / Rated operational current AC-15 acc. to IEC/EN 60947-5-1 for utilization category				
110-120 V	N.C., 95-96	3.00 A		
	N.O., 97-98	1.50 A		
220-230-240 V	N.C., 95-96	3.00 A		
	N.O., 97-98	1.20 A		
440 V	N.C., 95-96	0.75 A		
	N.O., 97-98	0.37 A		
480-500 V	N.C., 95-96	0.30 A		
	N.O., 97-98	0.25 A		
Ie / Rated operational current DC-13 acc. to IEC/EN 60947-5-1 for utilization category				
24 V	N.C., 95-96	1.25 A		
	N.O., 97-98	1.25 A		
60 V	N.C., 95-96	0.25 A		
	N.O., 97-98	0.25 A		
110-120-125 V	N.C., 95-96	0.25 A		
	N.O., 97-98	0.25 A		
250 V	N.C., 95-96	0.12 A		
	N.O., 97-98	0.04 A		
Minimum switching capacity		17 V / 3 mA		
Short-circuit protective device	N.C., 95-96	10 A, Fuse type gG		
	N.O., 97-98	6 A, Fuse type gG		
Rated impulse withstand voltage Uimp		6 kV		
Rated insulation voltage Ui		690 V		

Technical diagram – Intermittent periodic duty



ta: Motor starting time - TA80DU-M, TA110DU-M, TA200DU

Thermal overload relays TA80DU-M / TA110DU-M / TA200DU

Technical data

Main circuit – Utilization characteristics according to UL/CSA

Type	TA80DU-M	TA110DU-M	TA200DU
Standards	UL 508, CSA 22.2 No. 14		
Maximum operational voltage	600 V AC/DC		
Trip rating	125 % of FLA		
Full load amps (FLA)	See table "Full load amps and short-circuit protective device"		
Short-circuit rating RMS symmetrical	See table "Full load amps and short-circuit protective device"		
Short-circuit protective device	See table "Full load amps and short-circuit protective device"		

Auxiliary circuit according to UL/CSA

Type	TA80DU-M / TA110DU-M / TA200DU		
Contact rating	N.C., 95-96	B600	
	N.O., 97-98	C600	
Conventional free-air thermal current	5 A		

Full load amps and short-circuit protective device

Type	Full load amps (FLA)	Short-circuit protective device			
		480 / 600 V AC		480 / 600 V AC	
		Short circuit rating RMS symmetrical	Fuse K5 / RK5	Short circuit rating RMS symmetrical	Fuse J
TA80DU-42M	42 A	5000 A	150 A	50000 A	200 A
TA80DU-52M	52 A	5000 A	175 A	50000 A	200 A
TA80DU-63M	63 A	10000 A	200 A	50000 A	200 A
TA80DU-80M	80 A	10000 A	250 A	50000 A	200 A
TA110DU-90M	90 A	10000 A	250 A	65000 A	200 A
TA110DU-110M	110 A	10000 A	250 A	65000 A	200 A
TA200DU-90	90 A	10000 A	250 A	100000 A	250 A
TA200DU-110	110 A	10000 A	250 A	100000 A	250 A
TA200DU-135	135 A	10000 A	300 A	100000 A	250 A
TA200DU-150	150 A	10000 A	300 A	100000 A	250 A
TA200DU-175	175 A	10000 A	300 A	100000 A	300 A
TA200DU-200	200 A	10000 A	400 A	100000 A	400 A

Thermal overload relays TA80DU-M / TA110DU-M / TA200DU

Technical data

General technical data

Type		TA80DU-M	TA110DU-M	TA200DU
Pollution degree		3		
Phase loss sensitive		Yes		
Ambient air temperature				
Operation	Open - compensated	-25 ... +55 °C		
	Open	-25 ... +55 °C		
Storage		-40 ... +70 °C		
Ambient air temperature compensation		Acc. to IEC/EN60947-4-1		
Maximum operating altitude permissible		2000 m		
Resistance to shock acc. to IEC 60068-2-27		12g / 15 ms		
Mounting position		Position 1-6		
Mounting		Mount on the contactor and tighten the screws of the main circuit terminals or with single mounting kit		
Degree of protection	Housing	IP20		
	Main circuit terminals	IP10		

Electrical connection - Main circuit

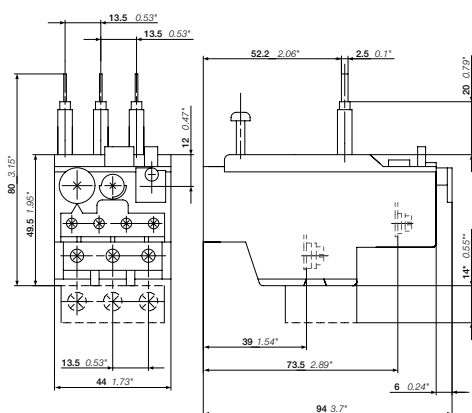
Type		TA80DU-M	TA110DU-M	TA200DU
Connecting capacity				
 Rigid	1 x	2.5 ... 25 mm ²	16 ... 35 mm ²	25 ... 120 mm ²
	2 x	2.5 ... 16 mm ²	-	-
 Flexible with ferrule	1 x	2.5 ... 25 mm ²	16 ... 35 mm ²	25 ... 120 mm ²
	2 x	2.5 ... 10 mm ²	-	-
 Lugs		-	-	L > 10 mm
	Stranded acc. to UL/CSA	1 x or 2 x	AWG 8-1	AWG 4-0000
	Flexible acc. to UL/CSA	1 x or 2 x	AWG 8-1	AWG 4-0000
Stripping length		14 mm	25 mm	-
Tightening torque		4.5 Nm / 40 lb.in	7.2 ... 9.6 Nm	25 Nm / 220 lb.in
Recommended screw driver		M6 (Pozidriv 2)	M8 (Hexagon 4)	Open bars

Auxiliary circuit

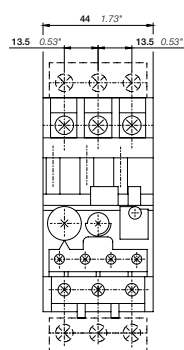
Type		TA80DU-M	TA110DU-M	TA200DU
Connecting capacity				
 Rigid	1 x or 2 x	0.75 ... 4 mm ²		
 Flexible with ferrule	1 x or 2 x	0.75 ... 2.5 mm ²		
 Flexible	1 x or 2 x	0.75 ... 2.5 mm ²		
	Stranded acc. to UL/CSA	1 x or 2 x	AWG 18-14	
	Flexible acc. to UL/CSA	1 x or 2 x	AWG 18-14	
Stripping length		9 mm		
Tightening torque		1 ... 1.3 Nm / 12 lb.in		
Recommended screw driver		M3.5 (Pozidriv 2)		

Thermal overload relays TA25DU-M / TA42DU-M / TA75DU-M

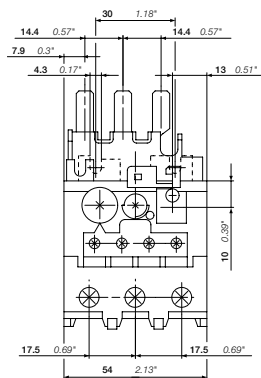
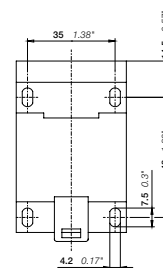
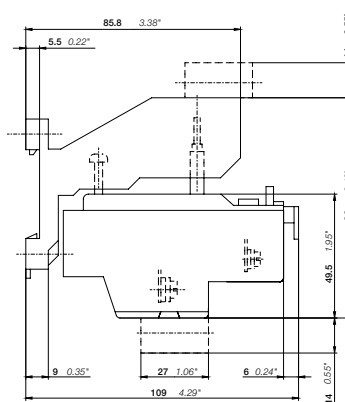
Dimensions



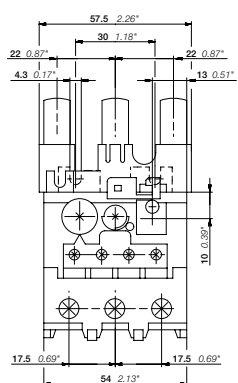
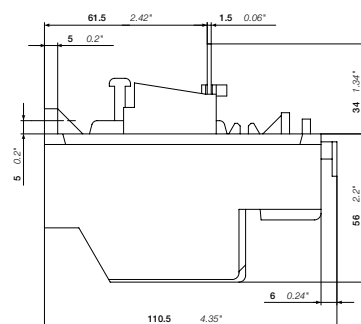
TA25DU-M + DX25



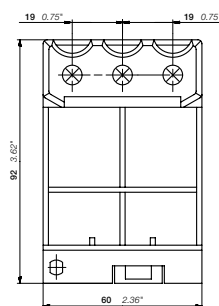
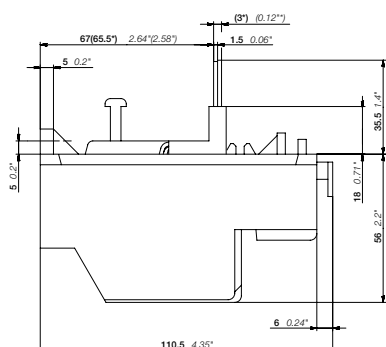
TA25DU-M + DB25 + DX25



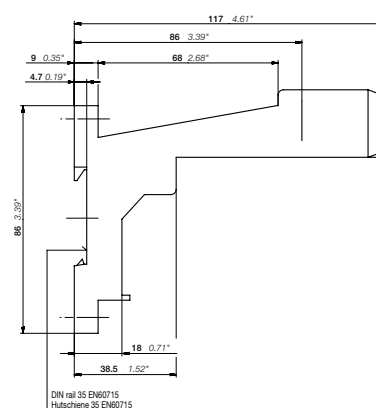
TA42DU-M



TA75DU-M



TA75DU-M + DB80
TA42DU-M + DB80



DIN rail 35 EN60715
Hutschiene 35 EN60715

EF205, EF370 electronic overload relays

63 to 380 A



EF205-210

2CDC233010V0012



EF370-380

2CDC233010V0012



KPR-101L

1SFC151402F0001

The EF205 and EF370 are self-supplied electronic overload relays, which means no extra external supply is needed. It offers reliable protection for motors in the event of overload or phase failure. Easy to use like a thermal overload relay and compatible with standard motor applications, the electronic overload relay is convincing, above all, due to its wide setting range, high accuracy, high operational temperature range and the possibility to select a trip class (10E, 20E, 30E). Further features are the temperature compensation, trip contact (N.C.), signal contact (N.O.), automatic or manual reset selectable, trip-free mechanism, STOP and TEST function and a trip indication. The overload relays are connected directly to the contactors. The EF205 and EF370 have ATEX and IECEx certification (1).

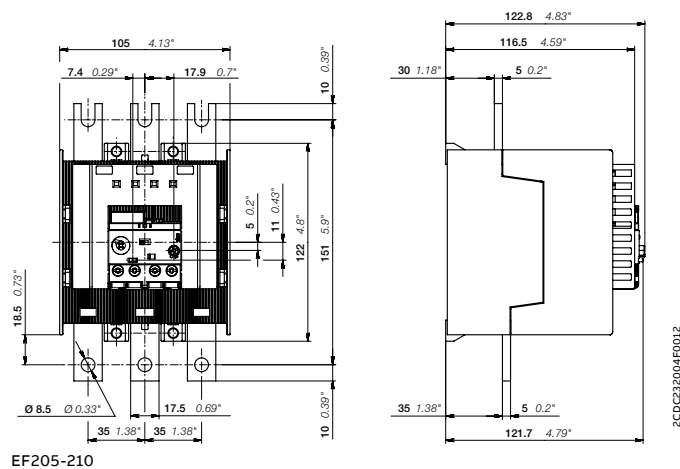
Setting range	Short-circuit protective device	Trip class	Type	Order code	Weight (1 pce) kg
A					
63 ... 210	1250 A, fuse type gG	10E, 20E, 30E	EF205-210	1SAX531001R1101	1.210
115 ... 380	1600 A, fuse type gG	10E, 20E, 30E	EF370-380	1SAX611001R1101	1.430

(1) ATEX is valid for products produced from week 42, 2015. IECEx is valid for products produced from week 15, 2017.

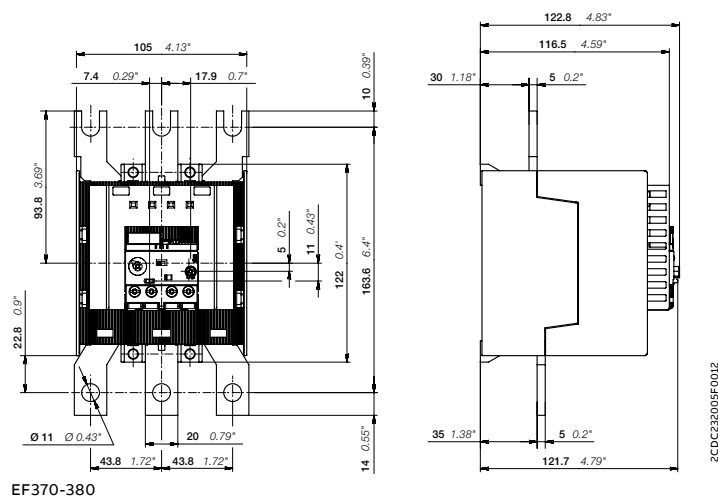
Ordering details accessories

Suitable for	Description	Type	Order code	Weight (1 pce) kg
EF205, EF370	Reset push button (2)	KPR-101L	1SFA616162R1014	0.027
EF205	Terminal shroud	LT200E	1SAX501904R0001	0.085
EF370	Terminal shroud	LT320E	1SAX601904R0001	0.105

(2) Note: for more information see catalog 1SFC151005C0201 rev B.



EF205-210



EF370-380

Main dimensions mm, inches

EF205, EF370 electronic overload relays

Technical data

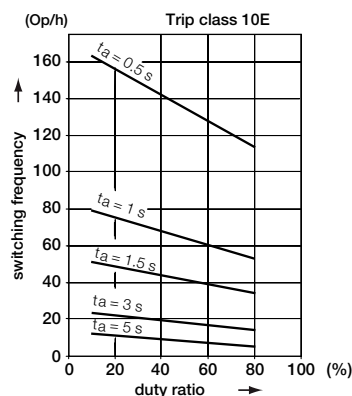
Main circuit – Utilization characteristics according to IEC/EN

Type	EF205, EF370
Standards	IEC/EN 60947-1, IEC/EN 60947-4-1, IEC/EN 60947-5-1
Rated operational voltage U_e	1000 V AC
Rated frequency	50/60 Hz – not suitable for DC applications
Trip class	10E, 20E, 30E, selectable
Number of poles	3
Duty time	100%
Operating frequency without early tripping	Up to 15 operations/h, see "Technical diagram – Intermittent periodic duty"
Rated impulse withstand voltage U_{imp}	8 kV
Rated insulation voltage U_i	1000 V

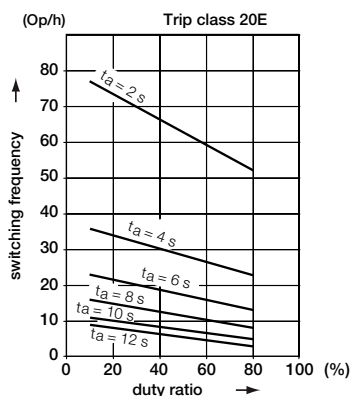
Auxiliary circuit according to IEC/EN

Type	EF205, EF370
Rated operational voltage U_e	600 V AC / DC
Conventional free air thermal current I_{th}	6 A
Rated frequency	DC, 50/60 Hz
Number of poles	1 N.C. + 1 N.O.
I_e / Rated operational current AC-15 acc. to IEC/EN 60947-5-1 for utilization category	
110-120 V	50/60 Hz 3.00 A
220-230-240 V	50/60 Hz 3.00 A
400 V	50/60 Hz 1.10 A
480-500 V	50/60 Hz 0.75 A
I_e / Rated operational current DC-13 acc. to IEC/EN 60947-5-1 for utilization category	
24 V	1.50 A
60 V	0.55 A
110-120-125 V	0.55 A
250 V	0.27 A
Minimum switching capacity	12 V / 3 mA
Short-circuit protective device	6 A, fuse type gG
Rated impulse withstand voltage U_{imp}	6 kV
Rated insulation voltage U_i	690 V

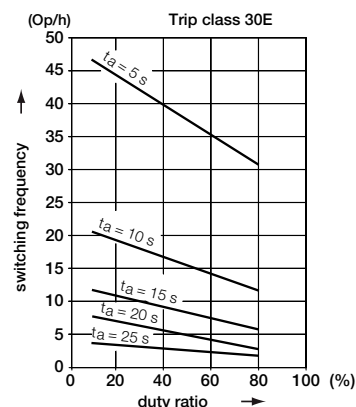
Technical diagram – Intermittent periodic duty



Trip class 10E



Trip class 20E



Trip class 30E

EF205, EF370 electronic overload relays

Technical data

Main circuit – Utilization characteristics according to UL/CSA

Type	EF205, EF370
Standards	UL 508, CSA 22.2 No. 14, UL 60947-4-1A
Maximum operational voltage	600 V AC
Trip rating	125% of FLA
Full load amps (FLA)	See table "Full load amps and short-circuit protective device"
Short-circuit rating RMS symmetrical	See table "Full load amps and short-circuit protective device"
Short-circuit protective device	See table "Full load amps and short-circuit protective device"

Auxiliary circuit according to UL/CSA

Type	EF205, EF370
Contact rating	N.C., 95-96 B600, Q600
	N.O., 97-98 B600, Q600
Conventional free-air thermal current	6 A

Full load amps and short-circuit protective device

Type	Full load amps (FLA)	Short-circuit protective device					
		480 V AC		600 V AC			
		SCCR	Fuse type	SCCR	Fuse type	SCCR	Fuse type
EF205-210	210 A	10 kA	400 A, R5/RK5	10 kA	400 A, R5/RK5	100 kA	400 A, J
EF370-380	380 A	18 kA	800 A, L/T	18 kA	800 A, L/T	-	-

EF205, EF370 electronic overload relays

Technical data

General technical data

Type	EF205	EF370
Pollution degree	3	
Phase loss sensitive	Yes	
Ambient air temperature		
Operation	Open - compensated	-25 ... +70 °C
Storage		-50 ... +85 °C
Ambient air temperature compensation	Acc. to IEC/EN 60947-4-1	
Maximum operating altitude permissible	2000 m	
Resistance to shock acc. to IEC 60068-2-27	25g / 11 ms	
Resistance to vibrations acc. to IEC 60068-2-6	5g / 3 ... 150 Hz	
Mounting position	Position 1-6	
Mounting	Mount on the contactor and tighten the screws of the main circuit terminals	
Degree of protection	Housing	IP20
	Main circuit terminals	IP20

Electrical connection - Main circuit

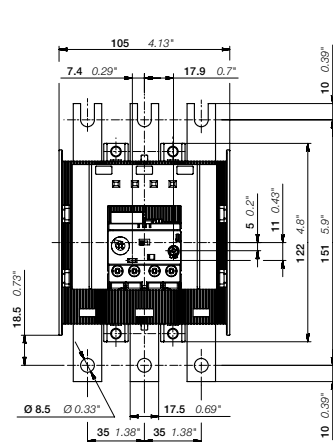
Type	EF205	EF370
Connecting capacity		
 Rigid	1 x 16 ... 185 mm ² 2 x 16 ... 120 mm ²	50 ... 240 mm ² 50 ... 150 mm ²
 Flexible	1 x 16 ... 185 mm ² 2 x 16 ... 120 mm ²	50 ... 240 mm ² 50 ... 150 mm ²
 Lugs	L ≤ 24 mm	32 mm
 Bars	Ø > 8 mm	10 mm
Stranded acc. to UL/CSA	1 x AWG 6-0000 2 x AWG 6-0000	AWG 1-500 kcmil AWG 1-500 kcmil
Flexible acc. to UL/CSA	1 x AWG 6-0000 2 x AWG 6-0000	AWG 1-500 kcmil AWG 1-500 kcmil
Stripping length	-	-
Tightening torque	18 Nm / 160 lb.in	28 Nm / 247 lb.in
Recommended screw driver	M8	M10

Auxiliary circuit

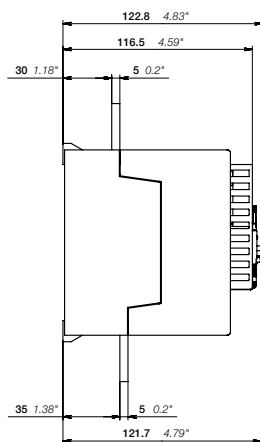
Type	EF205, EF370
Connecting capacity	
 Rigid	1 x or 2 x 1 ... 4 mm ²
 Flexible with ferrule	1 x or 2 x 0.75 ... 2.5 mm ²
 Flexible with insulated ferrule	1 x or 2 x 0.75 ... 2.5 mm ²
 Flexible	1 x or 2 x 0.75 ... 2.5 mm ²
Stranded acc. to UL/CSA	1 x or 2 x AWG 18-10
Flexible acc. to UL/CSA	1 x or 2 x AWG 18-10
Stripping length	9 mm
Tightening torque	0.8 ... 1.2 Nm / 7 ... 11 lb.in
Recommended screw driver	M3.5 (Pozidriv 2)

Electronic overload relays EF205, EF370

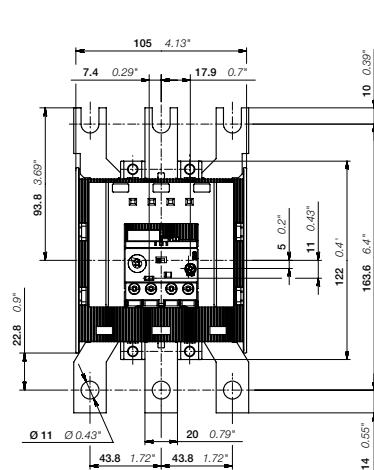
Dimensions



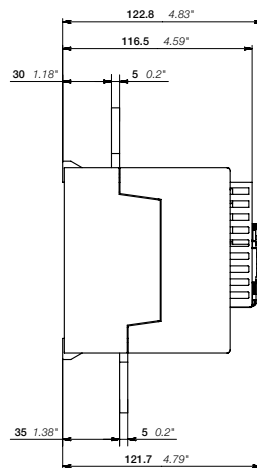
EF205



2CDD232004F0012



EF370



2CDD232005F0012

Main dimensions mm, inches

General technical data

118	Coordination with short-circuit protection devices
119	Terms and technical definitions
121	Standards and utilization categories
123	Degrees of protection

Coordination with short-circuit protection devices

In compliance with standards IEC 60947-4-1 and EN 60947-4-1, we define for the contactors and starters the type, rating and characteristics of the short-circuit protection devices SCPD which allow selective protection against overloads and ensure protection against short circuits.

Basic functions

Any starter is designed to:

- start motors,
- ensure continuous functioning of motors,
- disconnect motors from the supply line,
- guarantee protection of motors against overloads.

The starter is typically made up of a switching device (contactor) and an overload protection device (thermal overload relay or electronic overload relay).

These two devices **MUST** be coordinated with equipment capable of providing protection against short circuit (SCPD: short circuit protective device): typically a circuit breaker with magnetic release only or a switch fuse. These are not necessarily part of the starter.

Applicable standards

IEC 60947-4-1 (EN 60947-4-1) and UL 60947-4-1 between the branch circuit protective device and the motor starter precisely defines the different points to be considered in order to carry out correct coordination.

Complete coordination for a combination includes the following points:

- Selectivity test between the overload relay and the short-circuit protection device SCPD.
- Short-circuit condition tests:
- at prospective "r" currents - These currents depend on the rated operational current of the starter (I_e AC-3) and are given by the standard (Table 13). For example:
 - $r = 1 \text{ kA}$ for $I_e \text{ AC-3} < 16 \text{ A}$
 - $r = 3 \text{ kA}$ for $16 \text{ A} < I_e \text{ AC-3} < 63 \text{ A}$
 - $r = 5 \text{ kA}$ for $63 \text{ A} < I_e \text{ AC-3} < 125 \text{ A}$ etc.
- at the rated conditional short-circuit current " I_q " - This is the maximum prospective current that the combination can withstand, for example 50 kA.

Types of coordination

IEC 60947-4-1 (EN 60947-4-1) UL 60947-4-1 between the branch circuit protective device and the motor starter defines two types of coordination according to the expected level of service continuity. Acceptable extreme damage for the switchgear is divided into two types.

- Type 1: In short-circuit conditions, the contactor or starter does not endanger persons or installations and will not be able to then operate without being repaired or having parts replaced.
- Type 2: In short-circuit conditions, the contactor or starter does not endanger persons or installations and will be able to operate afterwards. The risk of contacts light welding is acceptable. In this case, the manufacturer must stipulate the measures to be taken with respect to maintenance of the equipment.

The complete ABB offer

ABB has acquired years of experience with respect to problems of coordination and is able to make a complete offer based on tests performed in its qualified laboratories.

This offer includes 400 V, 500 V, 690 V networks.

A complete data base of coordination tables, according to IEC 60947-4-1 (EN 60947-4-1), is available on the ABB Website.

In the coordination tables the following short-circuit protection devices are recommended:

- Moulded case circuit-breakers (MCCBs)
- Miniature circuit-breakers (MCBs)
- Switch-disconnector-fuses (aM, gG and BS)

General remarks applicable to all tables

- Each table is defined for a maximum ambient temperature of 40 °C. For higher temperatures, apply a derating factor according to the following rules:
 - Fuses: factor of 0.8 applied to I_n for an ambient temperature of 70 °C
 - MCCBs and MCBs: factor of 0.8 applied to I_n for an ambient temperature of 60 °C
 - The starter derating factor depends on the operating conditions of thermal overload relays:
 - Factor of 0.9 applied to I_n for an ambient temperature of 70 °C.
 - Each table is defined for motor currents: 3-phase motors, 4-pole
 - **Normal starting** means a starting time $< 2 \text{ s}$.
 - **Difficult starting** means an accelerating time $10 \text{ s} < t_s < 30 \text{ s}$
 - Tripping classes of thermal overload relays according to IEC 60947-4-1 (EN 60947-4-1): 10A and 10
 - Tripping classes of electronic overload relays according to IEC 60947-4-1 (EN 60947-4-1): 10E, 20E, 30E selectable
 - In the tables with MCCBs, these are fitted with the magnetic relay alone. Setting is always carried out at $> 12.3 I_e \text{ AC-3}$ so that the transient current peak occurring during starting does not lead to tripping.

Terms and technical definitions

Circuits

- auxiliary circuit: All the conductive parts of a contactor designed to be inserted in a different circuit from the main circuit and the contactor control circuits.
- control circuit: All the conductive parts of a contactor (other than the main circuit and the auxiliary circuit) used to control the contactor's closing operation or opening operation or both.
- main circuit: All the conductive parts of a contactor designed to be inserted in the circuit that it controls.

Thermal overload relay tripping classes

IEC 60947-4-1 defines tripping classes 10 A, 10, 20 and 30. Types 10 A, 10, etc. correspond to the maximum tripping time for a making current at 7.2 times the setting current. Furthermore, for each class the standard specifies the tripping time for 1.5 times the setting current and sets the non tripping condition at 1.05 times the setting current. All these data are summarized in the table below.

Extract from IEC 60947-4-1:

Tripping class		10 A	10	20	30
Max. tripping time for 1.5 times the setting current (warm state)	s	120	240	480	720
Tripping time for 7.2 times the setting current (cold state)	s	2 - 10	4 - 10	6 - 20	9 - 30
For 1.05 times the setting current		No tripping			

Electromagnetic compatibility

AF... contactors comply with IEC 60947-1, 60947-4-1 and EN 60947-1, 60947-4-1 standards.

• Definitions:

Environment A: "Mainly relates to low-voltage non public or industrial networks/locations/installations (EN 50082-2 article 4) including highly disturbing sources".

Environment B: "Mainly relates to low-voltage public networks (EN50082-1 article 5) such as residential, commercial and light industrial locations/installations. Highly disturbing sources such as arc welders are not covered by this environment".

Notice for AF09 ... AF38, AF116 ... AF2650 contactors and NF contactor relays: these products have been designed for environment

A. Use of this product in environment B may cause unwanted electromagnetic disturbances in which case the user may be required to take adequate mitigation measures.

AF40 ... AF96 have been designed for environment B.

Definitions according to SEMI F47-0706

SEMI F47-0706 defines the voltage sag immunity required for semiconductor processing, metrology and automated test equipment, and on subsystems and components which are used in the construction of semiconductor processing equipment including but not limited to:

- Power supplies
- Generators
- Robots and factory interface
- Chillers, pumps, blowers
- AC operated contactors and contactor relays

- Voltage sag: an rms reduction in the AC voltage, at the power frequency, for durations from a half cycle to a few seconds.

The IEC terminology for this phenomenon is voltage dip. voltage sag immunity: the ability of equipment to withstand momentary electrical power interruptions or sags.

Coordination of protections against short circuit

The goal here is to protect electromechanical starters and softstarters.

Any starter is designed to:

- start motors,
- ensure continuous functioning of motors,
- disconnect motors from the supply line,
- guarantee protection of motors against overloads.

The starter is typically made up of a switching device (contactor) and an overload protection device (thermal overload relay or electronic overload relay). These two devices MUST be coordinated with equipment capable of providing protection against short circuit (SCPD: short circuit protective device): typically a circuit breaker with magnetic release only or a switch fuse. These are not necessarily part of the starter.

The characteristics of the starter must comply with the international standard IEC 60947-4-1 which defines the above items as follows:

contactor: a mechanical switching device having only one position of rest, operated otherwise than by hand, capable of making, carrying and breaking currents under normal circuit conditions including overload conditions.

overload release: overload relay or release which operates in the case of overload and also in case of loss of phase.

circuit-breaker: defined by IEC 60947-2 as a mechanical switching device, capable of making, carrying and breaking currents under normal circuit conditions and also making, carrying for a specified time and breaking currents under specified abnormal circuit conditions.

IEC publication 60947-4-1 defines coordination types "1" and "2":

- Type "1" coordination requires that, in the event of a short-circuit, the contactor or starter does not endanger persons or installations and will not then be able to operate without being repaired or parts being replaced.
- Type "2" coordination requires that, in short-circuit conditions, the contactor or starter does not endanger persons or installations and will be able to operate afterwards. The risk of contacts being light welded is acceptable. In this case, the manufacturer must stipulate the measures to be taken with respect to maintenance of the equipment.

Rated operational current I_e.

Current rated by the manufacturer. It is mainly based on the rated operational voltage U_e, the rated frequency, the utilization category, the rated duty and the type of protective enclosure, if necessary.

Terms and technical definitions

Conventional free air thermal current I_{th}

Current that the contactor can withstand in free air for a duty time of 8 hours without the temperature rise of its various parts exceeding the maximum values given by the standard.

Operating cycle or cycle

Includes one making operation and one breaking operation.

Cycle time

This is the sum of the current flow time and the no-current time for given cycle.

Electrical durability

Number of on-load operating cycles that the contactor is able to carry out. It depends on the operational current, the operational voltage and the utilization category.

Mechanical durability

Number of no-current operating cycles that a contactor is able to carry out.

Assessed failure rate

Defined according to IEC 60947-5-4. This rate is given in standard industrial environments for the contactor relays and for the built-in auxiliary contact of contactors.

Load factor

Ratio of the on-load operating time to the total cycle time x 100.

Switching frequency

Number of switching cycles per hour.

Plugging

Stopping or fast reversal in rotation direction of a motor by two supply leads being interchanged while the motor is running.

Inching

Energization of a motor's circuit repeatedly or for short periods with the aim of obtaining small movements of the driven mechanism.

Coil operating limits

Expressed in multiples of the nominal control circuit voltage U_c for the upper and lower limits.

Mounting position

Comply with the manufacturer's instructions. Restrictions are to be taken into account for certain mounting positions.

Rated breaking or making capacity

Root mean square (r.m.s.) value of the current that the contactor is able to break or make at a given voltage according to the conditions specified by standards and for a given utilization category.

Intermittent duty

Duty during which the contactor is successively closed or open for periods which are too short to enable the contactor to achieve thermal balance.

Ambient temperature

Air temperature close to the contactor.

Time

- Time constant: Ratio of the inductance to the resistance ($L/R = \text{mH}/\Omega = \text{ms}$).
- Short-time withstand current: Current that the contactor is able to withstand in closed position for a short time interval and in specified conditions.
- Closing time: Time interval between the coil energization and the instant the contacts touch on all the poles.
- Opening time: Time interval between the coil de-energization and the instant the contacts separate on all the poles.

Rated control voltage U_c

Control voltage value for which the control circuit is sized.

Rated operational voltage U_e

Voltage to which the contactor's utilization characteristics refer. In three-phase it is the phase-to-phase voltage.

Rated insulation voltage U_i

Reference voltage for dielectric tests and creepage distances.

Rated impulse withstand voltage U_{imp}

Peak value of an impulse voltage, having a specified form and polarity, which does not cause breakdown in specific test conditions.

Shock withstand

Requirement for vehicles, crane drives, installations on board ships and plug-in equipment. For the acceptable "g" values, the contacts must not change position and the thermal overload relays must not trip.

Resistance to vibrations

Requirements for vehicles, boats and other means of transport. For the specified vibration amplitude and frequency values the device must remain able to operate.

Standards and utilization categories

Utilization categories:

A contactor's duty is characterised by the utilization category together with the rated operational voltage and current indicated.

Utilization categories for contactors according to IEC 60947-4-1:

Alternating current:	AC-1	Non-inductive or slightly inductive loads, resistance furnaces.
	AC-2	Slip-ring motors: starting, switching off.
	AC-3	Cage motors: starting, switching off running motors.
	AC-4	Cage motors: starting, plugging, inching.
	AC-5a	Discharge lamp switching.
	AC-5b	Incandescent lamp switching.
	AC-6a	Transformer switching.
	AC-6b	Capacitor bank switching.
	AC-8a	Hermetic refrigeration compressor motor control with manual resetting of overload releases.
Direct current:	AC-8b	Hermetic refrigeration compressor motor control with automatic resetting of overload releases.
	DC-1	Non inductive or slightly inductive loads, resistance furnaces.
	DC-3	Shunt motors: starting, plugging, inching, dynamic breaking of DC motors.
	DC-5	Series motors: starting, plugging, inching, dynamic breaking of DC motors.
	DC-6	Incandescent lamp switching.

Utilization categories for contactor relays according to IEC 60947-5-1:

Alternating current:	AC-12	Control of resistive loads and static loads with opto-coupler isolation.
	AC-13	Control of static loads with transformer isolation.
	AC-14	Control of weak electromagnetic loads (≤ 72 VA).
	AC-15	Control of electromagnetic loads (> 72 VA).
Direct current:	DC-12	Control of resistive loads and static loads with opto-coupler isolation.
	DC-13	Control of DC electromagnets.
	DC-14	Control of DC electromagnets having economy resistors.

In fact some applications, and the specific criteria characterizing the various loads controlled by contactors, may modify the utilization characteristics of the contactors. The main applications concerned are:

Capacitor bank switching

Account must be taken of high peaks when the current is made and of harmonic currents during continuous duty. For this application, IEC publication 60947-4-1 stipulates utilization category AC-6b. The operational currents or powers acceptable for the contactors are determined by our electrical tests; IEC publication 60947-4-1 gives the calculating formula for determining the operational current (Table 9).

Transformer switching

Account must be taken of the peaks due to magnetization phenomena when the current is made. For this application, IEC publication 60947-4-1 stipulates utilization category AC-6a. The operational currents or powers acceptable for the contactors are determined using the values obtained for AC-3 or AC-4 category tests and the calculating formula given in IEC 60947-4-1 (Table 9).

Lighting circuit switching

The current peaks occurring on energization of the circuit and the power factor depend on the type of lamps, the connection mode and whether or not there is compensation.

For this application, IEC publication 60947-4-1 stipulates two standard utilization categories:

AC-5a for discharge lamp switching.

AC-5b for incandescent lamp switching.

Slip-ring motor switching

The contactors used for short-circuiting rotor resistors can be used for rotor voltages up to 2 times the rated operational voltage.

The conditions of use of rotor contactors depend on the connection mode of the main poles. IEC 60947-4-1 stipulates AC-2 utilization category for startor contactor.

Standards and utilization categories

Utilization categories (cont.)

DC power circuit switching

Arc suppression is more difficult in direct current than in alternating current. Higher the time constant and voltage, heavier the breaking conditions: consequently several poles have to be connected in series.

AC high current circuit switching

Possibility of increasing performances by connecting poles in parallel.

Circuit switching during temporary and intermittent duty

In these cases higher operational currents are acceptable.

Influence of the length of the conductors used in the contactor control circuit

According to the operational voltages, the cross-sectional areas, the coil consumption and the control layout, difficulties due to line resistances and capacitances may appear during contactor closing and opening orders.

Making and breaking conditions for utilization categories

Utilization category	Durability test conditions						Occasional operation					
	Making conditions			Breaking conditions			Making and breaking capacities - 50 operating cycles					
	I/le	U/ue	Cos. ϕ or L/R (ms)	I/le	U/ue	Cos. ϕ or L/R (ms)	Ic/le	Ur/ue	Cos. ϕ or L/R (ms)	Ic/le	Ur/ue	Cos. ϕ or L/R (ms)

Contactors for AC circuit switching

AC-1		1	0.95	1	1	0.95	1.5	1.05	0.8	1.5	1.05	0.8
AC-2		2.5	0.65	2.5	1	0.65	4	1.05	0.65	4	1.05	0.65
AC-3	le < 17 A	6	0.65	1	0.17	0.65	10	1.05	0.45	8	1.05	0.45
	17 < le < 100 A	6	0.35	1	0.17	0.35	10	1.05	0.45	8	1.05	0.45
	le > 100 A	6	0.35	1	0.17	0.35	10	1.05	0.35	8	1.05	0.35
AC-4	le < 17 A	6	0.65	6	1	0.65	12	1.05	0.45	10	1.05	0.45
	17 < le < 100 A	6	0.35	6	1	0.35	12	1.05	0.45	10	1.05	0.45
	le > 100 A	6	0.35	6	1	0.35	12	1.05	0.35	10	1.05	0.35

Contactors for DC circuit switching

DC-1		1	1	1	1	1	1.5	1.05	1	1.5	1.05	1
DC-3		2.5	2	2.5	1	2	4	1.05	2.5	4	1.05	2.5
DC-5		2.5	7.5	2.5	1	7.5	4	1.05	15	4	1.05	15

Contactor relays for AC circuit switching

AC-14	(≤ 72 VA)	–	–	–	–	–	6	1.1	0.7	6	1.1	0.7
AC-15	(> 72 VA)	10	1	0.7	1	0.4	10	1.1	0.3	10	1.1	0.3

Contactor relays for DC circuit switching

Utilization category	Standard operation						Occasional operation					
	Making conditions			Breaking conditions			Making and breaking capacities - 50 operating cycles					
	I/le	U/ue	T0.95	I/le	U/ue	T0.95	Ic/le	Ur/ue	T0.95	Ic/le	Ur/ue	T0.95
DC-13	1	1	6 P(1)	1	1	6 P(1)	1.1	1.1	6 P(1)	1.1	1.1	6 P(1)
DC-14	–	–	–	–	–	–	10	1.1	15 ms	10	1.1	15 ms

(1) The value "6 x P" is the result of an empirical relation which is estimated to represent most DC magnetic loads up to the highest limit of P = 50 W (6 x P = 300 ms). It is accepted that loads having drawn energy above 50 W are made up of weaker loads in parallel. As a consequence, the 300 ms value must form the highest limit whatever the value of the power drawn.

Key:

U (I) = applied voltage (current)

Ur = recovery voltage

L/R = test circuit time constant

Ue (Ie) = rated operational voltage (current)

Ic = making and breaking current expressed in DC or in AC like the r.m.s. value of the symmetrical components

T0.95 = time required to reach 95 % of the current in steady-state conditions, expressed in milliseconds

Degrees of protection

General

In an installation, the degree of protection required for electrical equipment depends on the environmental characteristics. The degree of protection, ensured by the enclosure of equipment or by the cubicle containing the equipment is expressed by the IP code which gives the level of protection against access to hazardous parts, the ingress of foreign bodies and/or the ingress of water, in compliance with IEC 60529, IEC 60947-1.

Besides the IP symbol, the complete code has two figures followed (optionally) by two additional letters. A short description of the elements used in IP coding is given below.

IP... code	Figures or letters	Specifications for installation protection	Protection of persons
First figure		Against ingress of foreign bodies	Against access to hazardous parts with:
	0	No protection	No protection
	1	Diameter > 50 mm	Back of hand
	2	Diameter > 12.5 mm	Finger
	3	Diameter > 2.5 mm	Tool
	4	Diameter > 1 mm	Wire
	5	Limited protection against dust	Wire
	6	Total protection against dust	Wire
Second figure		Against entrance of water having a harmful effect	
	0	No protection	
	1	Vertical dripping	
	2	Dripping at a vertical angle of < 15°	
	3	Rain at a vertical angle of < 60°	
	4	Splashing	
	5	Low pressure water jet	
	6	Powerful water jets	
	7	Temporary immersion	
	8	Permanent immersion	
Additional letter (optional) for use with:		Against ingress of foreign bodies	Against access to hazardous parts with:
First figure 0	A	Stopped by a barrier with a 50 mm Ø sphere	Back of hand
First figure 0 or 1	B	Entrance of test finger limited to 80 mm	Finger
First figure 1 or 2	C	Wire with 2.5 mm Ø and length of 100 mm	Tool
First figure 2 or 3	D	Wire with 1 mm Ø and length of 100 mm	Wire
Additional letter (optional)		Specific additional information	
	H	High voltage apparatus	–
	M	Moving parts which are moving during water test	
	S	Moving parts which are stationary during water test	
	W	Specified atmospheric conditions	

Note: The type of enclosure or cubicle in which the equipment must be installed prevails with respect to the degree of protection.

Index

Order code	Type	Page
1SAM101923R0002	MSMN (3)	97
1SAM101923R0012	MSMNO (4)	97
1SAM201901R1001	HKF1-11	90
1SAM201901R1002	HKF1-20	90
1SAM201901R1003	HKF1-10	90
1SAM201901R1004	HKF1-01	90
1SAM201902R1001	HK1-11	90
1SAM201902R1002	HK1-20	90
1SAM201902R1003	HK1-02	90
1SAM201902R1004	HK1-20L	90
1SAM201903R1001	SK1-11	90
1SAM201903R1002	SK1-20	90
1SAM201903R1003	SK1-02	90
1SAM201904R1001	UA1-24	91
1SAM201904R1002	UA1-48	91
1SAM201904R1003	UA1-60	91
1SAM201904R1004	UA1-110	91
1SAM201904R1005	UA1-230	91
1SAM201904R1006	UA1-400	91
1SAM201904R1007	UA1-415	91
1SAM201904R1008	UA1-208	91
1SAM201904R1009	UA1-575	91
1SAM201904R1010	UA1-20	91
1SAM201906R1102	PS1-2-0-65	94
1SAM201906R1103	PS1-3-0-65	94
1SAM201906R1104	PS1-4-0-65	94
1SAM201906R1105	PS1-5-0-65	94
1SAM201906R1112	PS1-2-1-65	94
1SAM201906R1113	PS1-3-1-65	94
1SAM201906R1114	PS1-4-1-65	94
1SAM201906R1115	PS1-5-1-65	94
1SAM201906R1122	PS1-2-2-65	94
1SAM201906R1123	PS1-3-2-65	94
1SAM201906R1124	PS1-4-2-65	94
1SAM201906R1125	PS1-5-2-65	94
1SAM201907R1101	S1-M1-25	94
1SAM201907R1102	S1-M2-25	94
1SAM201907R1103	S1-M3-25	94
1SAM201908R1001	BS1-3	94
1SAM201909R1001	FS116	94
1SAM201909R1021	MSAH1	97
1SAM201910R1001	AA1-24	91
1SAM201910R1002	AA1-110	91
1SAM201910R1003	AA1-230	91
1SAM201910R1004	AA1-400	91
1SAM201911R1010	IB132-G	96
1SAM201911R1011	IB132-Y	96
1SAM201912R1010	DMS132-G	96
1SAM201912R1011	DMS132-Y	96
1SAM201913R1103	S1-M3-35	94
1SAM201914R1001	PB1-1-32	94
1SAM201914R1002	PS1-PB1-25	94
1SAM201916R1103	PS1-3-0-100	94
1SAM201916R1104	PS1-4-0-100	94
1SAM201916R1105	PS1-5-0-100	94
1SAM201916R1113	PS1-3-1-100	94
1SAM201916R1114	PS1-4-1-100	94
1SAM201916R1115	PS1-5-1-100	94
1SAM201916R1123	PS1-3-2-100	94
1SAM201920R1000	MSH-AR	97
1SAM201920R1001	MSHD-LB (1)	97
1SAM201920R1002	MSHD-LY (1)	97
1SAM201920R1011	MSHD-LTB (2)	97
1SAM201920R1012	MSHD-LTY (2)	97
1SAM250000R1001	MS116-0.16	85
1SAM250000R1002	MS116-0.25	85
1SAM250000R1003	MS116-0.4	85
1SAM250000R1004	MS116-0.63	85
1SAM250000R1005	MS116-1.0	85
1SAM250000R1006	MS116-1.6	85
1SAM250000R1007	MS116-2.5	85
1SAM250000R1008	MS116-4.0	85

Order code	Type	Page
1SAM250000R1009	MS116-6.3	85
1SAM250000R1010	MS116-10	85
1SAM250000R1011	MS116-16	85
1SAM250000R1012	MS116-12	85
1SAM250000R1013	MS116-20	85
1SAM250000R1014	MS116-25	85
1SAM250000R1015	MS116-32	85
1SAM250005R1001	MS116-0.16-HKF1-11	85
1SAM250005R1002	MS116-0.25-HKF1-11	85
1SAM250005R1003	MS116-0.4-HKF1-11	85
1SAM250005R1004	MS116-0.63-HKF1-11	85
1SAM250005R1005	MS116-1.0-HKF1-11	85
1SAM250005R1006	MS116-1.6-HKF1-11	85
1SAM250005R1007	MS116-2.5-HKF1-11	85
1SAM250005R1008	MS116-4.0-HKF1-11	85
1SAM250005R1009	MS116-6.3-HKF1-11	85
1SAM250005R1010	MS116-10.0-HKF1-11	85
1SAM250005R1011	MS116-16.0-HKF1-11	85
1SAM250005R1012	MS116-12.0-HKF1-11	85
1SAM250005R1013	MS116-20-HKF1-11	85
1SAM250005R1014	MS116-25-HKF1-11	85
1SAM250005R1015	MS116-32-HKF1-11	85
1SAX501904R0001	LT200E	112
1SAX531001R1101	EF205-210	112
1SAX601904R0001	LT320E	112
1SAX611001R1101	EF370-380	112
1SAZ201108R0001	DB25/25A	105
1SAZ201108R0002	DB25/32A	105
1SAZ201307R0002	DX25	105
1SAZ211201R2005	TA25DU-0.16M	101
1SAZ211201R2009	TA25DU-0.25M	101
1SAZ211201R2013	TA25DU-0.4M	101
1SAZ211201R2017	TA25DU-0.63M	101
1SAZ211201R2021	TA25DU-1.0M	101
1SAZ211201R2023	TA25DU-1.4M	101
1SAZ211201R2025	TA25DU-1.8M	101
1SAZ211201R2028	TA25DU-2.4M	101
1SAZ211201R2031	TA25DU-3.1M	101
1SAZ211201R2033	TA25DU-4.0M	101
1SAZ211201R2035	TA25DU-5.0M	101
1SAZ211201R2038	TA25DU-6.5M	101
1SAZ211201R2040	TA25DU-8.5M	101
1SAZ211201R2043	TA25DU-11M	101
1SAZ211201R2045	TA25DU-14M	101
1SAZ211201R2047	TA25DU-19M	101
1SAZ211201R2051	TA25DU-25M	101
1SAZ211201R2053	TA25DU-32M	101
1SAZ301110R0001	DB80	105, 106
1SAZ311201R2001	TA42DU-25M	101
1SAZ311201R2002	TA42DU-32M	101
1SAZ311201R2003	TA42DU-42M	101
1SAZ311201R2004	TA42DU-52M	101
1SAZ311201R2005	TA42DU-63M	101
1SAZ311201R2006	TA42DU-80M	101
1SAZ331201R2003	TA80DU-42M	106
1SAZ331201R2004	TA80DU-52M	106
1SAZ331201R2005	TA80DU-63M	106
1SAZ331201R2006	TA80DU-80M	106
1SAZ401110R0001	DB200	106
1SAZ401901R1001	LT200/A	106
1SAZ411201R2001	TA110DU-90M	106
1SAZ411201R2002	TA110DU-110M	106
1SAZ421201R1001	TA200DU-90	106
1SAZ421201R1002	TA200DU-110	106
1SAZ421201R1003	TA200DU-135	106
1SAZ421201R1004	TA200DU-150	106
1SAZ421201R1005	TA200DU-175	106
1SAZ421201R1006	TA200DU-200	106
1SBH901074R8022	NX22E-80	52

Order code	Type	Page
1SBH901074R8031	NX31E-80	52
1SBH901074R8040	NX40E-80	52
1SBH901074R8122	NX22E-81	52
1SBH901074R8131	NX31E-81	52
1SBH901074R8140	NX40E-81	52
1SBH901074R8422	NX22E-84	52
1SBH901074R8431	NX31E-84	52
1SBH901074R8440	NX40E-84	52
1SBH901074R8622	NX22E-86	52
1SBH901074R8631	NX31E-86	52
1SBH901074R8640	NX40E-86	52
1SBH901074R8822	NX22E-88	52
1SBH901074R8831	NX31E-88	52
1SBH901074R8840	NX40E-88	52
1SBL281074R8001	AX32-30-01-80	14
1SBL281074R8010	AX32-30-10-80	14
1SBL281074R8101	AX32-30-01-81	14
1SBL281074R8110	AX32-30-10-81	14
1SBL281074R8401	AX32-30-01-84	14
1SBL281074R8410	AX32-30-10-84	14
1SBL281074R8601	AX32-30-01-86	14
1SBL281074R8610	AX32-30-10-86	14
1SBL281074R8801	AX32-30-01-88	14
1SBL281074R8810	AX32-30-10-88	14
1SBL321074R8001	AX40-30-01-80	14
1SBL321074R8010	AX40-30-10-80	14
1SBL321074R8101	AX40-30-01-81	14
1SBL321074R8110	AX40-30-10-81	14
1SBL321074R8401	AX40-30-01-84	14
1SBL321074R8410	AX40-30-10-84	14
1SBL321074R8501	AX40-30-01-85	14
1SBL321074R8601	AX40-30-01-86	14
1SBL321074R8610	AX40-30-10-86	14
1SBL321074R8810	AX40-30-10-88	14
1SBL351074R8000	AX50-30-00-80	15
1SBL351074R8100	AX50-30-00-81	15
1SBL351074R8011	AX50-30-11-80	16
1SBL351074R8111	AX50-30-11-81	16
1SBL351074R8400	AX50-30-00-84	15
1SBL351074R8411	AX50-30-11-84	16
1SBL351074R8600	AX50-30-00-86	15
1SBL351074R8611	AX50-30-11-86	16
1SBL351074R8800	AX50-30-00-88	15
1SBL351074R8811	AX50-30-11-88	16
1SBL371074R8000	AX65-30-00-80	15
1SBL371074R8100	AX65-30-00-81	15
1SBL371074R8011	AX65-30-11-80	16
1SBL371074R8111	AX65-30-11-81	16
1SBL371074R8400	AX65-30-00-84	15
1SBL371074R8411	AX65-30-11-84	16
1SBL371074R8600	AX65-30-00-86	15
1SBL371074R8611	AX65-30-11-86	16
1SBL371074R8800	AX65-30-00-88	15
1SBL371074R8811	AX65-30-11-88	16
1SBL411074R8000	AX80-30-00-80	15
1SBL411074R8100	AX80-30-00-81	15
1SBL411074R8011	AX80-30-11-80	16
1SBL411074R8111	AX80-30-11-81	16
1SBL411074R8400	AX80-30-00-84	15
1SBL411074R8411	AX80-30-11-84	16
1SBL411074R8600	AX80-30-00-86	15
1SBL411074R8611	AX80-30-11-86	16
1SBL411074R8800	AX80-30-00-88	15
1SBL411074R8811	AX80-30-11-88	16
1SBL901074R8001	AX09-30-01-80	12
1SBL901074R8010	AX09-30-10-80	12
1SBL901074R8101	AX09-30-01-81	12
1SBL901074R8110	AX09-30-10-81	12
1SBL901074R8401	AX09-30-01-84	12
1SBL901074R8410	AX09-30-10-84	12
1SBL901074R8601	AX09-30-01-86	12
1SBL901074R8610	AX09-30-10-86	12

Index

Order code	Type	Page
1SBL901074R8801	AX09-30-01-88	12
1SBL901074R8810	AX09-30-10-88	12
1SBL911074R8001	AX12-30-01-80	12
1SBL911074R8010	AX12-30-10-80	12
1SBL911074R8101	AX12-30-01-81	12
1SBL911074R8110	AX12-30-10-81	12
1SBL911074R8401	AX12-30-01-84	12
1SBL911074R8410	AX12-30-10-84	12
1SBL911074R8501	AX12-30-01-85	12
1SBL911074R8601	AX12-30-01-86	12
1SBL911074R8610	AX12-30-10-86	12
1SBL911074R8810	AX12-30-10-88	12
1SBL921074R8001	AX18-30-01-80	13
1SBL921074R8010	AX18-30-10-80	13
1SBL921074R8101	AX18-30-01-81	13
1SBL921074R8110	AX18-30-10-81	13
1SBL921074R8401	AX18-30-01-84	13
1SBL921074R8410	AX18-30-10-84	13
1SBL921074R8601	AX18-30-01-86	13
1SBL921074R8610	AX18-30-10-86	13
1SBL921074R8801	AX18-30-01-88	13
1SBL921074R8810	AX18-30-10-88	13
1SBL931074R8001	AX25-30-01-80	13
1SBL931074R8010	AX25-30-10-80	13
1SBL931074R8101	AX25-30-01-81	13
1SBL931074R8110	AX25-30-10-81	13
1SBL931074R8401	AX25-30-01-84	13
1SBL931074R8410	AX25-30-10-84	13
1SBL931074R8501	AX25-30-01-85	13
1SBL931074R8601	AX25-30-01-86	13
1SBL931074R8610	AX25-30-10-86	13
1SBL931074R8810	AX25-30-10-88	13
1SBN010015R1001	CE5-01D0.1	60
1SBN010015R1010	CE5-10D0.1	60
1SBN010016R1001	CE5-01W0.1	60
1SBN010016R1010	CE5-10W0.1	60
1SBN010017R1001	CE5-01D2	60
1SBN010017R1010	CE5-10D2	60
1SBN010018R1001	CE5-01W2	60
1SBN010018R1010	CE5-10W2	60
1SBN019010R1001	CA5X-01	19, 58
1SBN019010R1010	CA5X-10	19, 58
1SBN019020R1011	CAL5X-11	19, 58
1SBN019040R1004	CA5X-04E	58
1SBN019040R1022	CA5X-22E	19, 58
1SBN019040R1031	CA5X-31E	58
1SBN019040R1040	CA5X-40E	58
1SBN019040R1104	CA5X-04M	58
1SBN019040R1113	CA5X-13M	58
1SBN019040R1122	CA5X-22M	19, 58
1SBN019040R1131	CA5X-31M	58
1SBN019040R1204	CA5X-04N	58
1SBN019040R1213	CA5X-13N	58
1SBN019040R1222	CA5X-22N	58
1SBN019040R1231	CA5X-31N	58
1SBN019040R1240	CA5X-40N	58
1SBN019040R1304	CA5X-04U	58
1SBN019040R1322	CA5X-22U	58
1SBN019040R1331	CA5X-31U	58
1SBN019040R1340	CA5X-40U	58
1SBN020312R1000	TEF5-ON	20, 63
1SBN020314R1000	TEF5-OFF	20, 63
1SBN030100R1000	VM5-1	19, 66
1SBN030110R1000	VE5-1	19, 66
1SBN030210R1000	VE5-2	19, 66
1SBN050010R1000	RV5/50	20, 72
1SBN050010R1001	RV5/133	20, 72
1SBN050010R1002	RV5/250	20, 72
1SBN050010R1003	RV5/440	20, 72
1SBN050100R1000	RC5-1/50	20, 72
1SBN050100R1001	RC5-1/133	20, 72
1SBN050100R1002	RC5-1/250	20, 72

Order code	Type	Page
1SBN050100R1003	RC5-1/440	20, 72
1SBN050200R1000	RC5-2/50	20, 72
1SBN050200R1001	RC5-2/133	20, 72
1SBN050200R1002	RC5-2/250	20, 72
1SBN050200R1003	RC5-2/440	20, 72
1SBN060300R1000	RA5-1	68
1SBN073508R1000	LD75	74
1SBN073552R1002	LK75-F	75
1SBN081406R1000	BEA16/116	20, 78
1SBN081411R1000	BER16V	78
1SBN082411R1000	BER40V	78
1SBN083501R1000	BEM75-30	78
1SBN083504R1000	BES75-30	79
1SBN110000R1000	BA5-50	75
1SBN060300T1000	RA5-1	68
1SBN089306T1000	BEA25/116	20, 78
1SBN089306T1001	BEA25/132	20, 78
1SCA101647R1001	OXS6X85	97
1SCA101655R1001	OXS6X130	97
1SCA101659R1001	OXS6X180	97
1SCA108043R1001	OXS6X105	97
1SFA616162R1014	KPR-101L	105, 106, 112
1SFL431074R8011	AX95-30-11-80	17
1SFL431074R8111	AX95-30-11-81	17
1SFL431074R8411	AX95-30-11-84	17
1SFL431074R8611	AX95-30-11-86	17
1SFL431074R8811	AX95-30-11-88	17
1SFL491074R8011	AX185-30-11-80	21
1SFL491074R8111	AX185-30-11-81	21
1SFL491074R8411	AX185-30-11-84	21
1SFL491074R8611	AX185-30-11-86	21
1SFL491074R8811	AX185-30-11-88	21
1SFL501074R8011	AX205-30-11-80	21
1SFL501074R8111	AX205-30-11-81	21
1SFL501074R8411	AX205-30-11-84	21
1SFL501074R8611	AX205-30-11-86	21
1SFL501074R8811	AX205-30-11-88	21
1SFL547074R8011	AX260-30-11-80	22
1SFL547074R8111	AX260-30-11-81	22
1SFL547074R8411	AX260-30-11-84	22
1SFL547074R8611	AX260-30-11-86	22
1SFL547074R8811	AX260-30-11-88	22
1SFL587074R8011	AX300-30-11-80	22
1SFL587074R8111	AX300-30-11-81	22
1SFL587074R8411	AX300-30-11-84	22
1SFL587074R8611	AX300-30-11-86	22
1SFL587074R8811	AX300-30-11-88	22
1SFL607074R8011	AX370-30-11-80	22
1SFL607074R8111	AX370-30-11-81	22
1SFL607074R8411	AX370-30-11-84	22
1SFL607074R8611	AX370-30-11-86	22
1SFL607074R8811	AX370-30-11-88	22
1SFL981074R8011	AX115-30-11-80	17
1SFL981074R8111	AX115-30-11-81	17
1SFL981074R8411	AX115-30-11-84	17
1SFL981074R8611	AX115-30-11-86	17
1SFL981074R8811	AX115-30-11-88	17
1SFL991074R8011	AX150-30-11-80	17
1SFL991074R8111	AX150-30-11-81	17
1SFL991074R8411	AX150-30-11-84	17
1SFL991074R8611	AX150-30-11-86	17
1SFL991074R8811	AX150-30-11-88	17
1SFN010820R1011	CAL19-11	25, 58
1SFN010820R3311	CAL19-11B	25, 58
1SFN019820R1011	CAL18X-11	19, 25, 58
1SFN019820R3311	CAL18X-11B	25, 58
1SFN030300R1000	VM19	25, 66
1SFN034700R1000	VM300H	19, 25, 66
1SFN034701R1000	VM300V	19, 25, 66
1SFN035003R1000	VM205/260	25, 66
1SFN035403R1000	VM370/400	25, 66
1SFN050300R1003	RC5-3/440	25, 72

Order code	Type	Page
1SFN074308R1000	LD110 (1)	74
1SFN074703R1000	LY185	77
1SFN074707R1000	LW185	25, 76
1SFN074710R1000	LX185	25, 76
1SFN074712R1000	LP185	77
1SFN075103R1000	LY300	77
1SFN075112R1000	LP300	77
1SFN075407R1000	LW370	25, 76
1SFN075410R1000	LX370	25, 76
1SFN084301R1000	BEM110-30 (1)	78
1SFN084304R1000	BES110 (1)	79
1SFN084503R1000	BED110 (1)	79
1SFN084701R1000	BEM185-30	78
1SFN084703R1000	BED145A	79
1SFN084704R1000	BES185	79
1SFN084706R1003	BEA185/T3	78
1SFN084903R1000	BED185	79
1SFN085406R1000	BEA370/T5	78
1SFN085411R1000	BER370-4	78
1SFN085414R1000	BEP370-30	79
1SFN085813R1000	BEY370-4	79
1SFN124701R1000	LT185-AC	25, 76
1SFN124703R1000	LT185-AL	25, 76
1SFN124704R1000	LT185-AY	25, 76
1SFN125401R1000	LT370-30C	25, 76
1SFN125403R1000	LT370-30L	25, 76
1SFN125404R1000	LT370-30Y	25, 76
1SNA235256R2700	BA5-50	75
1SNA235712R2400	HTP500-BA4	75
1SNA360010R1500	SPRC 1	75
FPEP407000R0001	LP16	77
FPEP407002R0001	LY16	77
FPTN372726R1001	WB75-A	20, 70
FPTN372726R1002	WB75-A	70
FPTN372726R1003	WB75-A	70
FPTN372726R1004	WB75-A	70
FPTN372726R1005	WB75-A	70
FPTN372726R1006	WB75-A	20, 70
FPTN372726R1007	WB75-A	70
FPTN372726R1008	WB75-A	70
GJF1101903R0001	SA1	94
GJF1101903R0002	SA2	94
GJF1101903R0003	SA3	94

Index

Type	Order code	Page
AA1-110	1SAM201910R1002	91
AA1-230	1SAM201910R1003	91
AA1-24	1SAM201910R1001	91
AA1-400	1SAM201910R1004	91
AX09-30-01-80	1SBL901074R8001	12
AX09-30-01-81	1SBL901074R8101	12
AX09-30-01-84	1SBL901074R8401	12
AX09-30-01-86	1SBL901074R8601	12
AX09-30-01-88	1SBL901074R8801	12
AX09-30-10-80	1SBL901074R8010	12
AX09-30-10-81	1SBL901074R8110	12
AX09-30-10-84	1SBL901074R8410	12
AX09-30-10-86	1SBL901074R8610	12
AX09-30-10-88	1SBL901074R8810	12
AX115-30-11-80	1SFL981074R8011	17
AX115-30-11-81	1SFL981074R8111	17
AX115-30-11-84	1SFL981074R8411	17
AX115-30-11-86	1SFL981074R8611	17
AX115-30-11-88	1SFL981074R8811	17
AX12-30-01-80	1SBL911074R8001	12
AX12-30-01-81	1SBL911074R8101	12
AX12-30-01-84	1SBL911074R8401	12
AX12-30-01-85	1SBL911074R8501	12
AX12-30-01-86	1SBL911074R8601	12
AX12-30-10-80	1SBL911074R8010	12
AX12-30-10-81	1SBL911074R8110	12
AX12-30-10-84	1SBL911074R8410	12
AX12-30-10-86	1SBL911074R8610	12
AX12-30-10-88	1SBL911074R8810	12
AX150-30-11-80	1SFL991074R8011	17
AX150-30-11-81	1SFL991074R8111	17
AX150-30-11-84	1SFL991074R8411	17
AX150-30-11-86	1SFL991074R8611	17
AX150-30-11-88	1SFL991074R8811	17
AX18-30-01-80	1SBL921074R8001	13
AX18-30-01-81	1SBL921074R8101	13
AX18-30-01-84	1SBL921074R8401	13
AX18-30-01-86	1SBL921074R8601	13
AX18-30-01-88	1SBL921074R8801	13
AX18-30-10-80	1SBL921074R8010	13
AX18-30-10-81	1SBL921074R8110	13
AX18-30-10-84	1SBL921074R8410	13
AX18-30-10-86	1SBL921074R8610	13
AX18-30-10-88	1SBL921074R8810	13
AX185-30-11-80	1SFL491074R8011	21
AX185-30-11-81	1SFL491074R8111	21
AX185-30-11-84	1SFL491074R8411	21
AX185-30-11-86	1SFL491074R8611	21
AX185-30-11-88	1SFL491074R8811	21
AX205-30-11-80	1SFL501074R8011	21
AX205-30-11-81	1SFL501074R8111	21
AX205-30-11-84	1SFL501074R8411	21
AX205-30-11-86	1SFL501074R8611	21
AX205-30-11-88	1SFL501074R8811	21
AX25-30-01-80	1SBL931074R8001	13
AX25-30-01-81	1SBL931074R8101	13
AX25-30-01-84	1SBL931074R8401	13
AX25-30-01-85	1SBL931074R8501	13
AX25-30-01-86	1SBL931074R8601	13
AX25-30-10-80	1SBL931074R8010	13
AX25-30-10-81	1SBL931074R8110	13
AX25-30-10-84	1SBL931074R8410	13
AX25-30-10-86	1SBL931074R8610	13
AX25-30-10-88	1SBL931074R8810	13
AX260-30-11-80	1SFL547074R8011	22
AX260-30-11-81	1SFL547074R8111	22
AX260-30-11-84	1SFL547074R8411	22
AX260-30-11-86	1SFL547074R8611	22
AX260-30-11-88	1SFL547074R8811	22
AX300-30-11-80	1SFL587074R8011	22
AX300-30-11-81	1SFL587074R8111	22
AX300-30-11-84	1SFL587074R8411	22

Type	Order code	Page
AX300-30-11-86	1SFL587074R8611	22
AX300-30-11-88	1SFL587074R8811	22
AX32-30-01-80	1SBL281074R8001	14
AX32-30-01-81	1SBL281074R8101	14
AX32-30-01-84	1SBL281074R8401	14
AX32-30-01-86	1SBL281074R8601	14
AX32-30-01-88	1SBL281074R8801	14
AX32-30-10-80	1SBL281074R8010	14
AX32-30-10-81	1SBL281074R8110	14
AX32-30-10-84	1SBL281074R8410	14
AX32-30-10-86	1SBL281074R8610	14
AX32-30-10-88	1SBL281074R8810	14
AX370-30-11-80	1SFL607074R8011	22
AX370-30-11-81	1SFL607074R8111	22
AX370-30-11-84	1SFL607074R8411	22
AX370-30-11-86	1SFL607074R8611	22
AX370-30-11-88	1SFL607074R8811	22
AX40-30-01-80	1SBL321074R8001	14
AX40-30-01-81	1SBL321074R8101	14
AX40-30-01-84	1SBL321074R8401	14
AX40-30-01-85	1SBL321074R8501	14
AX40-30-01-86	1SBL321074R8601	14
AX40-30-10-80	1SBL321074R8010	14
AX40-30-10-81	1SBL321074R8110	14
AX40-30-10-84	1SBL321074R8410	14
AX40-30-10-86	1SBL321074R8610	14
AX40-30-10-88	1SBL321074R8810	14
AX50-30-00-80	1SBL351074R8000	15
AX50-30-00-81	1SBL351074R8100	15
AX50-30-00-84	1SBL351074R8400	15
AX50-30-00-86	1SBL351074R8600	15
AX50-30-00-88	1SBL351074R8800	15
AX50-30-11-80	1SBL351074R8011	16
AX50-30-11-81	1SBL351074R8111	16
AX50-30-11-84	1SBL351074R8411	16
AX50-30-11-86	1SBL351074R8611	16
AX50-30-11-88	1SBL351074R8811	16
AX65-30-00-80	1SBL371074R8000	15
AX65-30-00-81	1SBL371074R8100	15
AX65-30-00-84	1SBL371074R8400	15
AX65-30-00-86	1SBL371074R8600	15
AX65-30-00-88	1SBL371074R8800	15
AX65-30-11-80	1SBL371074R8011	16
AX65-30-11-81	1SBL371074R8111	16
AX65-30-11-84	1SBL371074R8411	16
AX65-30-11-86	1SBL371074R8611	16
AX65-30-11-88	1SBL371074R8811	16
AX80-30-00-80	1SBL411074R8000	15
AX80-30-00-81	1SBL411074R8100	15
AX80-30-00-84	1SBL411074R8400	15
AX80-30-00-86	1SBL411074R8600	15
AX80-30-00-88	1SBL411074R8800	15
AX80-30-11-80	1SBL411074R8011	16
AX80-30-11-81	1SBL411074R8111	16
AX80-30-11-84	1SBL411074R8411	16
AX80-30-11-86	1SBL411074R8611	16
AX80-30-11-88	1SBL411074R8811	16
AX95-30-11-80	1SFL431074R8011	17
AX95-30-11-81	1SFL431074R8111	17
AX95-30-11-84	1SFL431074R8411	17
AX95-30-11-86	1SFL431074R8611	17
AX95-30-11-88	1SFL431074R8811	17
BA5-50	1SBN110000R1000	75
BA5-50	1SNA235256R2700	75
BEA16/116	1SBN081406R1000	20, 78
BEA185/T3	1SBN084706R1003	78
BEA25/116	1SBN089306T1000	20, 78
BEA25/132	1SBN089306T1001	20, 78
BEA370/T5	1SBN085406R1000	78
BED110 (1)	1SBN084503R1000	79
BED145A	1SBN084703R1000	79
BED185	1SBN084903R1000	79

Type	Order code	Page
BEM110-30 (1)	1SBN084301R1000	78
BEM185-30	1SBN084701R1000	78
BEM75-30	1SBN083501R1000	78
BEP370-30	1SBN085414R1000	79
BER16V	1SBN081411R1000	78
BER370-4	1SBN085411R1000	78
BER40V	1SBN082411R1000	78
BES110 (1)	1SBN084304R1000	79
BES185	1SBN084704R1000	79
BES75-30	1SBN083504R1000	79
BEY370-4	1SBN085813R1000	79
BS1-3	1SAM201908R1001	94
CA5X-01	1SBN019010R1001	19, 58
CA5X-04E	1SBN019040R1004	58
CA5X-04M	1SBN019040R1104	58
CA5X-04N	1SBN019040R1204	58
CA5X-04U	1SBN019040R1304	58
CA5X-10	1SBN019010R1010	19, 58
CA5X-13M	1SBN019040R1113	58
CA5X-13N	1SBN019040R1213	58
CA5X-22E	1SBN019040R1022	19, 58
CA5X-22M	1SBN019040R1122	19, 58
CA5X-22N	1SBN019040R1222	58
CA5X-22U	1SBN019040R1322	58
CA5X-31E	1SBN019040R1031	58
CA5X-31M	1SBN019040R1131	58
CA5X-31N	1SBN019040R1231	58
CA5X-31U	1SBN019040R1331	58
CA5X-40E	1SBN019040R1040	58
CA5X-40N	1SBN019040R1240	58
CA5X-40U	1SBN019040R1340	58
CAL18X-11	1SBN019820R1011	19, 25, 58
CAL18X-11B	1SBN019820R3311	25, 58
CAL19-11	1SBN010820R1011	25, 58
CAL19-11B	1SBN010820R3311	25, 58
CAL5X-11	1SBN019020R1011	19, 58
CE5-01D0.1	1SBN010015R1001	60
CE5-01D2	1SBN010017R1001	60
CE5-01W0.1	1SBN010016R1001	60
CE5-01W2	1SBN010018R1001	60
CE5-10D0.1	1SBN010015R1010	60
CE5-10D2	1SBN010017R1010	60
CE5-10W0.1	1SBN010016R1010	60
CE5-10W2	1SBN010018R1010	60
DB200	1SAZ401110R0001	106
DB25/25A	1SAZ201108R0001	105
DB25/32A	1SAZ201108R0002	105
DB80	1SAZ301110R0001	105, 106
DMS132-G	1SAM201912R1010	96
DMS132-Y	1SAM201912R1011	96
DX25	1SAZ201307R0002	105
EF205-210	1SAX531001R1101	112
EF370-380	1SAX611001R1101	112
FS116	1SAM201909R1001	94
HK1-02	1SAM201902R1003	90
HK1-11	1SAM201902R1001	90
HK1-20	1SAM201902R1002	90
HK1-20L	1SAM201902R1004	90
HKF1-01	1SAM201901R1004	90
HKF1-10	1SAM201901R1003	90
HKF1-11	1SAM201901R1001	90
HKF1-20	1SAM201901R1002	90
HTP500-BA4	1SNA235712R2400	75
IB132-G	1SAM201911R1010	96
IB132-Y	1SAM201911R1011	96
KPR-101L	1SFA616162R1014105, 106, 112	
LD110 (1)	1SBN074308R1000	74
LD75	1SBN073508R1000	74
LK75-F	1SBN073552R1002	75
LP16	FPEP407000R0001	77

Index

Type	Order code	Page
LP185	1SFN074712R1000	77
LP300	1SFN075112R1000	77
LT185-AC	1SFN124701R1000	25, 76
LT185-AL	1SFN124703R1000	25, 76
LT185-AY	1SFN124704R1000	25, 76
LT200/A	1SAZ401901R1001	106
LT200E	1SAX501904R0001	112
LT320E	1SAX601904R0001	112
LT370-30C	1SFN125401R1000	25, 76
LT370-30L	1SFN125403R1000	25, 76
LT370-30Y	1SFN125404R1000	25, 76
LW185	1SFN074707R1000	25, 76
LW370	1SFN075407R1000	25, 76
LX185	1SFN074710R1000	25, 76
LX370	1SFN075410R1000	25, 76
LY16	FPEP407002R0001	77
LY185	1SFN074703R1000	77
LY300	1SFN075103R1000	77
MS116-0.16	1SAM250000R1001	85
MS116-0.16-HKF1-11	1SAM250005R1001	85
MS116-0.25	1SAM250000R1002	85
MS116-0.25-HKF1-11	1SAM250005R1002	85
MS116-0.4	1SAM250000R1003	85
MS116-0.4-HKF1-11	1SAM250005R1003	85
MS116-0.63	1SAM250000R1004	85
MS116-0.63-HKF1-11	1SAM250005R1004	85
MS116-1.0	1SAM250000R1005	85
MS116-1.0-HKF1-11	1SAM250005R1005	85
MS116-1.6	1SAM250000R1006	85
MS116-1.6-HKF1-11	1SAM250005R1006	85
MS116-10	1SAM250000R1010	85
MS116-10.0-HKF1-11	1SAM250005R1010	85
MS116-12	1SAM250000R1012	85
MS116-12.0-HKF1-11	1SAM250005R1012	85
MS116-16	1SAM250000R1011	85
MS116-16.0-HKF1-11	1SAM250005R1011	85
MS116-2.5	1SAM250000R1007	85
MS116-2.5-HKF1-11	1SAM250005R1007	85
MS116-20	1SAM250000R1013	85
MS116-20-HKF1-11	1SAM250005R1013	85
MS116-25	1SAM250000R1014	85
MS116-25-HKF1-11	1SAM250005R1014	85
MS116-32	1SAM250000R1015	85
MS116-32-HKF1-11	1SAM250005R1015	85
MS116-4.0	1SAM250000R1008	85
MS116-4.0-HKF1-11	1SAM250005R1008	85
MS116-6.3	1SAM250000R1009	85
MS116-6.3-HKF1-11	1SAM250005R1009	85
MSAH1	1SAM201909R1021	97
MSH-AR	1SAM201920R1000	97
MSHD-LB (1)	1SAM201920R1001	97
MSHD-LTB (2)	1SAM201920R1011	97
MSHD-LTY (2)	1SAM201920R1012	97
MSHD-LY (1)	1SAM201920R1002	97
MSMN (3)	1SAM101923R0002	97
MSMNO (4)	1SAM101923R0012	97
NX22E-80	1SBH901074R8022	52
NX22E-81	1SBH901074R8122	52
NX22E-84	1SBH901074R8422	52
NX22E-86	1SBH901074R8622	52
NX22E-88	1SBH901074R8822	52
NX31E-80	1SBH901074R8031	52
NX31E-81	1SBH901074R8131	52
NX31E-84	1SBH901074R8431	52
NX31E-86	1SBH901074R8631	52
NX31E-88	1SBH901074R8831	52
NX40E-80	1SBH901074R8040	52
NX40E-81	1SBH901074R8140	52
NX40E-84	1SBH901074R8440	52
NX40E-86	1SBH901074R8640	52
NX40E-88	1SBH901074R8840	52
OXS6X105	1SCA108043R1001	97

Type	Order code	Page
OXS6X130	1SCA101655R1001	97
OXS6X180	1SCA101659R1001	97
OXS6X85	1SCA101647R1001	97
PB1-1-32	1SAM201914R1001	94
PS1-2-0-65	1SAM201906R1102	94
PS1-2-1-65	1SAM201906R1112	94
PS1-2-2-65	1SAM201906R1122	94
PS1-3-0-100	1SAM201916R1103	94
PS1-3-0-65	1SAM201906R1103	94
PS1-3-1-100	1SAM201916R1113	94
PS1-3-1-65	1SAM201906R1113	94
PS1-3-2-100	1SAM201916R1123	94
PS1-3-2-65	1SAM201906R1123	94
PS1-4-0-100	1SAM201916R1104	94
PS1-4-0-65	1SAM201906R1104	94
PS1-4-1-100	1SAM201916R1114	94
PS1-4-1-65	1SAM201906R1114	94
PS1-4-2-65	1SAM201906R1124	94
PS1-5-0-100	1SAM201916R1105	94
PS1-5-0-65	1SAM201906R1105	94
PS1-5-1-100	1SAM201916R1115	94
PS1-5-1-65	1SAM201906R1115	94
PS1-5-2-65	1SAM201906R1125	94
RA5-1	1SBN060300R1000	68
RA5-1	1SBN060300T1000	68
RC5-1/133	1SBN050100R1001	20, 72
RC5-1/250	1SBN050100R1002	20, 72
RC5-1/440	1SBN050100R1003	20, 72
RC5-1/50	1SBN050100R1000	20, 72
RC5-2/133	1SBN050200R1001	20, 72
RC5-2/250	1SBN050200R1002	20, 72
RC5-2/440	1SBN050200R1003	20, 72
RC5-2/50	1SBN050200R1000	20, 72
RC5-3/440	1SBN050300R1003	25, 72
RV5/133	1SBN050010R1001	20, 72
RV5/250	1SBN050010R1002	20, 72
RV5/440	1SBN050010R1003	20, 72
RV5/50	1SBN050010R1000	20, 72
S1-M1-25	1SAM201907R1101	94
S1-M2-25	1SAM201907R1102	94
S1-M3-25	1SAM201907R1103	94
S1-M3-35	1SAM201913R1103	94
S1-PB1-25	1SAM201914R1002	94
SA1	GJF1101903R0001	94
SA2	GJF1101903R0002	94
SA3	GJF1101903R0003	94
SK1-02	1SAM201903R1003	90
SK1-11	1SAM201903R1001	90
SK1-20	1SAM201903R1002	90
SPRC 1	1SNA360010R1500	75
TA110DU-110M	1SAZ411201R2002	106
TA110DU-90M	1SAZ411201R2001	106
TA200DU-110	1SAZ421201R1002	106
TA200DU-135	1SAZ421201R1003	106
TA200DU-150	1SAZ421201R1004	106
TA200DU-175	1SAZ421201R1005	106
TA200DU-200	1SAZ421201R1006	106
TA200DU-90	1SAZ421201R1001	106
TA25DU-0.16M	1SAZ211201R2005	101
TA25DU-0.25M	1SAZ211201R2009	101
TA25DU-0.4M	1SAZ211201R2013	101
TA25DU-0.63M	1SAZ211201R2017	101
TA25DU-1.0M	1SAZ211201R2021	101
TA25DU-1.4M	1SAZ211201R2023	101
TA25DU-1.8M	1SAZ211201R2025	101
TA25DU-11M	1SAZ211201R2043	101
TA25DU-14M	1SAZ211201R2045	101
TA25DU-19M	1SAZ211201R2047	101
TA25DU-2.4M	1SAZ211201R2028	101
TA25DU-25M	1SAZ211201R2051	101
TA25DU-3.1M	1SAZ211201R2031	101
TA25DU-32M	1SAZ211201R2053	101

Type	Order code	Page
TA25DU-4.0M	1SAZ211201R2033	101
TA25DU-5.0M	1SAZ211201R2035	101
TA25DU-6.5M	1SAZ211201R2038	101
TA25DU-8.5M	1SAZ211201R2040	101
TA42DU-25M	1SAZ311201R2001	101
TA42DU-32M	1SAZ311201R2002	101
TA42DU-42M	1SAZ311201R2003	101
TA75DU-25M	1SAZ321201R2001	101
TA75DU-32M	1SAZ321201R2002	101
TA75DU-42M	1SAZ321201R2003	101
TA75DU-52M	1SAZ321201R2004	101
TA75DU-63M	1SAZ321201R2005	101
TA75DU-80M	1SAZ321201R2006	101
TA80DU-42M	1SAZ331201R2003	106
TA80DU-52M	1SAZ331201R2004	106
TA80DU-63M	1SAZ331201R2005	106
TA80DU-80M	1SAZ331201R2006	106
TEF5-OFF	1SBN020314R1000	20, 63
TEF5-ON	1SBN020312R1000	20, 63
UA1-110	1SAM201904R1004	91
UA1-20	1SAM201904R1010	91
UA1-208	1SAM201904R1008	91
UA1-230	1SAM201904R1005	91
UA1-24	1SAM201904R1001	91
UA1-400	1SAM201904R1006	91
UA1-415	1SAM201904R1007	91
UA1-48	1SAM201904R1002	91
UA1-575	1SAM201904R1009	91
UA1-60	1SAM201904R1003	91
VE5-1	1SBN030110R1000	19, 66
VE5-2	1SBN030210R1000	19, 66
VM19	1SFN030300R1000	25, 66
VM205/260	1SFN035003R1000	25, 66
VM300H	1SFN034700R1000	19, 25, 66
VM300V	1SFN034701R1000	19, 25, 66
VM370/400	1SFN035403R1000	25, 66
VM5-1	1SBN030100R1000	19, 66
WB75-A	FPTN372726R1001	20, 70
WB75-A	FPTN372726R1002	70
WB75-A	FPTN372726R1003	70
WB75-A	FPTN372726R1004	70
WB75-A	FPTN372726R1005	70
WB75-A	FPTN372726R1006	20, 70
WB75-A	FPTN372726R1007	70
WB75-A	FPTN372726R1008	70



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